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# A survey on electronic health record driven multimodal representation learning

Yutao Dou[a], Yaoyu Liu[a], Haitao Zou[a], Wenwu Zeng[a], Kele Xu[b], Shaoliang Peng[a,*]

[a] College of Computer Science and Electronic Engineering, Hunan University, Changsha, 410082, China
[b] College of Computer Science and Technology, National University of Defense Technology, Changsha, 410073, China

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ABSTRACT

Electronic health records (EHRs) capture extensive multimodal information, such as clinical narratives, imaging, physiological waveforms and laboratory results, yielding richly detailed yet heterogeneous portraits of patient status. Leveraging these complementary sources is pivotal for precision medicine, and recent advances in multimodal machine learning have made such integration increasingly tractable. We present a critical survey of multimodal learning for healthcare, organized around the predominant "text + X" paradigm that arises because imaging, waveform and laboratory findings are routinely encoded or summarized in free-text reports (e.g., radiology or ECG summaries). From this vantage, we review: representation learning, fusion strategies, and clinical downstream tasks, highlighting how each technical choice aligns or conflicts with the underlying semantics and statistical properties of the modalities involved. Throughout, we emphasize concepts such as cross-modal semantic consistency (e.g., lesion descriptors matching visual regions) and modality-specific noise characteristics, drawing connections to current modeling techniques. By exposing mismatches between data properties and algorithmic assumptions, we outline open challenges and propose directions toward more unified, clinically deployable multimodal frameworks. Addressing these gaps, we argue, is essential for translating methodological innovation into reliable decision support systems in real-world medical settings.

## 1. Introduction

Electronic health records (EHRs) lie at the core of modern clinical practice and biomedical research, serving as the principal repositories of patient information [1,2]. By design, EHRs are inherently *multimodal*: they encompass unstructured clinical narratives and diagnostic reports, high-resolution medical images (e.g., radiographs, CT, MRI), and diverse physiological waveforms (e.g., ECG, EMG). Integrating these complementary data streams yields a holistic view of patient status and underpins personalized decision-making.

Historically, each modality was analyzed in isolation natural language processing for text, computer vision for images, and digital signal processing for waveforms. Recent breakthroughs in artificial intelligence have catalyzed multimodal learning frameworks that jointly model heterogeneous inputs, mirroring the brain's ability to fuse sensory cues. This paradigm shift has invigorated research into cross-modal interactions and their translational impact on clinical care.

In routine EHR workflows, many non-textual findings are retained as free-text summaries (e.g., radiology or cardiology reports), positioning text as the *pivot* modality around which imaging and signal data provide corroborative evidence. Each modality offers unique diagnostic value: imaging reveals anatomical or pathological morphology; signals capture dynamic physiological function; and text encodes clinician reasoning and longitudinal context. Their concerted use promises deeper mechanistic insight and more accurate prediction of complex disease trajectories. Fig. 1 illustrates the potential of AI in integrating diverse modalities within EHR.

This review systematically dissects progress in EHR-centric multimodal learning. We (i) survey representation learning for individual modalities, contrasting supervised approaches that rely on labeled datasets with self-supervised paradigms that exploit abundant unlabeled records, as well as the neural architectures that underpin them; (ii) evaluate fusion strategies tailored to varying clinical objectives; and (iii) catalogue downstream applications, grouping them into *discriminative* and *generative* tasks. Our comparative analysis reveals recurring principles that link data characteristics to algorithmic design, thereby clarifying when particular techniques succeed or fail. We further distill open challenges and prospective research directions.

* Corresponding author.
*E-mail addresses:* ytdou@hnu.edu.cn (Y. Dou), slpeng@hnu.edu.cn (S. Peng).

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# A 综述： 基于 电子 健康 档案 驱动的 多模态 表示学习

Yutao Dou[a], Yaoyu Liu[a], Haitao Zou[a], Wenwu Zeng[a], Kele Xu[b], Shaoliang Peng[a,*]

[a] 湖南 大学 计算机 科学 与 电子 工程学院， 中国 长沙， 410082 [b]国防 科技 大学 计算机 学院， 中国 长沙， 410073

文章 信息

关键词：电子 健康 档案 LLM 多模态 表示 学习 多模态 融合 跨模态 生成

摘要

电子健康记录（EHRs）捕获了广泛的多模态信息，如临床叙述、影像、生理波形和实验室结果，从而生成丰富详细但异质的患者状态画像。利用这些互补来源对于精准医学至关重要，而多模态机器学习的最新进展使这种整合变得越来越可行。我们提出了一项关于医疗健康多模态学习的批判性综述，围绕主流的“文本+ X”范式展开，该范式的产生是因为影像、波形和实验室发现通常被编码或总结在自由文本报告中（例如，放射学或ECG摘要）。从这个视角出发，我们回顾了：表示学习、融合策略和临床下游任务，重点强调每种技术选择如何与所涉及模态的底层语义和统计特性相一致或相冲突。贯穿全文，我们强调诸如跨模态语义一致性（例如，病灶描述符与视觉区域匹配）和模态特定噪声特征等概念，并将其与当前建模技术相联系。通过揭示数据属性与算法假设之间的不匹配，我们概述了开放挑战，并提出了迈向更统一、临床可部署的多模态框架的方向。我们认为，解决这些差距对于将方法学创新转化为现实医疗环境中的可靠决策支持系统至关重要。

## 1. 引言

电子健康记录（EHRs）处于现代临床实践和生物医学研究的核心，是患者信息[1,2]的主要存储库。EHRs 在设计上本质上是多模态的：它们包含非结构化的临床叙述和诊断报告、高分辨率的医学图像（例如 X射线图像、CT、MRI），以及各种生理波形（例如 心电图、EMG）。整合这些互补的数据流可全面了解患者状态，并支撑个性化决策。

历史上，每种模态都是孤立分析的——文本用自然语言处理，图像用计算机视觉，波形用数字信号处理。人工智能领域的最新突破催生了多模态学习框架，能够联合建模异构输入，模拟大脑融合感官线索的能力。这一范式转变激发了跨模态交互及其对临床护理转化影响的研究。

在 常规 EHR 工作流程中， 许多 非文本 发现 被 保留 为自由文本 摘要 （例如， 放射学 或 心脏病学 报告）， 将文本定位为枢纽模态，影像和信号数据围绕其提供佐证。每种模态都具有独特的诊断价值：影像揭示解剖或病理形态；信号捕捉动态生理功能；文本编码临床医生的推理和纵向背景。它们的协同使用有望带来更深入的机制洞察，并更准确地预测复杂疾病轨迹。图1展示了AI在EHR中整合多种模态的潜力。

本综述系统性地剖析了以电子健康记录为中心的多模态学习进展。我们（i）综述了针对单个模态的表示学习，对比了依赖标注数据集的监督方法与利用大量无标注记录的自监督范式，以及支撑它们的神经网络架构；（ii）评估了针对不同临床目标量身定制的融合策略；（iii）对下游应用进行了编目，将其分为判别任务和生成任务。我们的比较分析揭示了将数据特征与算法设计联系起来的反复出现的规律，从而阐明了特定技术何时成功、何时失败。我们进一步提炼了开放性挑战和前瞻性研究方向。

*通讯 作者。 电子邮箱 地址： ytdou@hnu.edu.cn （Y. Dou），slpeng@hnu.edu.cn （S. Peng）。

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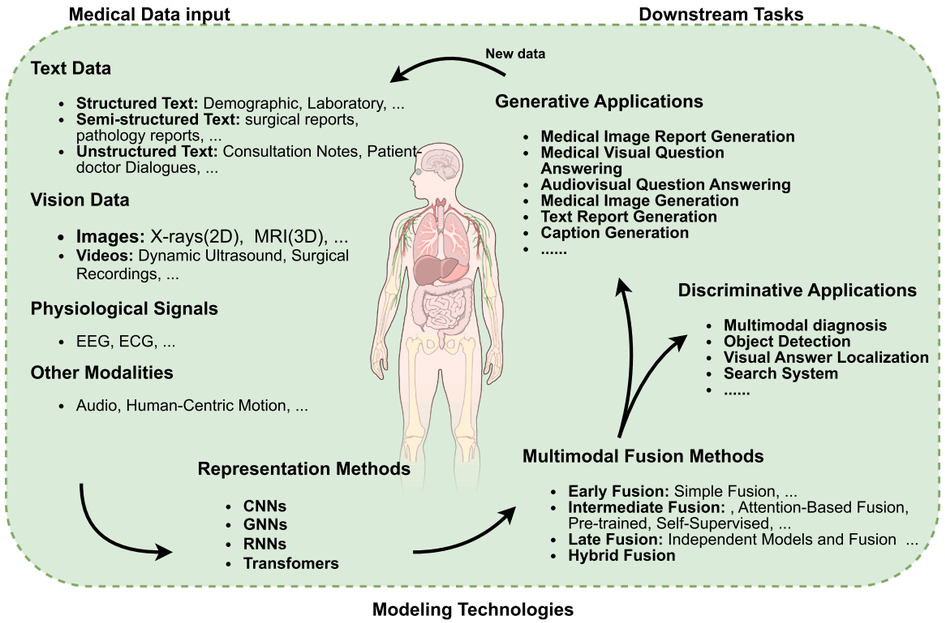



**Fig. 1.** AI technology can effectively leverage multimodal EHR data.

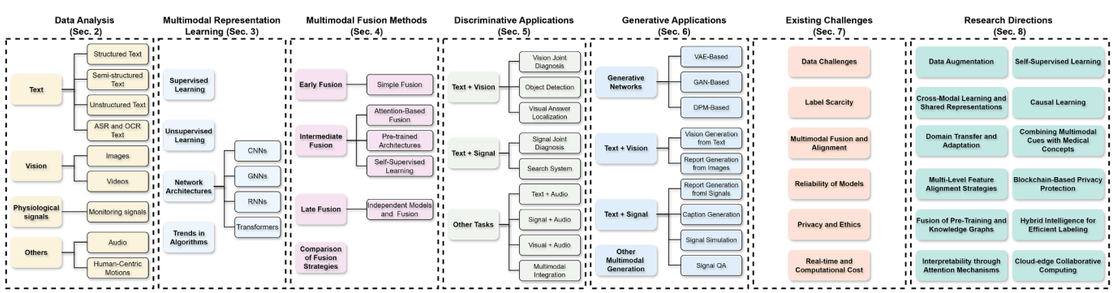



**Fig. 2.** The overall structure of our survey is as follows: First, we introduce common EHR data types. Next, it covers multimodal representation learning, categorizing studies by approach and discussing algorithm trends. Then, data fusion methods are explored. Applications are highlighted, divided into discriminative and generative tasks, classified by modality combinations. Finally, challenges and future research directions are addressed.

*Contributions.*

1. **Systematic review.** We catalogue about 200 studies spanning early data-level fusion to state-of-the-art large multimodal models, providing the most up-to-date panorama of multimodal learning in EHRs.
2. **Modality-task mapping.** We articulate how specific modality combinations align with clinical questions, elucidating the role multimodal learning plays in medical reasoning and decision support.
3. **Road-map for progress.** We identify six pressing challenges and propose twelve concrete avenues for future work, furnishing actionable guidance for the field.

Distinct from prior surveys [3–7], our discussion is anchored in *real-world* EHR practice. We argue that (i) exploiting modality-specific properties can resolve niche clinical bottlenecks, while (ii) embracing shared cross-modal structures paves the way toward unified, collaborative frameworks with higher predictive fidelity and clinical utility.

*Paper Organization.* Fig. 2 outlines the review. Section 2 characterizes prevalent EHR modalities. Section 3 surveys representation methods and trends. Section 4 details fusion techniques. Sections 5 and 6 discuss discriminative and generative applications, respectively, organized by modality pairing. We summarize the challenges encountered in Section 7 and outline potential future directions for their improvement in Section 8. Finally, Section 9 provides a summary of the overall contributions of this work.

## 2. Data analysis

In this section, we reveal the intrinsic nature of different data modalities in EHRs by deeply analyzing their features and commonalities. Table 1 presents commonly-used multimodal EHR datasets and competition benchmarks, including detailed descriptions of their modalities and a list of applicable research domains.

### 2.1. Text

Text modality is one of the most critical information carriers in EHRs. As a core component of healthcare information systems, text data often





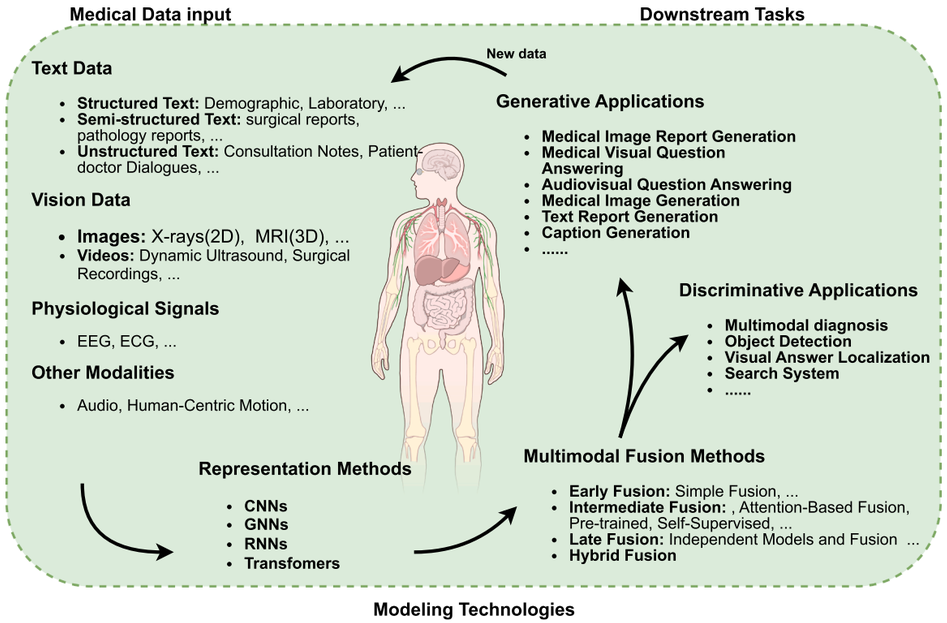



**图 1.** 人工智能 技术 能够 有效地 利用 多模态 电子健康记录 数据。

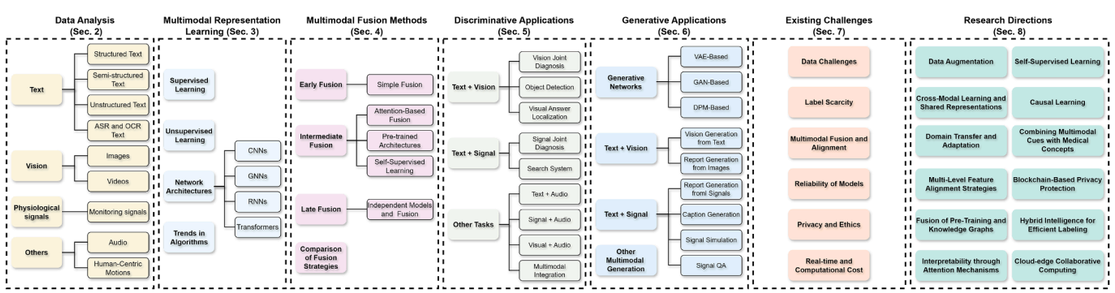



图2. 本综述的整体结构如下：首先，我们介绍常见的电子健康记录数据类型。接着，涵盖多模态表示学习，按方法对研究进行分类并讨论算法趋势。然后，探讨数据融合方法。重点介绍应用，分为判别任务和生成任务，按模态组合进行分类。最后，讨论挑战和未来研究方向。

贡献。

1. 系统综述。我们编目了约200项研究，涵盖从早期数据级融合到最先进的大型多模态模型，提供了电子健康记录中多模态学习的最新全景。2. 模态-任务映射。我们阐述了特定模态组合如何与临床问题对齐，阐明了多模态学习在医学推理和决策支持中所发挥的作用。3. 进展路线图。我们识别了六项紧迫挑战，并提出了十二条具体的未来工作方向，为该领域提供了可操作的指导。

与以往的综述不同，[3–7]，我们的讨论立足于真实世界的电子健康记录实践。我们认为：(i) 利用模态特有的属性可以解决特定的临床瓶颈，而 (ii) 拥抱共享的跨模态结构为构建具有更高预测保真度和临床实用性的统一协作框架铺平了道路。

论文 组织结构。 图 2 概述了 本文的 综述内容。 第 2 节描述了 常见的 电子健康记录 模态。 第 3 节综述了 表示 方法 及趋势。第4节详细介绍了融合技术。第5节和第6节分别讨论了判别式和生成式应用，并按模态配对进行组织。我们在第7节总结了所遇到的挑战，并在第8节概述了改进这些挑战的潜在未来方向。最后，第9节对本工作整体贡献进行了总结。

## 2. 数据分析

在本节中，我们通过深入分析EHR中不同数据模态的特征和共性，揭示其内在本质。表1列出了常用的多模态EHR数据集和竞赛基准，包括对其模态的详细描述以及适用研究领域的列表。

### 2.1. 文本

文本 模态 是 EHRs 中 最 关键 的 信息 载体 之一。 作为 医疗健康 信息 系统 的 核心 组成部分， 文本 数据 通常

records the patient's diagnosis and treatment process, the content of medical advice, and multidimensional health information in a highly condensed manner. Compared with the original high-dimensional image information in the visual modality, text is usually more compact, clearly structured and semantically explicit. According to the different structural levels and data generation methods, text data in EHRs can be roughly divided into three categories: structured text, semi-structured text, and unstructured text. Different types of textual modalities have significant differences in their representation structures and modeling approaches, imposing different technical requirements in multimodal learning tasks.

#### 2.1.1. Structured text

Structured text in EHRs is organized in predefined formats, such as tables or fields, allowing direct retrieval and parsing by database systems. Examples include demographic data, laboratory results, and genomic information. Unlike high-dimensional data like images, structured text provides a standardized, quantitative representation of clinical indicators, such as age or blood glucose levels, ensuring high interpretability and efficiency from data generation.

Traditional machine learning methods process structured text through manual feature engineering or statistical analysis, followed by predictive algorithms such as XGBoost. These approaches offer interpretability because features like importance scores or decision tree paths reveal the model's reasoning, which helps build trust in diagnostic and predictive tasks. ITFG [8] is a tree-based feature enhancement method that enhances clinical data in an interpretable manner and integrates with an XGBoost classifier to enable early disease detection. ETCXGB [9] is a hybrid model that combines ExtraTreesClassifier (ETC) with XGBoost, where ETC first generates prediction probabilities to enhance the feature space, and XGBoost then performs the learning.

The development of deep neural networks and large-scale data integration methods has been widely applied. However, missing data due to varying patient tests poses challenges for model development [10]. Structured text supports discriminative tasks, such as disease prediction [11], diagnosis [12], and classification [13]. Emerging generative models also use structured data to infer other modalities, expanding its role in multimodal medical systems [14].

#### 2.1.2. Semi-structured text

Semi-structured text combines fixed fields with free-text descriptions, common in surgical or pathology reports. For example, radiology reports pair quantitative metrics, like lesion size, with narrative descriptions of tissue conditions, balancing standardization and flexibility. This format captures comprehensive clinical details, supporting natural language processing (NLP) and information extraction.

Historically, semi-structured text analysis relied on rule-based methods and manual feature engineering, using dictionaries or regular expressions to extract medical terms. These were combined with structured data for classification or clustering via traditional machine learning. Modern pre-trained language models, such as BERT [15] and GPT [16], leverage attention mechanisms to model relationships between fixed fields and free text, improving analysis accuracy. Semi-structured EHRs are widely used in information extraction, NLP, and multimodal learning. They support clinical decision-making by extracting medical entities and relationships. Generative models further extend their application by generating cross-modal data, enhancing multimodal medical systems.

#### 2.1.3. Unstructured text

Unstructured text, found in consultation notes or patient-doctor dialogues, is highly flexible and diverse. It captures rich semantic information, including symptoms, disease progression, lifestyle, and psychological factors. In dialogues, coherence across conversational turns is critical for accurate clinical reasoning.

Processing unstructured text traditionally involved rule-based or template-based methods to identify medical entities, followed by sequential models like Recurrent Neural Networks (RNNs) or Long Short-Term Memory networks (LSTMs) [17,18]. These models use memory states to capture word relationships within sentences. Large language models (LLMs), using attention mechanisms [19], now excel in tasks such as classification, relationship extraction, and question answering. Their pretraining on vast corpora enhances multi-task learning and contextual understanding [20].

#### 2.1.4. ASR and OCR text

Automatic speech recognition (ASR) and optical character recognition (OCR) texts connect directly to audio and visual modalities. Unlike high-level semantic captions, ASR and OCR provide precise text-modality correspondence, supporting mature multimodal research [21]. OCR also extracts text from visual corpora, broadening its utility in medical informatics.

### 2.2. Vision

Medical visual data, including images and videos, are classified by imaging methods and dimensionality [22]. Images range from two-dimensional (2D) formats, like X-rays, to three-dimensional (3D) formats, such as CT scans. Videos include dynamic recordings, like ultrasound or surgical footage. These data are critical for diagnostics, surgical planning, and research, requiring high precision and security due to their complex, multi-source nature. They support discriminative tasks (e.g., classification, segmentation) and generative tasks (e.g., synthesis, cross-modal transformations).

#### 2.2.1. Images

Medical images, prevalent in clinical and research settings, are divided into 2D and 3D categories based on dimensionality. 2D images, such as X-rays or pathology slides, capture planar or surface views of organs or lesions. 3D images, like CT or MRI scans, represent internal structures and spatial relationships. Medical images exhibit transformation invariance (e.g., translation, rotation), enabling robust analysis through classical image processing and deep learning methods, which extract valuable diagnostic features in varied clinical contexts.

*2D Images:* Common 2D images include chest X-rays, ultrasound stills, pathology slides, and dermatological photographs. These images, capturing single-perspective views, support large datasets and diverse applications. Early analysis used manual feature extraction with algorithms like SIFT or HOG. Deep convolutional neural networks, such as U-Net, have advanced tasks like classification, lesion detection, and organ segmentation. Generative neural networks further enable image synthesis and cross-modal data generation, expanding their utility.

*3D Images:* 3D images, such as CT or MRI scans, compile multiple slices or voxels to depict organ structures and spatial relationships, offering richer detail than 2D images. This complexity demands greater computational resources and analysis of voxel or slice relationships. Extending 2D convolutional neural networks to 3D architectures, like 3D U-Net, supports tasks such as volume segmentation and organ reconstruction. Core tasks mirror 2D applications, including classification and segmentation, with growing interest in generative tasks like 3D data synthesis.

#### 2.2.2. Videos

Medical videos, including dynamic ultrasound and surgical recordings, capture temporal dynamics, such as organ motion or procedural steps, essential for clinical diagnosis and surgical assistance. Unlike static images, videos require temporal analysis, as seen in cardiac cycles in ultrasound or sequential steps in surgical footage. Traditional image processing methods are inadequate, necessitating architectures that incorporate the temporal dimension.





以高度凝练的方式记录患者的诊断和治疗过程、医嘱内容以及多维度的健康信息。与视觉模态中原始的高维图像信息相比，文本通常更加紧凑、结构清晰且语义明确。根据不同的结构层次和数据生成方式，EHRs中的文本数据大致可分为三类：结构化文本、半结构化文本和非结构化文本。不同类型的文本模态在表示结构和建模方法上存在显著差异，对多模态学习任务提出了不同的技术要求。

#### 2.1.1. 结构化 文本

电子健康记录中的结构化文本以预定义格式组织，例如表格或字段，允许数据库系统直接检索和解析。示例包括人口统计数据、实验室结果和基因组信息。与图像等高维数据不同，结构化文本提供了临床指标（如年龄或血糖水平）的标准化、定量表示，确保从数据生成中获得高可解释性和效率。

传统机器学习方法通过手动特征工程或统计分析处理结构化文本，然后使用XGBoost等预测算法。这些方法具有可解释性，因为特征重要性分数或决策树路径等特征揭示了模型的推理过程，有助于在诊断和预测任务中建立信任。ITFG [8] 是一种基于树的特征增强方法，以可解释的方式增强临床数据，并与XGBoost分类器集成，以实现早期疾病检测。ETCXGB [9] 是一种混合模型，将极端随机树分类器（ETC）与XGBoost相结合，其中ETC首先生成预测概率以增强特征空间，然后XGBoost执行学习。

深度神经网络与大规模数据集成方法的发展已得到广泛应用。然而，由于患者检查项目不同而导致的数据缺失，给模型开发带来了挑战[10]。结构化文本支持判别任务，如疾病预测[11]，诊断[12]，和分类[13]。新兴的生成模型也利用结构化数据来推断其他模态，从而扩展了其在多模态医疗系统中的作用[14]。

#### 2.1.2. 半结构化 文本

半结构化文本将固定字段与自由文本描述相结合，常见于外科或病理学报告中。例如，放射学报告将病灶大小等定量指标与组织状况的叙述性描述配对，兼顾标准化与灵活性。这种格式能够捕捉全面的临床细节，支持自然语言处理（NLP）和信息抽取。

历史上，半结构化文本分析依赖于基于规则的方法和人工特征工程，使用词典或正则表达式来提取医学术语。这些方法与结构化数据相结合，通过传统机器学习进行分类或聚类。现代预训练语言模型，如BERT[15]和GPT[16]，，利用注意力机制来建模固定字段与自由文本之间的关系，从而提高分析准确率。半结构化电子健康记录广泛应用于信息抽取、NLP和多模态学习。它们通过提取医学实体和关系来支持临床决策。生成模型通过生成跨模态数据进一步扩展了其应用，增强了多模态医疗系统。

#### 2.1.3. 非结构化 文本

非结构化 文本， 存在于 问诊 记录 或 医患 对话 中， 具有 高度 灵活 性 和 多样 性。 它 捕捉 丰富的 语义 信息， 包括 症状、 疾病 进展、 生活方式 以及 心理 因素。 在 对话 中， 对话 轮次 之间的 连贯 性 对于 准确 的 临床 推理 至关重要。

传统上，处理非结构化文本涉及基于规则或基于模板的方法来识别医学实体，随后使用循环神经网络（RNN）或长短期记忆网络（LSTM）[17,18]等序列模型。这些模型利用记忆状态来捕捉句子中的词语关系。大语言模型（LLM）利用注意力机制[19]，，目前在分类、关系抽取和问答等任务中表现出色。其在大规模语料上的预训练增强了多任务学习和上下文理解能力[20]。

#### 2.1.4. ASR 和 OCR 文本

自动语音识别（ASR）和光学字符识别（OCR）文本直接关联音频和视觉模态。与高层语义描述不同，ASR和OCR提供精确的文本模态对应关系，支持成熟的多模态研究[21]。OCR还能从视觉语料库中提取文本，拓宽了其在医学信息学中的用途。

### 2.2. 视觉

医学视觉数据，包括图像和视频，按成像方法和维度进行分类 [22]。图像涵盖从二维（2D）格式（如X光）到三维（3D）格式（如CT扫描）。视频包括动态记录，如超声或手术影像。这些数据对于诊断、手术规划和科研至关重要，因其复杂的多源性，需要高精确率和安全性。它们支持判别任务（如分类、分割）和生成任务（如合成、跨模态转换）。

#### 2.2.1. 图像

医学图像在临床和研究环境中十分常见，根据维度可分为2D和3D两类。2D图像，如X射线或病理切片，捕捉器官或病灶的平面或表面视图。3D图像，如CT或MRI扫描，表示内部结构和空间关系。医学图像具有变换不变性（例如，平移、旋转），使得通过经典图像处理和深度学习方法进行稳健分析成为可能，这些方法能够在不同的临床情境中提取有价值的诊断特征。

2D图像：常见的2D图像包括胸部X射线、超声静态图像、病理切片和皮肤科照片。这些图像捕捉单一视角视图，支持大规模数据集和多样化应用。早期分析使用人工特征提取，配合SIFT或HOG等算法。深度卷积神经网络，如U-Net，已推进了分类、病灶检测和器官分割等任务。生成神经网络进一步实现了图像合成和跨模态数据生成，扩展了其效用。

3D图像：3D图像，如CT或MRI扫描，通过整合多个切片或体素来描绘器官结构和空间关系，提供比2D图像更丰富的细节。这种复杂性要求更多的计算资源以及对体素或切片关系的分析。将2D卷积神经网络扩展到3D架构（如3D U-Net）可支持体积分割和器官重建等任务。核心任务与2D应用类似，包括分类和分割，同时对3D数据合成等生成任务的兴趣日益增长。

#### 2.2.2. 视频

医学视频，包括动态超声和手术记录，捕捉时间动态，如器官运动或手术步骤，这对临床诊断和手术辅助至关重要。与静态图像不同，视频需要时序分析，如超声中的心动周期或手术影像中的连续步骤。传统的图像处理方法不足以应对，需要纳入时间维度的架构。

**Table 1**
Multimodal medical datasets (*: Competition Benchmark).

| Datasets | Text | Vision | Signal | Others | Typical Applications | Field |
|---|---|---|---|---|---|---|
| VQA-RAD [24] | QA Pairs | Image | – | – | VQA | Radiology |
| SLAKE [25] | QA Pairs | Image | – | – | VQA | Radiology |
| PathVQA [26] | QA Pairs | Image | – | – | VQA | Pathology |
| IU X-ray [27] | Reports | Image | – | – | Report Generation/VQA | Radiology |
| MIMIC-CXR [28] | Reports | Image | – | – | Report Generation/Classification/Image-Text Retrieval/VQA | Radiology |
| MIMIC-IV [29] | Clinical Notes/Reports/Demographics | Image | ECG/Vital Signs | Microbiology/Procedures/Medications | Prediction/Diagnosis/Classification | Comprehensive |
| Peir Gross [27] | Notes | Image | – | – | VQA | Radiology |
| AVEC2013* [30] | Speech Transcripts | Video | – | Audio | Classification | Mental Health |
| AVEC2014* [31] | Speech Transcripts | Video | – | Audio | Classification/Severity Estimation | Mental Health |
| DAIC-WOZ [32] | Speech Transcripts | Video | – | Audio | Classification | Mental Health |
| MODMA [33] | Speech Transcripts/Demographics | Video | EEG | Audio | Classification | Mental Health |
| EATD [34] | Text | Video | – | Audio | Classification | Mental Health |
| CMDC [35] | Speech Transcripts | Video | – | Audio | Classification | Mental Health |
| MedICaT [36] | Description | Image | – | – | Image Captioning/Image Retrieval | Radiology |
| PIC [37] | Clinical Notes/Reports/Demographics | Image | ECG/Vital Signs | Microbiology/Procedures/Medications | Prediction/Diagnosis/Classification | Paediatric |
| TCGA [38] | Clinical Notes/Reports/Demographics | Image | – | Multi-omics (Genomics/Proteomics/Methylation) | Classification/Prediction/Multi-modal integration | Oncology |
| AMOS-MM 2024* [39] | Reports | Image | – | – | Report Generation/ Segmentation | Abdominal Imaging |
| FLARE 2025* [40] | Reports/ Structured Text | Image | – | – | Classification/ Counting/ Regression | Abdominal Imaging |
| MuSe 2021* [41] | Text | Video | EDA/ECG/RESP/BPM | Audio | Classification/ Regression | Mental Health |
| George B. Moody PhysioNet Challenge 2023* [42] | Clinical Notes/Reports/Demographics | – | EEG/ECG/EMG/EOG | – | Classification/ Regression | Cardiovascular/Neurology |
| M4TAGSV (NLPCC 2025)* [43] | Text | Video | – | – | VQA | Medical Teaching |







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4

A common approach extends 2D convolutional neural networks to 3D models, like the I3D model [23], adapted for medical video tasks such as segmentation, classification (e.g., surgical scene recognition), and video generation. Large-scale generative models, like OpenAI's Sora, enable video synthesis, data augmentation, and remote teaching. Multimodal models also support cross-modal tasks, such as real-time subtitle generation for surgical videos, opening new research avenues.

### 2.3. Physiological signals

Physiological signals are data from human body activities, vital for health monitoring and clinical diagnosis. Collected non-invasively, they reflect organ and neuromuscular function. Common signals include electrocardiograms (ECG), electroencephalograms (EEG) and electromyograms (EMG), which track dynamic changes over time and frequency. Sensitive to environmental noise, these signals require advanced noise reduction and multi-channel fusion for accurate analysis.

Traditional analysis uses time-frequency transformations, such as wavelet transforms for transient features or short-time Fourier transforms for localized frequency data. These features support classification or regression with conventional machine learning. Deep learning advances, including convolutional neural networks (CNN) and graph convolutional networks (GCN), model multi-channel signal relationships. In GCNs, channels are nodes in a graph, with edges representing spatiotemporal connections, enhancing analysis accuracy. Processed signals support tasks like anomaly detection (e.g., arrhythmias, EEG irregularities), disease diagnosis (e.g., epilepsy, neuromuscular disorders), and functional evaluation (e.g., motor rehabilitation, cardiac assessment). They also enable cross-modal applications, such as generating diagnostic reports from ECGs or integrating EEGs with images or text in brain-computer interfaces.

### 2.4. Other modalities

Multimodal learning includes data types beyond visuals, signals, and text. This section focuses on clinically relevant modalities, such as audio and human-centric motion, and examines their connections to primary modalities for a clearer understanding of their roles.

#### 2.4.1. Audio

Audio data captures health-related sounds, such as doctor-patient conversations, breath sounds, or heart sounds, typically stored as voice recordings. These data support diagnosis, document medical interactions, and aid health monitoring. Audio is valuable for emotion analysis in mental health assessments and for analyzing breath sounds to diagnose conditions like asthma or chronic obstructive pulmonary disease.

#### 2.4.2. Human-centric motion

Human-centric motion data tracks patient body movements, collected via wearable devices, sensors, or motion-tracking technologies. It includes gait, activity levels, and physical abilities. This data monitors physical health, informs diagnosis, guides treatment plans, and tracks rehabilitation progress.

## 3. Multimodal representation learning

This section reviews recent advances in multimodal representation learning for medical data analysis, organized into four main parts. The first two parts introduce supervised and unsupervised learning methods and their roles in multimodal medical data analysis. The third part presents commonly used network architectures. The fourth part discusses emerging trends and key challenges in medical multimodal representation learning.

### 3.1. Supervised learning

Supervised representation learning uses labeled data from multiple modalities to guide the training of models [44]. It is one of the most established approaches in multimodal learning.

Supervised methods can be broadly divided into two categories. The first category is a staged approach, which extracts features separately from each modality before combining them. Typically, it involves two steps: first, a fixed neural network is used to extract features from each modality independently; then, these features are mapped into a common representation [44–46]. The second category employs an end-to-end strategy, integrating feature extraction and modality fusion into a single unified model. This method dynamically optimizes representations and directly learns cross-modal interactions.

An example of the staged approach is the work by Zhou et al. [44], who proposed a multimodal fusion model for diagnosing Alzheimer's disease by independently learning modality-specific features and then combining them into a joint representation. Similarly, Ye et al. [46] developed MedCoSS, a continuous self-supervised framework. MedCoSS divides modality data into distinct training phases, effectively reducing modality conflicts and avoiding forgetting previous knowledge by employing k-means sampling and feature distillation.

The end-to-end approach includes models like the Transformer based network by Saeed et al. [47], which processes multimodal data directly for tasks like cancer segmentation and survival prediction. Another example is the hybrid early fusion attention network by Hemker et al. [48], which captures interactions across modalities using a dual-branch architecture.

The staged approach works best for situations involving high differences among modalities, limited training data, or when explicit control over modality integration is needed. In contrast, the end-to-end approach excels with large datasets and highly integrated features, proving particularly effective in diagnostic and generative tasks involving combined image and text data [49].

### 3.2. Unsupervised learning

In medical data analysis, supervised learning often faces limitations due to the scarcity and cost of labeled data. Annotating medical data is expensive and requires specialized expertise. Consequently, unsupervised, weakly supervised, and self-supervised learning methods have gained increasing attention. These methods efficiently train models by exploring relationships between modalities without fully relying on labeled data. Their main strategy is to leverage complementary and correlated information across modalities, such as linking medical images with clinical reports or patient histories.

Recently, large-scale pre-trained models have emerged as powerful tools in unsupervised multimodal learning [50]. These models address the shortage of labeled medical data and exhibit strong generalization capabilities across diverse tasks. They aim to build joint representations from multiple modalities, enhancing prediction and analysis without explicit annotations. For instance, ConVIRT [51] demonstrates strong performance in medical image classification tasks by learning visual features from naturally associated image-text pairs. Similarly, MedCoSS [52] introduces a self-supervised approach where different modalities are trained sequentially, effectively managing conflicts between modalities and preventing knowledge loss. These advancements highlight the significant potential of unsupervised multimodal learning in the medical domain.

### 3.3. Network architectures

Several neural network architectures are commonly used for analyzing medical data from multiple modalities, including text, images, and physiological signals. These architectures effectively extract and combine features, serving as the foundation for multimodal learning.





一种常见的方法是将2D卷积神经网络扩展到3D模型，例如I3D模型[23]，被适配用于医学视频任务，如分割、分类（例如手术场景识别）和视频生成。大规模生成模型，如OpenAI的Sora，支持视频合成、数据增强和远程教学。多模态模型还支持跨模态任务，例如手术视频的实时字幕生成，开辟了新的研究方向。

### 2.3. 生理 信号

生理信号是来自人体活动的数据，对健康监测和临床诊断至关重要。它们以非侵入性方式采集，反映器官和神经肌肉功能。常见的信号包括心电图（ECG）、脑电图（EEG）和肌电图（EMG），它们追踪随时间和频率的动态变化。这些信号对环境噪声敏感，需要先进的降噪和多通道融合技术才能进行准确分析。

传统分析使用时频变换，例如用于瞬态特征的小波变换或用于局部频率数据的短时傅里叶变换。这些特征支持使用传统机器学习进行分类或回归。深度学习的进展，包括卷积神经网络（CNN）和图卷积网络（GCN），能够对多通道信号关系进行建模。在GCN中，通道是图中的节点，边表示时空连接，从而提升分析准确率。处理后的信号支持异常检测（如心律失常、脑电图异常）、疾病诊断（如癫痫、神经肌肉疾病）和功能评估（如运动康复、心脏评估）等任务。它们还支持跨模态应用，例如从心电图生成诊断报告，或在脑机接口中将脑电图与图像或文本集成。

### 2.4. 其他 模态

多模态 学习 包括 视觉、 信号 和 文本 之外 的 数据 类型。 本节 聚焦 于 临床 相关 模态， 如 音频 和 以人为中心 的运动， 并 考察 它们 与 主要 模态 的 联系 ， 以 更清晰 地 理解 其作用。

#### 2.4.1. 音频

音频数据捕捉与健康相关的声音，如医患对话、呼吸音或心音，通常以语音记录的形式存储。这些数据支持诊断、记录医疗互动，并有助于健康监测。音频在心理健康评估中的情感分析方面具有重要价值，也可用于分析呼吸音以诊断哮喘或慢性阻塞性肺疾病等病症。

#### 2.4.2. 以人为中心的 运动

以人为中心的 运动 数据 追踪 患者的 身体 运动， 通过 可穿戴 设备、 传感器 或 运动追踪 技术 采集。 它 包括 步态、 活动 水平 和 身体 能力。 这些 数据 监测 身体健康， 为 诊断 提供信息， 指导 治疗 方案， 并 追踪 康复 进展。

## 3. 多模态 表示 学习

本节回顾了用于医学数据分析的多模态表示学习的最新进展，内容分为四个主要部分。前两部分介绍监督学习和无监督学习方法及其在多模态医学数据分析中的作用。第三部分介绍常用的网络架构。第四部分讨论医学多模态表示学习中的新兴趋势和关键挑战。

### 3.1. 监督 学习

监督 表示 学习 利用 来自 多个 模态 的 标注 数据 来 指导 模型 的 训练 [44]。 它 是 多模态 学习 中 最 成熟 的 方法 之一 。

监督方法可大致分为两类。第一类是分阶段方法，先从每个模态中分别提取特征，然后再将它们组合起来。通常，它包括两个步骤：首先，使用固定的神经网络独立地从每个模态中提取特征；然后，将这些特征映射到一个公共表示 [44–46] 中。第二类采用端到端策略，将特征提取和模态融合整合到一个统一的模型中。这种方法动态地优化表示，并直接学习跨模态交互。

分阶段方法的一个例子是Zhou等人[44]，的工作，他们提出了一种用于诊断阿尔茨海默病的多模态融合模型，该模型独立学习模态特定特征，然后将其组合成联合表示。类似地，Ye等人[46] 开发了MedCoSS，一个连续自监督框架。MedCoSS将模态数据划分为不同的训练阶段，通过采用k-means采样和特征蒸馏，有效减少了模态冲突并避免了遗忘先前的知识。

端到端方法包括Saeed等人提出的基于Transformer的网络模型[47]，，该模型直接处理多模态数据以完成癌症分割和生存预测等任务。另一个例子是Hemker等人提出的混合早期融合注意力网络[48]，，该网络利用双分支架构捕获跨模态的交互关系。

分阶段方法最适合模态间差异较大、训练数据有限、或需要对模态整合进行显式控制的情况。相比之下，端到端方法在大数据集和高度整合特征方面表现优异，在涉及图像与文本数据结合的诊断和生成任务中尤为有效[49]。

### 3.2. 无监督 学习

在医学数据分析中，监督学习常因标注数据的稀缺性和高成本而面临局限。标注医疗数据费用高昂且需要专业知识。因此，无监督、弱监督和自监督学习方法日益受到关注。这些方法通过探索模态之间的关系来高效训练模型，而无需完全依赖标注数据。其主要策略是利用模态间的互补和相关信息，例如将医学图像与临床报告或患者病史相关联。

近年来，大规模预训练模型已成为无监督多模态学习中的强大工具[50]。这些模型解决了标注医疗数据短缺的问题，并在多种任务中展现出强大的泛化能力。它们旨在从多模态中构建联合表征，在无需显式标注的情况下增强预测与分析。例如，ConVIRT[51] 通过学习自然关联的图像-文本对中的视觉特征，在医学图像分类任务中展现出强劲性能。类似地，MedCoSS[52] 引入了一种自监督方法，其中不同模态按顺序训练，有效管理模态间的冲突并防止知识丢失。这些进展凸显了无监督多模态学习在医学领域的巨大潜力。

### 3.3. 网络 架构

几种 神经网络 架构 通常 用于 分析 来自多种模态的医学数据， 包括 文本、 图像 和 生理信号。 这些 架构 有效地提取 并组合 特征，成为多模态学习的基础。

#### 3.3.1. Convolutional neural networks (CNNs)

CNNs are widely applied in visual and signal processing tasks. They extract features through convolutional layers followed by activation functions:

$$h^{(l)} = \sigma\left(W^{(l)} * h^{(l-1)} + b^{(l)}\right), \tag{1}$$

where $h^{(l)}$ denotes the feature map of the $l$th layer, $W^{(l)}$ is the convolution kernel, $*$ represents convolution, $b^{(l)}$ is the bias, and $\sigma$ is the activation function. CNNs progressively extract deeper features, making them ideal for processing image and signal data, typically trained with cross-entropy loss. For example, Jiang et al. [53] proposed a multi-resolution parallel residual CNN (MPR-CNN) for denoising CXR images to obtain higher-quality image representations, which were then applied to COVID-19 diagnosis. SVMVGGNet-16 [54] is a CNN-based hybrid model that extracts high-level features from CT images using VGGNet-16 and classifies them with SVM for lung cancer detection. But existing CNN-based methods often suffer from limited ability to capture long-range dependencies in data.

#### 3.3.2. Graph neural networks (GNNs)

GNNs process graph-structured data effectively [55], suitable for medical signals like EEG, where sensors or brain regions form graph nodes. GNNs use message-passing to aggregate information from neighboring nodes, capturing local and global relationships. They are widely used for tasks like classification and anomaly detection, employing supervised and self-supervised methods. Alsarhan et al. [56] proposed PH-GCN, which combines multi-level granularity feature modeling with pair-wise hyper GCN to capture human keypoint relationships and enhance spatio-temporal feature representation. High-dimensional graph structures may contain noise and irrelevant information. Xing et al. [45] proposed EMO-GCN, which uses graph pooling to remove redundant and irrelevant information in high-dimensional graphs while retaining task-relevant features to improve efficiency and performance. RA-MGNN [57] is a GNN that integrates multi-graph modeling with a resource-aware mechanism, performing parallel learning and feature fusion on heterogeneous graphs while considering computation and communication constraints to improve efficiency and deployability without compromising accuracy.

Despite these advancements, current GNN-based methods still encounter limitations, including dependency on precise graph construction, and reduced adaptability to large-scale or dynamically evolving graphs.

#### 3.3.3. Recurrent neural networks (RNNs)

RNNs specialize in sequential data but initially struggled with long sequences due to gradient issues. LSTMs resolved this using gates, namely forget, input, and output gates, to manage long-term dependencies. Thus, LSTMs effectively model time-series medical data, such as ECG, EEG, and continuous textual records. BiGRU is a bidirectional variant of GRU [58] that captures both forward and backward context in sequence data, offering a simpler structure, fewer parameters, and faster training than LSTM with similar long-term dependency modeling.

Even with these improvements, RNN-based methods often face challenges such as difficulty in modeling extremely long-term dependencies, and susceptibility to noise in complex medical time-series data.

#### 3.3.4. Transformers

Transformers process sequences simultaneously using a self-attention mechanism, overcoming RNN limitations. Their core operation is scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V, \tag{2}$$

where $Q, K, V$ denote queries, keys, and values, respectively, and $\sqrt{d_k}$ normalizes the scores. Transformers efficiently handle variable-length sequences with parallel computations. Initially used in language tasks, they have since demonstrated success with medical texts, imaging sequences, and speech signals. Recently, architectures like Mamba [59] further improve handling very long sequences by combining selective mechanisms with state-space models. For text-related tasks, such as language modeling, Transformers typically predict the next token using cross-entropy loss. Alternative objectives have also been introduced to enhance learning performance.

Transformer-based models can still be constrained by intensive computational and memory requirements, sensitivity to hyperparameter choices, and dependence on large-scale labeled datasets for optimal performance in medical domains.

### 3.4. Trends in algorithms

Traditional approaches that rely solely on a single modality or representation method often face inherent limitations. By integrating multiple methods, these weaknesses can be mitigated and overall performance enhanced. For instance, Li et al. [60] observed that in iris segmentation, the restricted receptive field of CNNs limits the modeling of global semantic relations and reduces robustness against noise. To overcome this challenge, they introduced a dual-stream hybrid model in which CNNs capture local details while a Transformer-based module learns global context. The integration of these complementary features improves both robustness and segmentation accuracy. Building on this foundation, Wang et al. [61] developed an attention-driven Cross-Domain Fusion framework. This approach not only supports segmentation and recognition tasks simultaneously but also enhances versatility and generalization across domains. Further extending this line of research, Ali et al. [62] employed GCNs to model complex network topologies, combined with an attention-based BiGRU to capture temporal dependencies. . This design emphasizes salient patterns, thereby boosting prediction accuracy and improving adaptability to intricate spatio-temporal dynamics.

Beyond these approaches, another major line of research focuses on foundation models, which are driven by the availability of massive datasets and powerful computational resources. Unlike task-specific architectures, these models aim to construct unified knowledge frameworks through large-scale pretraining on heterogeneous modalities. Typically grounded in Transformer architectures and comprising billions of parameters, they can capture deep associations across diverse data sources. For example, such models are able to integrate medical images, textual records, and physiological signals, while employing contrastive learning strategies to ensure generalization. Consequently, they can be fine-tuned with minimal labeled data, which is particularly beneficial for tasks such as diagnosing rare diseases [5]. Compared to traditional models, foundation models have an advantage because their unified structure reveals relationships between different data types. This enables them to adapt easily to various clinical tasks, such as classifying images, generating reports, and predicting treatments.

Despite successes in tasks like image retrieval and classification (e.g., BiomedCLIP [63], Prov-GigaPath [64], MUSK [65]), foundation models have limitations. Their performance often relies on more data and computing power, not new architectural ideas. Also, larger datasets can amplify biases, leading to unfair predictions [66]. Moreover, their probabilistic nature and lack of transparency pose risks in medical use [67]. Therefore, while foundation models are promising for multimodal learning, future research must address fairness, interpretability, and data privacy to ensure their safety and reliability in medical applications.

## 4. Multimodal fusion methods

The rapid digitization of healthcare data has made multimodal fusion techniques crucial for analyzing EHRs. By combining diverse data sources like clinical narratives and medical imaging, these methods provide comprehensive patient representations. Initially explored in





#### 3.3.1. 卷积 神经 网络 （CNN）

CNN 广泛应用于视觉 和信号处理任务。 它们 通过卷积层提取特征，随后经过激活函数：

$$h^{(l)} = \sigma\left(W^{(l)} * h^{(l-1)} + b^{(l)}\right), \tag{1}$$

其中$h^{(l)}$ 表示第$l$层的特征图，$W^{(l)}$ 是卷积核，$*$ 表示卷积，$b^{(l)}$ 是偏置，$\sigma$ 是激活函数。CNN逐步提取更深层的特征，使其非常适合处理图像和信号数据，通常使用交叉熵损失进行训练。例如，Jiang等[53] 提出了一种多分辨率并行残差CNN（MPR-CNN）用于CXR图像去噪，以获得更高质量的图像表示，随后将其应用于COVID-19诊断。SVMVGGNet-16 [54] 是一种基于CNN的混合模型，它使用VGGNet-16从CT图像中提取高层特征，并用SVM对其进行分类以进行肺癌检测。但现有的基于CNN的方法往往在捕获数据中的长距离依赖关系方面能力有限。

#### 3.3.2. 图 神经 网络 （GNN）

GNN 能有效处理图结构数据 [55]，适用于 EEG 等医学信号，其中传感器或脑区构成图节点。GNN 通过消息传递聚合邻居节点的信息，捕获局部和全局关系。它们广泛用于分类和异常检测等任务，采用监督和自监督方法。Alsarhan 等 [56] 提出了 PH-GCN，它将多级粒度特征建模与成对超图GCN相结合，以捕获人体关键点关系并增强时空特征表示。高维图结构可能包含噪声和无关信息。Xing 等 [45] 提出了 EMO-GCN，它利用图池化去除高维图中的冗余和无关信息，同时保留任务相关特征以提高效率和性能。RA-MGNN [57] 是一种 GNN，它将多图建模与资源感知机制相结合，在异构图 上执行并行学习和特征融合，同时考虑计算和通信约束，在不影响准确率的情况下提高效率和可部署性。

尽管 有 这些进展， 当前 基于GNN的 方法 仍然 面临 局限， 包括 依赖 于 精确的 图 构建， 以及 对大规模 或 动态 演化 图的适应性降低。

#### 3.3.3. 循环 神经 网络 （RNN）

RNN 擅长处理序列数据，但最初因梯度问题而难以处理长序列。LSTM 通过门控机制解决了这一问题，即遗忘门、输入门和输出门，以管理长期依赖关系。因此，LSTM 能够有效建模时间序列医学数据，如 ECG、EEG 和连续文本记录。BiGRU 是 GRU 的双向变体，[58] 能够捕获序列数据中的前向和后向上下文，与 LSTM 相比，其结构更简单、参数更少、训练更快，同时具有相似的长期依赖建模能力。

即使 有 这些 改进， 基于RNN的 方法 仍然 面临 挑战， 例如 难以建模 极长期的 依赖关系， 以及 易受 复杂 医疗 时间序列 数据中 噪声 的影响。

#### 3.3.4. Transformer

Transformer 利用 自注意力 机制 同时 处理 序列， 克服了 RNN 的局限性。 其 核心 运算 是 缩放点积注意力：

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V, \tag{2}$$

其中 $Q$， $K$， $V$ 分别表示 查询、 键 和 值， 而 $\sqrt{d_k}$ 对 分数进行归一化。 Transformer 能够高效地 处理 变长 序列， 并开展 并行 计算。它最初 被用于 语言 任务，

此后，它在医学文本、影像序列和语音信号方面也展现出良好的效果。最近，Mamba [59] 等架构通过将选择机制与状态空间模型相结合，进一步提升了对超长序列的处理能力。对于语言建模等文本相关任务，Transformer 通常使用交叉熵损失来预测下一个词元。此外，人们还引入了其他目标函数以提升学习性能。

基于Transformer的 模型 在 医学 领域 中 仍可能 受到 密集计算 与 内存需求 、 对超参数选择的sitivity 以及 对大规模标注数据集 依赖的制约， 从而难以实现 最优 性能 。 为 实现 最佳效果 ， 仍需 针对 医疗场景 进行优化。

### 3.4. 算法趋势

仅依赖单一模态或表示方法的传统方法往往面临固有的局限性。通过整合多种方法，可以缓解这些弱点并提升整体性能。例如，Li等[60] 观察到，在虹膜分割中，CNN受限的感受野限制了对全局语义关系的建模，并降低了对噪声的鲁棒性。为克服这一挑战，他们引入了双流混合模型，其中CNN捕获局部细节，而基于Transformer的模块学习全局上下文。这些互补特征的整合提升了鲁棒性和分割精度。在此基础上，Wang等 [61] 开发了注意力驱动的跨域融合框架。该方法不仅同时支持分割和识别任务，还增强了跨领域的多功能性和泛化能力。进一步扩展这一研究方向，Ali等[62] 采用GCN对复杂网络拓扑进行建模，并结合基于注意力的BiGRU来捕获时间依赖。该设计强调显著模式，从而提升预测准确率并改善对复杂时空动态的适应性。

除上述方法外，另一条主要研究路线聚焦于基础模型，其发展得益于大规模数据集的可用性和强大的计算资源。与任务特定架构不同，这些模型旨在通过对异构模态进行大规模预训练来构建统一的知识框架。它们通常基于Transformer架构，包含数十亿参数，能够捕捉跨多样数据源的深层关联。例如，此类模型能够整合医学图像、文本记录和生理信号，同时采用对比学习策略以确保泛化。因此，它们可以用极少的标注数据进行微调，这对诊断罕见病[5]等任务尤为有益。与传统模型相比，基础模型的优势在于其统一结构揭示了不同数据类型之间的关系。这使它们能够轻松适应各种临床任务，如图像分类、生成报告和预测治疗。

尽管在图像检索和分类等任务中取得了成功（例如BiomedCLIP [63]，、Prov-GigaPath[64]，、MUSK[65]），基础模型仍有局限性。它们的性能往往依赖于更多数据和计算能力，而非新的架构思路。此外，更大的数据集可能放大偏见，导致不公平预测[66]。而且，其概率性质和缺乏透明度在医疗应用中带来风险[67]。因此，虽然基础模型在多模态学习方面前景广阔，未来研究必须解决公平性、可解释性和数据隐私问题，以确保其在医疗应用中的安全性和可靠性。

## 4. 多模态 融合 方法

医疗健康数据的 快速 数字化 使 多模态 融合 技术 成为 分析 电子健康记录的关键。 通过 结合 临床 叙述 和 医学 影像等 多种 数据源，这些 方法 能够提供 全面的 患者 表示。 这些方法最初在

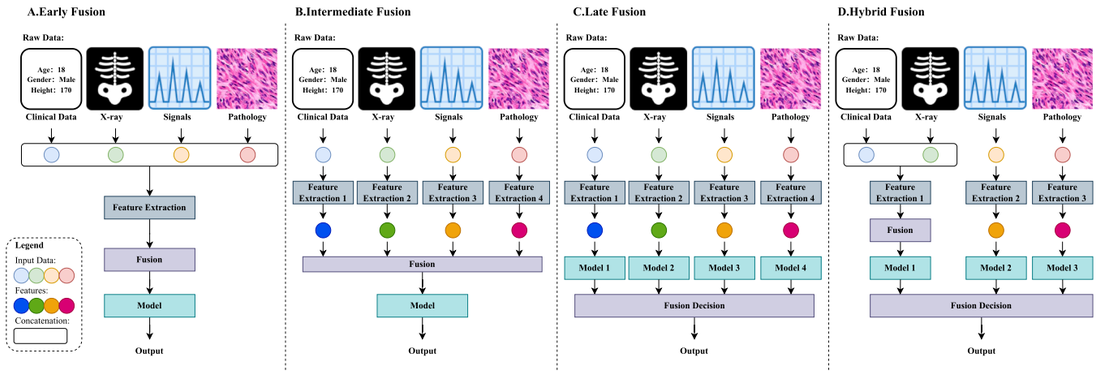



**Fig. 3.** Multimodal fusion can be roughly categorized into four strategies: (A) Early Fusion, (B) Intermediate Fusion, (C) Late Fusion, and (D) Hybrid Fusion.

computer vision and natural language processing, multimodal fusion enhances joint decision-making and cross-modal reasoning. Fusion methods include early fusion, intermediate fusion, late fusion, and hybrid fusion (Fig. 3).

### 4.1. Early fusion

Multimodal fusion can be performed at the early stage of data processing, where raw inputs from different modalities are directly combined without prior feature extraction. This method, commonly referred to as data-level fusion, operates on unprocessed input streams to form a unified representation across modalities. The workflow is shown in Fig. 3(A). By merging raw data before feature learning, data-level fusion enables the model to capture cross-modal correlations and interactions from the outset. This early fusion can strengthen the model's capacity to discover joint patterns across heterogeneous data sources.

The main advantages of this method are its simplicity and computational efficiency, particularly when modalities are well-aligned and share similar structures. However, it also presents challenges since the absence of initial feature extraction may lead to high dimensionality, greater sensitivity to noise, and difficulties in aligning diverse data formats.

#### 4.1.1. Simple fusion

Early-stage simple fusion refers to combining features from different modalities at shallow model layers using straightforward operations, such as feature concatenation, element-wise (weighted) summation, or concatenation after linear projection. These methods are computationally efficient and easy to implement. When modalities are well-aligned and exhibit roughly linear relationships, they can yield acceptable performance.

However, their ability to model complex cross-modal interactions is limited. They struggle with distribution, scale, and dimensional mismatches across modalities, as well as with temporal or semantic misalignment. Additionally, they are less robust when one modality is noisy or missing, making them less suitable for tasks requiring rich cross-modal reasoning.

### 4.2. Intermediate fusion

Intermediate Fusion is a type of feature-level fusion positioned between data-level fusion and decision-level fusion. It offers the benefit of retaining complementary information from different modalities while maintaining abstraction at the semantic level. This helps improve the expressive power of the model. On the downside, it requires well-designed mechanisms for feature alignment and fusion, involves relatively high computational complexity, and may result in information loss when feature representations are inadequate. The workflow is shown in Fig. 3(B).

#### 4.2.1. Attention-based fusion

Attention-based multimodal fusion models have attracted widespread attention since 2015 [19]. Compared with traditional fusion methods such as simple weighting or concatenation, these models introduce adaptive weighting strategies across modalities or features, effectively addressing the shortcomings of static fusion in feature selection and information integration. By dynamically adjusting the importance of each modality according to task demands or contextual cues, attention-based fusion highlights the most relevant information for final prediction. For instance, Xu and Kelvin [68] proposed a representative multimodal captioning model that selectively attends to different image regions during word generation, thereby capturing visual details more precisely and enhancing both accuracy and interpretability of captions.

In recent years, the cross-attention mechanism has demonstrated particular advantages in multimodal fusion. When handling incomplete multimodal MRI data for brain tumor segmentation, the CMAF-Net [69] model leverages a cross-attention fusion module to capture complementary information between modalities, thereby improving segmentation accuracy. Similarly, MFCA-Net [70] surpasses 97 % classification accuracy by integrating acceleration and surface electromyography (sEMG) signals through cross-modal attention, significantly outperforming single-modality models in distinguishing Parkinson's disease from idiopathic tremor.

Attention-based fusion not only enables dynamic modeling of modality importance but also supports fine-grained cross-modal alignment, enhancing the system's capacity to reason over complex relationships within medical data. Its advantages include stronger interpretability, robustness to noise, and broader generalization across tasks. Nevertheless, its effectiveness depends heavily on high-quality data and reliable annotations, which may restrict performance in resource-constrained or noisy settings.

#### 4.2.2. Pre-trained architectures

Transformer-based pre-trained architectures provide unified multimodal representations through large-scale training data. Models such as UNITER [71], VL-BERT [72], and LXMERT [73] leverage hierarchical Transformer structures to progressively fuse features. These models demonstrate strong generalization across tasks such as visual question answering and image-text retrieval, largely due to their modularity and scalability. They are particularly well-suited to complex medical applications where deep semantic understanding and cross-modal alignment are essential.





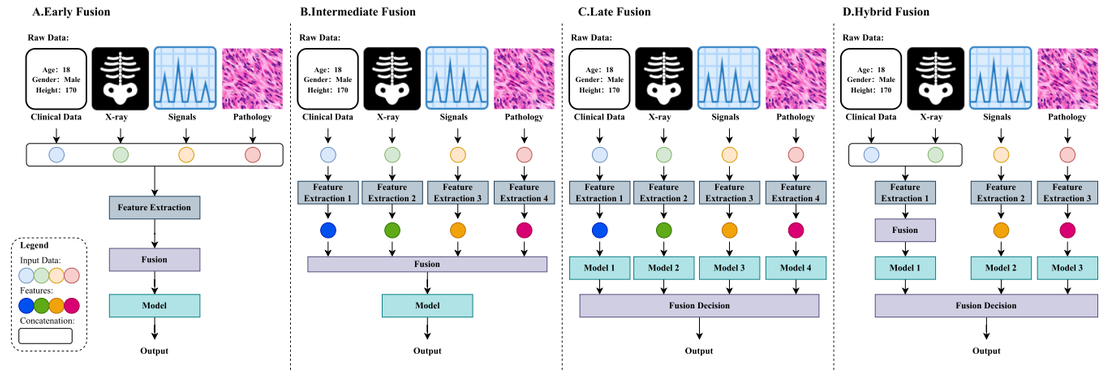



**图 3.** 多模态 融合 可 大致 分为 四 种 策略： (A) 早期 融合， (B) 中间 融合， (C) 晚期 融合， 以及 (D) 混合 融合。

计算机视觉和自然语言处理领域得到探索，多模态融合增强了联合决策和跨模态推理能力。融合方法包括早期融合、中间融合、晚期融合和混合融合（图3）。

### 4.1. 早期 融合

多模态融合可以在数据处理的早期阶段进行，此时来自不同模态的原始输入无需事先进行特征提取即可直接组合。这种方法通常被称为数据级融合，它作用于未处理的输入流，以形成跨模态的统一表示。其工作流程如图3(A)所示。通过在特征学习之前合并原始数据，数据级融合使模型能够从一开始就捕获跨模态相关性和交互。这种早期融合可以增强模型发现异构数据源之间联合模式的能力。

这种方法的主要优势在于其简单性和计算效率，尤其是当模态对齐良好且具有相似结构时。然而，它也带来了挑战，因为缺乏初始特征提取可能导致高维度、对噪声更敏感，以及难以对齐多样化的数据格式。

#### 4.1.1. 简单 融合

早期阶段的简单融合是指在模型浅层使用直接的操作来组合来自不同模态的特征，例如特征拼接、逐元素（加权）求和或线性投影后拼接。这些方法计算效率高且易于实现。当模态对齐良好且呈现大致线性关系时，它们可以获得可接受的性能。

然而，它们对复杂跨模态交互的建模能力有限。它们难以应对模态间的分布、尺度和维度不匹配，以及时间或语义错位问题。此外，当某一模态存在噪声或缺失时，它们的鲁棒性较差，使其不太适合需要丰富跨模态推理的任务。

### 4.2. 中间 融合

中间融合是一种特征级融合，介于数据级融合与决策级融合之间。它的优势在于能够保留来自不同模态的互补信息，同时保持语义层面的抽象。这有助于提升模型的表达能力。其缺点在于需要精心设计的特征对齐与融合机制，涉及相对较高的计算 复杂度， 并且 可能 导致 在 特征 表示 不充分 时 出现 信息 损失。 该 工作流程 如 图 3(B)所示。

#### 4.2.1. 基于注意力的 融合

自 2015 [19]以来，基于注意力的多模态融合模型引起了广泛关注。与简单加权或拼接等传统融合方法相比，这些模型引入了跨模态或跨特征的自适应加权策略，有效解决了静态融合在特征选择和特征整合方面的不足。通过根据任务需求或上下文线索动态调整每个模态的重要性，基于注意力的融合突出了对最终预测最相关的信息。例如，Xu和Kelvin [68]提出了一种具有代表性的多模态图像描述模型，该模型在词语生成过程中选择性地关注不同的图像区域，从而更精确地捕捉视觉细节，并提高描述的准确率和可解释性。

近年来，交叉注意力机制在多模态融合中展现出独特优势。在处理用于脑肿瘤分割的不完整多模态MRI数据时，CMAF-Net [69] 模型利用交叉注意力融合模块捕捉模态之间的互补信息，从而提高分割精度。类似地，MFCA-Net [70] 通过跨模态注意力整合加速度与表面肌电（s EMG）信号，分类准确率超过97%，在区分帕金森病与特发性震颤方面显著优于单模态模型。

基于注意力的融合不仅能够对模态重要性进行动态建模，还支持细粒度跨模态对齐，增强系统对医疗数据中复杂关系进行推理的能力。其优势包括更强的可解释性、噪声鲁棒性以及跨任务更广泛的泛化能力。然而，其有效性在很大程度上依赖于高质量数据和可靠标注，这可能限制其在资源受限或噪声环境中的表现。

#### 4.2.2. 预训练 架构

基于Transformer的预训练架构通过大规模训练数据提供统一的多模态表示。UNITER [71]，VL-BERT [72]，和LXMERT [73] 等模型利用分层Transformer结构逐步融合特征。这些模型在视觉问答和图文检索等任务中展现出强大的泛化能力，这主要归功于其模块化和可扩展性。它们尤其适用于需要深度语义理解和跨模态对齐的复杂医疗应用。

AMIE [74] is an LLM built on a pretrained Transformer architecture, PaLM 2 [75], specifically optimized for diagnostic reasoning and fine-tuned with domain-specific medical data. Flamingo-CXR [76] represents another specialized application: a chest X-ray report generation system that integrates vision and language. It takes chest X-ray images as input, interleaves them with textual data, and generates detailed reports. Med-BERT [14], in contrast, focuses on structured EHR data; its contextual embeddings substantially enhance disease prediction accuracy through targeted fine-tuning.

Different pretraining strategies shape cross-modal alignment and downstream generalization in distinct ways. Text encoders based on the BERT family provide strong domain-specific semantics, improving alignment stability and transferability in text-intensive tasks when combined with other modality encoders [77]. On the vision side, CNN backbones extract robust local features, while Vision Transformers pretrained with masked autoencoding [78] excel at capturing global, category-level patterns. CNNs often require subsequent fine-grained fusion modules to support region-level grounding in medical images. Beyond text and images, physiological signal encoders pretrained as time-series foundation models offer resilience against noise and missing channels, enabling reliable alignment of waveforms with clinical text or imaging results. Recent ECG foundation models that combine local and global features with lead-agnostic masking further enhance robustness under varying lead configurations [79].

In summary, modality-specific pretraining introduces complementary inductive biases that strengthen semantic consistency across modalities and facilitate the transferability of joint representations, thereby supporting broader generalization in downstream medical tasks. Despite these strengths, the approach entails high model complexity, substantial training requirements, and dependence on large-scale, high-quality multimodal datasets. Additional fine-tuning is often required to prevent task-specific performance bottlenecks.

#### 4.2.3. Self-supervised learning

Self-supervised learning utilizes automatically generated supervision signals through pretext tasks, thereby reducing the dependency on labeled datasets. This approach is particularly advantageous in medical scenarios where annotated data is scarce. Techniques such as contrastive learning and masked reconstruction are widely applied, as they not only capture cross-modal relationships but also preserve modality-specific characteristics.

These methods influence cross-modal alignment in distinct ways. Contrastive learning strengthens pairwise connections between modalities, promoting generalization across diverse downstream tasks. In contrast, masked reconstruction emphasizes intra-modal recovery, ensuring feature completeness and improving the quality of multimodal integration. Contrastive learning, exemplified by CLIP [80] and PMC-CLIP [50], highlights cross-modal alignment, whereas masked reconstruction, as proposed by Zhou et al. [81], focuses on recovering critical medical features. In the domain of physiological signals, contrastive learning applied to EEG and other biosignals enhances temporal synchronization and alignment with complementary modalities, thereby improving task generalization in areas such as emotion recognition and neurological diagnosis [82].

Self-supervised strategies are particularly suited to medical image interpretation and clinical text analysis, where annotated data remain limited. TransEHR [83] exemplifies this by employing three pretext tasks on unlabeled structured EHR data, followed by fine-tuning for downstream prediction tasks to address challenges in clinical time-series analysis. Holste et al. [84] proposed a self-supervised contrastive learning framework for echocardiography videos, combining multi-instance contrastive learning with frame re-ranking to improve cardiac disease classification. HierSSL [85], on the other hand, adopts a hierarchical self-supervised learning strategy. By integrating dual-scale mechanisms with graph contrast learning, it enhances robustness and predictive accuracy of GNN models under noisy data conditions.

Self-Supervised Learning-based fusion methods offer the advantage of reducing reliance on labeled data while capturing both cross-modal alignments and modality-specific features. Compared with pre-trained architectures, they are generally more data-efficient and adaptable to new modalities, but may provide less stable alignment and weaker performance on highly complex tasks without additional fine-tuning.

### 4.3. Late fusion

Late fusion belongs to decision-level fusion, where models are trained independently on each modality, and their outputs are combined to produce the final prediction. It offers strong flexibility, supports heterogeneous data types, and reduces the requirement for aligned multimodal data during training. Nevertheless, it may overlook interactions between modalities at earlier stages, which can limit the model's ability to capture deeper cross-modal relationships. It also relies heavily on the performance of individual models, making it sensitive to weaknesses in any single modality. The workflow of this strategy is illustrated in Fig. 3(C).

#### 4.3.1. Independent models and fusion

In late fusion, data from different modalities are processed independently by dedicated models or encoders, each designed to suit the characteristics of the corresponding modality. For example, CNNs are commonly used for feature extraction from image data, while RNNs or Transformers are applied to represent sequential data. At this stage, each modality learns its representation separately, without explicit cross-modal interaction. Fusion takes place at the decision-level, typically by combining the prediction results from individual models, such as class probabilities, confidence scores, or task-specific outputs.

A major advantage of late fusion is its flexibility. Each modality can use the most appropriate model architecture and training method. Additionally, the overall system can remain relatively robust when one modality is missing or affected by noise. However, because cross-modal dependencies are not explicitly modeled during feature learning, late fusion may struggle to capture complex interactions that require fine-grained alignment or semantic correspondence across modalities.

### 4.4. Comparison of fusion strategies

This section presents benchmark evaluations of different fusion methods across various tasks to highlight their differences. Table 2 summarizes the quantitative results for three datasets with six clinical prediction tasks, providing a comparative analysis of the multimodal fusion strategies. Following Wang et al. [86], the MIMIC-IV family (MIMIC-IV, MIMIC-IV-Note, and MIMIC-CXR) processes structured time-series data, free-text notes, and chest X-rays with the bidirectional LSTM, Doc2Vec [87], and ResNet [88], respectively. The extracted features are concatenated through a fully connected layer to predict in-hospital mortality, length of stay, and readmission. Models that integrate multiple modalities consistently outperform single-modality baselines because the inputs provide complementary information. Among the fusion schemes, intermediate fusion achieves the best metrics. Its refined joint representation and smaller parameter count also mitigate overfitting relative to late fusion. Dai et al. [89] integrated 44 public datasets to build CLIMB, which spans Graph (Brain Networks, Molecules), 1D features (ECG, EEG, Genomics, Gait, Text), 2D features (X-ray, Mammography, Dermoscopy, Fundus, Pathology), and 3D features (CT, MRI, Ultrasound, Endoscopy). These modalities support models that predict length of stay and in-hospital mortality. Each modality is handled by a dedicated encoder. The study compared Intermediate and late fusion multilayer perceptrons with a cross-attention model and found patterns similar to those reported by Wang et al. [86]. Cahan et al. [90] used clinical variables, laboratory results, and CTPA images from Sheba Medical Center to predict pulmonary-embolism mortality. Their pipeline combined TabNet,





AMIE[74] 是一个基于预训练Transformer架构构建的大语言模型，具体为PaLM 2 [75]，，专门针对诊断推理进行了优化，并使用领域特定的医疗数据进行了微调。Flamingo-CXR[76] 代表了另一种专门应用：一个集成视觉与语言的胸部X光报告生成系统。它以胸部X射线图像作为输入，将其与文本数据交错排列，并生成详细报告。相比之下，Med-BERT [14]，专注于结构化电子健康记录数据；其上下文嵌入通过有针对性的微调显著提升了疾病预测准确率。

不同的预训练策略以不同方式塑造跨模态对齐和下游泛化。基于BERT系列的文本编码器提供强大的领域特定语义，当与其他模态编码器[77]结合时，可提升文本密集型任务中的对齐稳定性和可迁移性。在视觉方面，CNN骨干网络提取稳健的局部特征，而使用掩码自编码[78] 预训练的视觉Transformer擅长捕获全局的类别级模式。CNN通常需要后续的细粒度融合模块来支持医学图像中的区域级定位。除文本和图像之外，作为时间序列基础模型预训练的生理信号编码器对噪声和缺失通道具有韧性，能够实现波形与临床文本或影像结果的可靠对齐。近期的心电图基础模型将局部和全局特征与导联无关掩码相结合，进一步增强了在不同导联配置[79]下的鲁棒性。

总之，模态特定预训练引入了互补的归纳偏置，增强了跨模态的语义一致性，并促进了联合表示的可迁移性，从而支持下游医学任务中更广泛的泛化。尽管具有这些优势，该方法仍带来高模型复杂度、大量训练需求以及对大规模高质量多模态数据集的依赖。通常还需要额外的微调以避免任务特定的性能瓶颈。

#### 4.2.3. 自监督 学习

自监督学习通过前置任务利用自动生成的监督信号，从而减少对标注数据集的依赖。这种方法在标注数据稀缺的医疗场景中尤为有利。对比学习和掩码重建等技术被广泛应用，因为它们不仅能捕获跨模态关系，还能保留模态特定特征。

这些方法以不同的方式影响跨模态对齐。对比学习加强模态之间的成对连接，促进在多样化下游任务中的泛化。相比之下，掩码重建强调模态内恢复，确保特征完整性并提升多模态集成质量。以CLIP[80] 和PMC-CLIP[50]，为代表的对比学习突出跨模态对齐，而Zhou等人[81]，提出的掩码重建则专注于恢复关键医疗特征。在生理信号领域，将对比学习应用于脑电图和其他生物信号可增强时间同步以及与互补模态的对齐，从而改善情感识别和神经诊断等领域的任务泛化 [82]。

自监督策略尤其适用于医学图像解读和临床文本分析，因为这些领域的标注数据仍然有限。TransEHR [83] 通过在无标注结构化EHR数据上采用三个前置任务，随后针对下游预测任务进行微调，以应对临床时间序列分析中的挑战，体现了这一点。Holste等 [84] 提出了一种用于超声心动图视频的自监督对比学习框架，将多实例对比学习与帧重排序相结合，以改进心脏疾病分类。另一方面，HierSSL [85]，采用分层自监督学习策略。通过将双尺度机制与图对比学习相结合，它增强了GNN模型在噪声数据条件下的鲁棒性和预测准确性。

基于自监督学习的融合方法具有减少对标注数据依赖的优势，同时能够捕捉跨模态对齐和模态特定特征。与预训练架构相比，它们通常更具数据高效性，并能适应新模态，但在没有额外微调的情况下，可能提供较不稳定的对齐，并在高度复杂的任务上表现较弱。

### 4.3. 晚期 融合

晚期融合属于决策级融合，其中模型在各模态上独立训练，然后将其输出组合以产生最终预测。它提供了很强的灵活性，支持异构数据类型，并降低了训练时对对齐多模态数据的要求。然而，它可能会忽略早期阶段模态之间的交互，这可能限制模型捕捉更深层次跨模态关系的能力。它还严重依赖于单个模型的性能，使其对任何单一模态的弱点都很敏感。该策略的工作流程如图3(C)所示。

#### 4.3.1. 独立 模型 与 融合

在晚期融合中，来自不同模态的数据由专用模型或编码器独立处理，每个模型或编码器都设计为适应相应模态的特征。例如，CNN通常用于从图像数据中提取特征，而RNN或Transformer则用于表示序列数据。在此阶段，每个模态分别学习其表示，没有显式的跨模态交互。融合发生在决策级，通常通过组合来自各个模型的预测结果，如类别概率、置信度分数或任务特定输出。

晚期融合的一个主要优势在于其灵活性。每个模态都可以使用最合适的模型架构和训练方法。此外，当某个模态缺失或受到噪声影响时，整个系统仍能保持相对稳健。然而，由于在特征学习过程中未显式建模跨模态依赖关系，晚期融合可能难以捕捉需要细粒度对齐或跨模态语义对应的复杂交互。

### 4.4. 融合 策略 的 比较

本节展示了不同融合方法在各种任务上的基准评估，以突出它们之间的差异。表2总结了三个数据集在六项临床预测任务上的定量结果，提供了多模态融合策略的比较分析。参照Wang等[86]，，MIMIC-IV系列（MIMIC-IV、MIMIC-IV-Note和MIMIC-CXR）分别使用双向LSTM、Doc2Vec[87]，和ResNet[88]，处理结构化时间序列数据、自由文本笔记和胸部X光。提取的特征通过全连接层进行拼接，以预测院内死亡率、住院时间和再入院。整合多个模态的模型始终优于单模态基线，因为这些输入提供了互补信息。在融合方案中，中间融合取得了最佳指标。其精细的联合表示和更少的参数数量也相对于晚期融合减轻了过拟合。Dai等[89] 整合了44个公共数据集来构建CLIMB，涵盖图（脑网络、分子）、1D特征（心电图、脑电图、基因组学、步态、文本）、2D特征（X光、乳腺X线摄影、皮肤镜、眼底、病理学）和3D特征（CT、MRI、超声、内窥镜）。这些模态支持预测住院时间和院内死亡率的模型。每个模态由专用编码器处理。该研究比较了中间融合和晚期融合多层感知机与交叉注意力模型，发现了与Wang等[86]所报道相似的模式。Cahan等[90] 使用来自Sheba医疗中心的临床变量、实验室结果和CTPA图像来预测肺栓塞死亡率。他们的流程结合了TabNet，

**Table 2**
Benchmark for multi-task evaluation of multimodal EHR datasets. Intermediate fusion operates at the feature level, avoiding the information loss of early fusion and allowing richer interactions than late ones, thus leading to better performance.

| Dataset | Task | Modal | Modal Description | Fusion Stage | Result |
|---|---|---|---|---|---|
| MIMIC-IV | In-hospital mortality prediction | T | Struct. Record | | AUC: 0.763, Accuracy: 0.84, Recall: 0.40, Precision: 0.39, F1: 0.40, AUPRC: 0.448 |
| | | T | Note | Partial | AUC: 0.675, Accuracy: 0.81, Recall: 0.30, Precision: 0.29, F1: 0.29, AUPRC: 0.255 |
| | | V | CXR | | AUC: 0.534, Accuracy: 0.77, Recall: 0.14, Precision: 0.14, F1: 0.14, AUPRC: 0.139 |
| | | T+V | Struct. Record + CXR | | AUC: 0.799, Accuracy: 0.86, Recall: 0.44, Precision: 0.46, F1: 0.45, AUPRC: 0.487 |
| | | T+T | Struct. Record + Note | Late | AUC: 0.813, Accuracy: 0.86, Recall: 0.49, Precision: 0.46, F1: 0.48, AUPRC: 0.480 |
| | | T+T+V | Struct. Record + CXR + Note | | AUC: 0.823, Accuracy: 0.86, Recall: 0.48, Precision: 0.45, F1: 0.47, AUPRC: 0.500 |
| | | T+V | Struct. Record + CXR | | AUC: 0.809, Accuracy: 0.86, Recall: 0.43, Precision: 0.46, F1: 0.45, AUPRC: 0.502 |
| | | T+T | Struct. Record + Note | **Intermediate** | AUC: 0.821, Accuracy: 0.86, Recall: 0.46, Precision: 0.46, F1: 0.46, AUPRC: 0.508 |
| | | T+T+V | Struct. Record + CXR + Note | | **AUC: 0.825, Accuracy: 0.86, Recall: 0.49, Precision: 0.46, F1: 0.47, AUPRC: 0.508** |
| | | T+V | Struct. Record + CXR | | AUC: 0.770, Accuracy: 0.84, Recall: 0.43, Precision: 0.39, F1: 0.41, AUPRC: 0.447 |
| | | T+T | Struct. Record + Note | Early | AUC: 0.788, Accuracy: 0.85, Recall: 0.45, Precision: 0.43, F1: 0.44, AUPRC: 0.438 |
| | | T+T+V | Struct. Record + CXR + Note | | AUC: 0.790, Accuracy: 0.85, Recall: 0.45, Precision: 0.44, F1: 0.44, AUPRC: 0.440 |
| MIMIC-IV | Long length of stay prediction | T | Struct. Record | | AUC: 0.704, Accuracy: 0.65, Recall: 0.71, Precision: 0.71, F1: 0.71, AUPRC: 0.790 |
| | | T | Note | Partial | AUC: 0.649, Accuracy: 0.62, Recall: 0.69, Precision: 0.69, F1: 0.69, AUPRC: 0.744 |
| | | V | CXR | | AUC: 0.565, Accuracy: 0.57, Recall: 0.64, Precision: 0.65, F1: 0.64, AUPRC: 0.659 |
| | | T+V | Struct. Record + CXR | | AUC: 0.721, Accuracy: 0.66, Recall: 0.71, Precision: 0.73, F1: 0.72, AUPRC: 0.804 |
| | | T+T | Struct. Record + Note | Late | AUC: 0.735, Accuracy: 0.67, Recall: 0.70, Precision: 0.75, F1: 0.72, AUPRC: 0.812 |
| | | T+T+V | Struct. Record + CXR + Note | | AUC: 0.735, Accuracy: 0.67, Recall: 0.71, Precision: 0.74, F1: 0.72, AUPRC: 0.812 |
| | | T+V | Struct. Record + CXR | | AUC: 0.730, Accuracy: 0.66, Recall: 0.71, Precision: 0.73, F1: 0.72, AUPRC: 0.810 |
| | | T+T | Struct. Record + Note | **Intermediate** | AUC: 0.730, Accuracy: 0.67, Recall: 0.71, Precision: 0.73, F1: 0.72, AUPRC: 0.816 |
| | | T+T+V | Struct. Record + CXR + Note | | **AUC: 0.735, Accuracy: 0.68, Recall: 0.72, Precision: 0.74, F1: 0.73, AUPRC: 0.812** |
| | | T+V | Struct. Record + CXR | | AUC: 0.713, Accuracy: 0.66, Recall: 0.70, Precision: 0.72, F1: 0.71, AUPRC: 0.794 |
| | | T+T | Struct. Record + Note | Early | AUC: 0.714, Accuracy: 0.67, Recall: 0.72, Precision: 0.73, F1: 0.72, AUPRC: 0.798 |
| | | T+T+V | Struct. Record + CXR + Note | | AUC: 0.718, Accuracy: 0.67, Recall: 0.73, Precision: 0.73, F1: 0.73, AUPRC: 0.801 |
| MIMIC-IV | Hospital readmission prediction | T | Struct. Record | | AUC: 0.544, Accuracy: 0.92, Recall: 0.10, Precision: 0.11, F1: 0.11, AUPRC: 0.072 |
| | | T | Note | Partial | AUC: 0.551, Accuracy: 0.92, Recall: 0.01, Precision: 0.01, F1: 0.01, AUPRC: 0.048 |
| | | V | CXR | | AUC: 0.525, Accuracy: 0.87, Recall: 0.06, Precision: 0.03, F1: 0.04, AUPRC: 0.047 |
| | | T+V | Struct. Record + CXR | | AUC: 0.580, Accuracy: 0.92, Recall: 0.10, Precision: 0.10, F1: 0.10, AUPRC: 0.060 |
| | | T+T | Struct. Record + Note | Late | AUC: 0.595, Accuracy: 0.91, Recall: 0.12, Precision: 0.09, F1: 0.10, AUPRC: 0.065 |
| | | T+T+V | Struct. Record + CXR + Note | | AUC: 0.606, Accuracy: 0.91, Recall: 0.10, Precision: 0.08, F1: 0.09, AUPRC: 0.066 |
| | | T+V | Struct. Record + CXR | **Intermediate** | AUC: 0.587, Accuracy: 0.91, Recall: 0.12, Precision: 0.09, F1: 0.10, AUPRC: 0.073 |
| | | T+T | Struct. Record + Note | | **AUC: 0.629, Accuracy: 0.92, Recall: 0.10, Precision: 0.09, F1: 0.10, AUPRC: 0.071** |
| | | T+T+V | Struct. Record + CXR + Note | | AUC: 0.513, Accuracy: 0.91, Recall: 0.06, Precision: 0.05, F1: 0.06, AUPRC: 0.057 |
| | | T+V | Struct. Record + CXR | | AUC: 0.578, Accuracy: 0.91, Recall: 0.12, Precision: 0.10, F1: 0.11, AUPRC: 0.062 |
| | | T+T | Struct. Record + Note | Early | AUC: 0.546, Accuracy: 0.91, Recall: 0.01, Precision: 0.01, F1: 0.01, AUPRC: 0.047 |
| | | T+T+V | Struct. Record + CXR + Note | | AUC: 0.544, Accuracy: 0.91, Recall: 0.01, Precision: 0.01, F1: 0.01, AUPRC: 0.047 |
| CLIMB | Length of stay prediction | T+V+S | Graph + 1D Features + 2D Features + 3D Features | Late | MAE: 4.71 |
| | | T+V+S | Graph + 1D + 2D + 3D Feature | **Intermediate** | **MAE: 2.61** |
| CLIMB | In-hospital mortality prediction | T+V+S | Graph + 1D + 2D + 3D Feature | Late | AUC: 0.859, Sens: 0.017, Spec: 0.983 |
| | | T+V+S | Graph + 1D + 2D + 3D Feature | **Intermediate** | **AUC: 0.961, Sens: 0.824, Spec: 0.975** |
| Sheba Medical Center | pulmonary embolism mortality prediction | T | Struct. Record | | AUC: 0.87, Accuracy: 0.81, Specificity: 0.79, Sensitivity: 0.90, PPV: 0.41, NPV: 0.98 |
| | | V | CT | Partial | AUC: 0.82, Accuracy: 0.79, Specificity: 0.79, Sensitivity: 0.80, PPV: 0.38, NPV: 0.96 |
| | | T+V | Struct. Record + CT | Late | AUC: 0.92, Accuracy: 0.89, Specificity: 0.90, Sensitivity: 0.89, PPV: 0.56, NPV: 0.98 |
| | | T+V | Struct. Record + CT | **Intermediate** | **AUC: 0.96, Accuracy: 0.93, Specificity: 0.90, Sensitivity: 0.94, PPV: 0.69, NPV: 0.98** |
| | | T+V | Struct. Record + CT | Early | AUC: 0.90, Accuracy: 0.81, Specificity: 0.90, Sensitivity: 0.79, PPV: 0.41, NPV: 0.98 |





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| | | T+T+V | Struct. Record + CXR + Note | | AUC: 0.823, Accuracy: 0.86, Recall: 0.48, Precision: 0.45, F1: 0.47, AUPRC: 0.500 |
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| | | T+T+V | Struct. Record + CXR + Note | | AUC: 0.718, Accuracy: 0.67, Recall: 0.73, Precision: 0.73, F1: 0.73, AUPRC: 0.801 |
| MIMIC-IV | Hospital readmission prediction | T | Struct. Record | | AUC: 0.544, Accuracy: 0.92, Recall: 0.10, Precision: 0.11, F1: 0.11, AUPRC: 0.072 |
| | | T | Note | Partial | AUC: 0.551, Accuracy: 0.92, Recall: 0.01, Precision: 0.01, F1: 0.01, AUPRC: 0.048 |
| | | V | CXR | | AUC: 0.525, Accuracy: 0.87, Recall: 0.06, Precision: 0.03, F1: 0.04, AUPRC: 0.047 |
| | | T+V | Struct. Record + CXR | | AUC: 0.580, Accuracy: 0.92, Recall: 0.10, Precision: 0.10, F1: 0.10, AUPRC: 0.060 |
| | | T+T | Struct. Record + Note | Late | AUC: 0.595, Accuracy: 0.91, Recall: 0.12, Precision: 0.09, F1: 0.10, AUPRC: 0.065 |
| | | T+T+V | Struct. Record + CXR + Note | | AUC: 0.606, Accuracy: 0.91, Recall: 0.10, Precision: 0.08, F1: 0.09, AUPRC: 0.066 |
| | | T+V | Struct. Record + CXR | **Intermediate** | AUC: 0.587, Accuracy: 0.91, Recall: 0.12, Precision: 0.09, F1: 0.10, AUPRC: 0.073 |
| | | T+T | Struct. Record + Note | | **AUC: 0.629, Accuracy: 0.92, Recall: 0.10, Precision: 0.09, F1: 0.10, AUPRC: 0.071** |
| | | T+T+V | Struct. Record + CXR + Note | | AUC: 0.513, Accuracy: 0.91, Recall: 0.06, Precision: 0.05, F1: 0.06, AUPRC: 0.057 |
| | | T+V | Struct. Record + CXR | | AUC: 0.578, Accuracy: 0.91, Recall: 0.12, Precision: 0.10, F1: 0.11, AUPRC: 0.062 |
| | | T+T | Struct. Record + Note | Early | AUC: 0.546, Accuracy: 0.91, Recall: 0.01, Precision: 0.01, F1: 0.01, AUPRC: 0.047 |
| | | T+T+V | Struct. Record + CXR + Note | | AUC: 0.544, Accuracy: 0.91, Recall: 0.01, Precision: 0.01, F1: 0.01, AUPRC: 0.047 |
| CLIMB | Length of stay prediction | T+V+S | Graph + 1D Features + 2D Features + 3D Features | Late | MAE: 4.71 |
| | | T+V+S | Graph + 1D + 2D + 3D Feature | **Intermediate** | **MAE: 2.61** |
| CLIMB | In-hospital mortality prediction | T+V+S | Graph + 1D + 2D + 3D Feature | Late | AUC: 0.859, Sens: 0.017, Spec: 0.983 |
| | | T+V+S | Graph + 1D + 2D + 3D Feature | **Intermediate** | **AUC: 0.961, Sens: 0.824, Spec: 0.975** |
| Sheba Medical Center | pulmonary embolism mortality prediction | T | Struct. Record | | AUC: 0.87, Accuracy: 0.81, Specificity: 0.79, Sensitivity: 0.90, PPV: 0.41, NPV: 0.98 |
| | | V | CT | Partial | AUC: 0.82, Accuracy: 0.79, Specificity: 0.79, Sensitivity: 0.80, PPV: 0.38, NPV: 0.96 |
| | | T+V | Struct. Record + CT | Late | AUC: 0.92, Accuracy: 0.89, Specificity: 0.90, Sensitivity: 0.89, PPV: 0.56, NPV: 0.98 |
| | | T+V | Struct. Record + CT | **Intermediate** | **AUC: 0.96, Accuracy: 0.93, Specificity: 0.90, Sensitivity: 0.94, PPV: 0.69, NPV: 0.98** |
| | | T+V | Struct. Record + CT | Early | AUC: 0.90, Accuracy: 0.81, Specificity: 0.90, Sensitivity: 0.79, PPV: 0.41, NPV: 0.98 |

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XGBoost, SANet [90], and Swin-UNETR [91]. The results confirmed the same advantages of multimodal learning.

In summary, early fusion concatenates raw signals from all modalities at the input layer. This approach preserves every detail of the data, but it also lets noise flow directly into the network. Intermediate fusion aligns modal representations at the feature layer, often through shared-attention or gating modules, allowing the model to exchange complementary cues while adding only modest parameters. For most clinical prediction tasks, this level of fusion yields the highest accuracy. Late fusion combines separate modality outputs at the decision layer. It is simple to implement and easy to extend, yet its coarse integration limits the discovery of subtle cross-modal relationships.

The multimodal hybrid fusion strategy (schematic diagram shown in Fig. 3(D)) proposes a unique method that integrates the advantages of various fusion approaches. This method aims to maximize the information utilization efficiency of multimodal data while fully considering the interaction and synergy between different modalities. Wollmer et al. [92] illustrate this idea. They first apply a bidirectional long short-term memory (BLSTM) network to fuse audio and visual features, an instance of intermediate fusion. They then merge this fused representation with the output of a text classifier at the decision layer, exemplifying late fusion. In this way, the model benefits from feature-level interaction without discarding the specialized expertise of individual modality classifiers.

Although hybrid fusion often improves robustness and accuracy, it also increases training complexity, reduces interpretability, and demands more computational resources. It is therefore best suited to tasks that require very high precision and have ample hardware support.

## 5. Discriminative applications

In this section, we discuss the applications of multimodal learning in discriminative tasks. We categorize these applications based on specific combinations of data modalities.

The overall framework of discriminative tasks consists of three main steps. First, data representation is performed, typically relying on general representation learning networks discussed in Section 3. Additional modules are added to satisfy task-specific requirements. The next step is multimodal fusion, with common fusion methods introduced in Section 4. Finally, task-specific outputs are produced. Details of specific tasks are discussed in the following subsections.

### 5.1. Text + Vision

In this section, we review related works on the integration of medical text and visual information. Table 3 summarizes their fusion stages, architectural designs, employed algorithms, target tasks, and the datasets used, along with the performance achieved in the respective tasks.

#### 5.1.1. Vision joint diagnosis

One common diagnostic application is leveraging complementary image and text information to improve diagnostic accuracy. These methods typically involve two critical components: 1) effective encoders for images and text to identify lesion-related features; and 2) fusion modules to unify text and image features with a focus on diagnostic information.

To further enhance diagnostic task performance, some studies incorporate multi-scale information. For example, ViLa-MIL [93] proposes a dual-scale approach that generates text prompts at both the whole-slide image (WSI) and patch levels, where a context-guided text decoder refines lesion-specific text representations. Prototype-guided modules then progressively aggregate image features, achieving AUCs of 0.926 and 0.747 on the public datasets TCGA-RCC and TCGA-Lung, respectively, and 0.843 on the real-world dataset TIHD-RCC. Berthop [94], another example, uses specialized text and visual feature extractors for medical applications. It effectively captures implicit relationships between H&E pathology images and text, with an AUC of 0.899 on 5000 real-world H&E pathology image-report pairs.

Typically, text provides direct descriptions of lesions, including locations and morphology. To exploit this, some methods use bidirectional attention mechanisms [96] to enhance interactions between image and text features. Other approaches extract specialized medical knowledge from textual data. For instance, Zhang et al. [130] integrated domain-specific text knowledge into pathology image models, significantly improving classification performance. Similarly, Zheng et al. [97] collected real-world doctor-patient dialogues and medical images from COVID-19 and control groups to construct a knowledge graph. This graph was then utilized to guide image feature matching, achieving a diagnostic accuracy of 0.99. Compared with multi-scale fusion that captures fine-grained tissue patterns, knowledge-driven methods may be preferred in settings with rich clinical records, because they enhance interpretability and align more closely with clinical reasoning.

#### 5.1.2. Object detection

Medical object detection combines textual descriptions from clinical records or annotations to identify lesions (e.g., tumors, nodules) in medical images, aiding diagnosis and segmentation. With the advancement of pretrained models, text-guided methods have become prevalent, particularly for weakly or unsupervised segmentation tasks [99,103,104,121].

LViT [99] leverages medical text annotations to overcome image quality limitations, demonstrating strong performance on lung infection, COVID-19, and real-world esophageal cancer CT datasets. It enhances segmentation via image-text modeling, improves pseudo label quality through text guidance, and reports results across multiple model sizes, showing that appropriately sized models perform best, with strong potential for deployment. ConTEXTual Net [103] extracts language features from free-form radiology reports, employing cross-attention with visual features to guide image analysis. SimTxtSeg [104] generates high-quality segmentation labels from simple text prompts, further improving segmentation performance.

In contrast to these methods, CLIP-based frameworks offer both scalability and adaptability, which makes them suitable for multi-domain clinical settings that require handling diverse imaging modalities and diagnostic tasks simultaneously. This encoded knowledge can significantly boost segmentation tasks requiring annotations. Huang et al. [111] proposed a CLIP-based multi-stage adaptive framework that jointly handles medical image classification and segmentation. It outperforms existing methods on most of six tasks, with notable advantages in cross-domain generalization and task transfer, eliminating the need for separate models. MedCLIP-SAM [121] combines CLIP and SAM to enable precise medical image segmentation via text prompts, and is extensively evaluated on breast tumor ultrasound, brain tumor MRI, and chest X-ray tasks, demonstrating the advantage of CLIP-based methods in multi-task settings.

#### 5.1.3. Visual answer localization (VAL)

VAL, also known as Video Moment Localization, aims to identify relevant time segments within videos or time series data based on textual activity descriptions. Unlike static image localization, which focuses on spatial features, VAL requires temporal reasoning and precise cross-modal matching. Existing methods [131,132] have shown strong performance on medical video datasets such as MedVidQA [129].

For effective medical VAL, models must accurately correlate visual data with text and locate critical segments within continuous video streams or time series. Achieving this typically involves multimodal fusion modules tailored to medical contexts as well as specialized decoding architectures. For example, Gupta et al. [127] proposed a method that identifies video segments corresponding to natural language queries through cycle consistency between modalities. In addition, their approach enhances visual responses in medical videos, enabling more accurate answers to public health-related questions. Similarly, Li et al. [128] introduced Visually Prompted Text Span Localization (VPTSL),





XGBoost、 SANet [90], 和 Swin-UNETR [91]。 结果 证实 了 多模态 学习 的 相同 优势。

总之，早期融合在输入层将所有模态的原始信号拼接起来。这种方法保留了数据的每一个细节，但同时也让噪声直接流入网络。中间融合在特征层对齐模态表示，通常通过共享注意力或门控模块实现，使模型能够交换互补线索，同时仅增加少量参数。对于大多数临床预测任务，这种融合层次能获得最高的准确率。晚期融合在决策层结合各模态的独立输出。它实现简单且易于扩展，但其粗粒度的整合限制了对细微跨模态关系的发现。

多模态混合融合策略（示意图见图3(D)）提出了一种独特的方法，综合了各种融合方法的优势。该方法旨在最大化多模态数据的信息利用效率，同时充分考虑不同模态之间的交互与协同。Wollmer等人[92] 阐述了这一思路。他们首先应用双向长短期记忆（BLSTM）网络来融合音频和视觉特征，这是中间融合的一个实例。然后，他们在决策层将这一融合表示与文本分类器的输出合并，体现了晚期融合。通过这种方式，模型既受益于特征级的交互，又不丢弃各模态分类器的专门知识。

尽管 混合 融合 通常 能提升 鲁棒性 和 准确率， 但它 也 增加了 训练 复杂度， 降低了 可解释性， 且 需要 更多的 计算 资源。 因此，它 最适合 那些 需要 极高 精确率 且具备 充足 硬件 支持 的 任务。

## 5. 判别 应用

在本节中， 我们 讨论 多模态 学习 在 判别 任务中的应用。 我们根据 数据模态的特定组合 对这些应用进行分类。

判别任务的总体框架由三个主要步骤组成。首先，进行数据表示，通常依赖于第3节中讨论的通用表示学习网络。添加额外模块以满足任务特定需求。下一步是多模态融合，第4节介绍了常见的融合方法。最后，生成任务特定输出。具体任务的细节将在以下小节中讨论。

### 5.1. 文本 + 视觉

在本节中，我们回顾了关于医学文本和视觉信息整合的相关工作。表3总结了它们的融合阶段、架构设计、所采用的算法、目标任务以及所使用的数据集，并附有在各自任务中取得的性能。

#### 5.1.1. 视觉联合诊断

一种常见的诊断应用是利用互补的图像和文本信息来提高诊断准确性。这些方法通常涉及两个关键组成部分：1）有效的图像和文本编码器，用于识别病变相关特征；2）融合模块，用于将文本和图像特征统一起来，并聚焦于诊断信息。

为了进一步提升诊断任务的性能，一些研究引入了多尺度信息。例如，ViLa-MIL [93] 提出了一种双尺度方法，在全切片图像（WSI）和图块级别上生成文本提示，其中上下文引导文本解码器细化病变特异性文本表示。原型引导模块随后逐步聚合图像特征，在公共数据集TCGA-RCC和TCGA-Lung上分别达到了0.926和0.747的AUC，在真实世界数据集TIHD-RCC上达到了0.843。Berthop [94]，是另一个例子，它使用专门的文本和视觉特征提取器用于医疗应用。它有效地捕捉了隐式关系，在 H&E 病理学 图像 与 文本 之间， 其 AUC 为 0.899 ， 基于 5000 个真实世界H&E 病理学 图像-报告对 。

通常，文本提供对病灶的直接描述，包括位置和形态。为利用这一点，一些方法使用双向注意力机制[96] 来增强图像特征与文本特征之间的交互。其他方法则从文本数据中提取专门的医学知识。例如，Zhang等人[130] 将领域特定的文本知识整合到病理图像模型中，显著提高了分类性能。类似地，Zheng等人[97] 收集了来自COVID-19组和对照组的真实医患对话及医学图像，以构建知识图谱。随后，该图谱被用于指导图像特征匹配，实现了0.99的诊断准确率。与捕捉细粒度组织模式的多尺度融合相比，在临床记录丰富的场景中，知识驱动方法可能更受青睐，因为它们增强了可解释性，并且更贴近临床推理。

#### 5.1.2. 目标 检测

医学目标检测结合临床记录或标注中的文本描述来识别医学图像中的病灶（例如肿瘤、结节），辅助诊断与分割。随着预训练模型的发展，文本引导方法已变得普遍，尤其是在弱监督或无监督分割任务中[99,103,104,121]。

LViT[99] 利用医学文本标注克服图像质量限制，在肺部感染、COVID-19和真实世界食管癌CT数据集上展现出强劲性能。它通过图像-文本建模增强分割，通过文本引导提高伪标签质量，并报告了多种模型规模的结果，表明适当规模的模型表现最佳，具有很强的部署潜力。ConTEXTual Net[103] 从自由形式的放射学报告中提取语言特征，采用交叉注意力与视觉特征结合来引导图像分析。SimTxtSeg[104] 从简单的文本提示中生成高质量的分割标签，进一步提升分割性能。

与这些方法相比，基于CLIP的框架兼具可扩展性和适应性，使其适用于需要同时处理多种成像模态和诊断任务的多域临床环境。这种编码知识可显著提升需要标注的分割任务。Huang等人[111] 提出了一种基于CLIP的多阶段自适应框架，能够联合处理医学图像分类和分割。该框架在六项任务中的大多数上优于现有方法，在跨域泛化和任务迁移方面具有显著优势，且无需单独的模型。MedCLIP-SAM[121] 结合了CLIP和SAM，通过文本提示实现精确的医学图像分割，并在乳腺肿瘤超声、脑肿瘤MRI和胸部X光任务上进行了广泛评估，展示了基于CLIP的方法在多任务场景中的优势。

#### 5.1.3. 视觉 答案 定位 (VAL)

VAL，又称视频时刻定位，旨在根据文本活动描述识别视频或时间序列数据中的相关时间段。与侧重于空间特征的静态图像定位不同，VAL需要时间推理和精确的跨模态匹配。现有方法[131,132] 在MedVidQA[129]等医学视频数据集上已展现出强劲的性能。

为实现有效的医学VAL，模型必须准确地将视觉数据与文本相关联，并定位连续视频流或时间序列中的关键片段。实现这一点通常涉及针对医学情境定制的多模态融合模块以及专门的解码架构。例如，Gupta等人[127] 提出了一种方法，通过模态间的循环一致性来识别与自然语言查询对应的视频片段。此外，他们的方法增强了医学视频中的视觉响应，从而能够更准确地回答公共卫生相关问题。类似地，Li等人[128] 引入了视觉提示文本片段定位（VPTSL），

**Table 3**
Summarizing different methods in Text + Vision modality medical tasks (Early fusion: data-level, Intermediate fusion: feature-level, Late fusion: decision-level). *represents non-public datasets.

| Task Type | Methods | Fusion Stage | Model Architectures | Fusion Algorithm | Target Tasks | Datasets | Performances |
|---|---|---|---|---|---|---|---|
| Joint Diagnosis | ViLa-MIL [93] | Feature-level (Intermediate) | Attention-Based Fusion | Dual-Attention | WSI Subtyping Classification | TIHD-RCC* | AUC: 84.3, F1: 68.7, ACC: 68.8 |
| | | | | | | TCGA-RCC | AUC: 92.6, F1: 78.3, ACC: 80.3 |
| | | | | | | TCGA-Lung | AUC: 74.7, F1: 67.0, ACC: 67.7 |
| | Berthop [94] | Feature-level (Intermediate) | Attention-Based Fusion | Dual-Attention | CXR Disease Diagnosis | OpenI [95] | AUC: 98.2 |
| | TandemNet [96] | Feature-level (Intermediate) | Attention-Based Fusion | Dual-Attention | CXR Disease Diagnosis | BCIDR* | ACC: 89.9 |
| | Zheng et al. [97] | Feature-level (Intermediate) | Attention-Based Fusion | Knowledge Graph Attention | COVID-19 Diagnosis | Multi-Modal COVID-19 Pneumonia* | AUC: 98.1, F1: 98.7, ACC: 99.0 |
| | Qiu et al. [98] | Decision-level (Late) | Independent Models | Voting | Mild Cognitive Impairment Diagnosis | NACC* | ACC: 0.909, F1: 0.944, MCC: 0.719 |
| Object Detection | LViT[99] | Feature-level (Intermediate) | And Fusion Attention-Based Fusion | Element-Wise Addition + Pixel-Level Attention | Infection Segmentation | MosMedData+ [100,101] | Dice: 74.57, mIoU: 61.33 |
| | | | | | | QaTa-COV19 [102] | Dice: 83.66, mIoU: 75.11 |
| | ConTEXTualNet [103] | Feature-level (Intermediate) | Attention-Based Fusion | Cross-Attention | Pneumothorax Segmentation | CANDID-PTX | Dice: 71.6 |
| | SimTxtSeg [104] | Feature-level (Intermediate) | Attention-Based Fusion | Cross-Attention | Colonic Polyp Segmentation | CVC-ClinicDB [105], CVC-ColonDB [106],ETIS-LaribPolypDB [107], Kvasir [108], PolypGen [109], | Dice: 88.2, mIoU: 82.4 |
| | | | | | MRI Brain Tumor Segmentation | MRI Brain Tumor (LGG Segmentation Dataset [110]) | Dice: 81.7, mIoU: 71.3 |
| | Huang et al. [111] | Feature-level (Intermediate) | Self-Supervised Learning | CLIP | Medical Anomaly Detection (Brain MRI) | Brain MRI [112–114] | AUC: 92.4 |
| | | | | | Medical Anomaly Detection (Liver CT) | Liver CT [115,116] | AUC: 81.1 |
| | | | | | Medical Anomaly Detection (Retinal OCT) | Retinal OCT [117,118] | AUC: 96.1 |
| | | | | | Medical Anomaly Detection (Retinal OCT) | OCT17 [118] | AUC: 99.3 |
| | | | | | Medical Anomaly Detection (Chest X-Ray) | ChestXray [119] | AUC: 81.9 |
| | | | | | Medical Anomaly Detection (Digital Histopathology) | HIS [120] | AUC: 82.7 |
| | MedCLIP-SAM [121] | Feature-level (Intermediate) | Self-Supervised Learning | CLIP | Breast Ultrasound Segmentation | Breast Tumor Ultrasound [122,123] | Dice: 67.8, mIoU: 57.9, AUC: 79.3 |
| | | | | | Brain Tumor Segmentation | Brain Tumor MRI [124] | Dice: 66.7, mIoU: 50.3, AUC: 81.3 |
| | | | | | COVID-19 Radiography Segmentation | Lung Chest X-Ray [125,126] | Dice: 49.0, mIoU: 64.4, AUC: 78.5 |
| Visual Answer Localization | CCAL [127] | Feature-level (Intermediate) | Attention-Based Fusion | Cross-Attention | Medical VQA | HealthVidQA-CRF [127], | mIoU: 52.9 |
| | | | | | | HealthVidQA-Prompt [127] | |
| | VPTSL [128] | Feature-level (Intermediate) | Attention-Based Fusion | Context Query Attention | Medical VQA + Temporal Answer Localization | MedVidQA [129] | mIoU: 57.8 |







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| | | | | | | TCGA-RCC | AUC: 92.6, F1: 78.3, ACC: 80.3 |
| | | | | | | TCGA-Lung | AUC: 74.7, F1: 67.0, ACC: 67.7 |
| | Berthop [94] | Feature-level (Intermediate) | Attention-Based Fusion | Dual-Attention | CXR Disease Diagnosis | OpenI [95] | AUC: 98.2 |
| | TandemNet [96] | Feature-level (Intermediate) | Attention-Based Fusion | Dual-Attention | CXR Disease Diagnosis | BCIDR* | ACC: 89.9 |
| | Zheng et al. [97] | Feature-level (Intermediate) | Attention-Based Fusion | Knowledge Graph Attention | COVID-19 Diagnosis | Multi-Modal COVID-19 Pneumonia* | AUC: 98.1, F1: 98.7, ACC: 99.0 |
| | Qiu et al. [98] | Decision-level (Late) | Independent Models | Voting | Mild Cognitive Impairment Diagnosis | NACC* | ACC: 0.909, F1: 0.944, MCC: 0.719 |
| Object Detection | LViT[99] | Feature-level (Intermediate) | And Fusion Attention-Based Fusion | Element-Wise Addition + Pixel-Level Attention | Infection Segmentation | MosMedData+ [100,101] | Dice: 74.57, mIoU: 61.33 |
| | | | | | | QaTa-COV19 [102] | Dice: 83.66, mIoU: 75.11 |
| | ConTEXTualNet [103] | Feature-level (Intermediate) | Attention-Based Fusion | Cross-Attention | Pneumothorax Segmentation | CANDID-PTX | Dice: 71.6 |
| | SimTxtSeg [104] | Feature-level (Intermediate) | Attention-Based Fusion | Cross-Attention | Colonic Polyp Segmentation | CVC-ClinicDB [105], CVC-ColonDB [106],ETIS-LaribPolypDB [107], Kvasir [108], PolypGen [109], | Dice: 88.2, mIoU: 82.4 |
| | | | | | MRI Brain Tumor Segmentation | MRI Brain Tumor (LGG Segmentation Dataset [110]) | Dice: 81.7, mIoU: 71.3 |
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| | | | | | Medical Anomaly Detection (Chest X-Ray) | ChestXray [119] | AUC: 81.9 |
| | | | | | Medical Anomaly Detection (Digital Histopathology) | HIS [120] | AUC: 82.7 |
| | MedCLIP-SAM [121] | Feature-level (Intermediate) | Self-Supervised Learning | CLIP | Breast Ultrasound Segmentation | Breast Tumor Ultrasound [122,123] | Dice: 67.8, mIoU: 57.9, AUC: 79.3 |
| | | | | | Brain Tumor Segmentation | Brain Tumor MRI [124] | Dice: 66.7, mIoU: 50.3, AUC: 81.3 |
| | | | | | COVID-19 Radiography Segmentation | Lung Chest X-Ray [125,126] | Dice: 49.0, mIoU: 64.4, AUC: 78.5 |
| Visual Answer Localization | CCAL [127] | Feature-level (Intermediate) | Attention-Based Fusion | Cross-Attention | Medical VQA | HealthVidQA-CRF [127], | mIoU: 52.9 |
| | | | | | | HealthVidQA-Prompt [127] | |
| | VPTSL [128] | Feature-level (Intermediate) | Attention-Based Fusion | Context Query Attention | Medical VQA + Temporal Answer Localization | MedVidQA [129] | mIoU: 57.8 |

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**Table 4**
Summarize different methods in Text + Signal modality medical tasks (Early fusion: data-level, Intermediate fusion: feature-level, Late fusion: decision-level). *represents non-public datasets.

| Task Type | Methods | Fusion Stage | Model Architectures | Fusion Algorithm | Target Tasks | Datasets | Performances |
|---|---|---|---|---|---|---|---|
| Joint Diagnosis | Hernandez et al. [133] | Data-level (Early) | Simple Fusion | – | Adverse Event Prediction | Michigan Medicine* | AUC: 87.0 |
| | sEHR-BERT [134] | Feature-level (Intermediate) | Self-Supervised Learning | CLIP | Cardiovascular Diseases Classification | PhysioNet2020 [135] | AUC: 91.5 |
| | | | | | Cardiovascular Diseases Classification | Chapman [136] | AUC: 99.0 |
| | ECG-LM [137] | Feature-level (Intermediate) | Pre-Trained Architectures | BioMedGPT-LM | Cardiovascular Diseases Classification | PTB-XL [138] | ACC: 69.3, F1: 0.647 |
| | C-MELT [139] | Feature-level (Intermediate) | Attention-Based Fusion | Cross-Attention | Cardiac Arrhythmia Classification | MIMIC-IV-ECG [140], PhysioNet 2021 [141] | ACC: 85.7 |
| Search System | Qiu et al. [142] | Feature-level (Intermediate) | Attention-Based Fusion | Cross-Attention | Cross-Modal Retrieval | PTBXL [138] | RSUM: 133.5 |

which integrates visual features into a pretrained language model. This integration strengthens semantic alignment between modalities and, compared with the previous method, places greater emphasis on temporal answer grounding, requiring precise identification of an answer's location within the video.

Overall, intermediate fusion has become the mainstream strategy for medical vision-language tasks. Among these, attention-based multimodal interaction is the most widely adopted. Its primary advantage lies in promoting deep semantic alignment and dynamic feature selection, thereby improving the model's ability to capture complex relationships between medical images and text.

### 5.2. Text + Signal

This section reviews prior studies on the fusion of medical texts and physiological signals, following the structure of the previous section. A summary of these studies and their outcomes is presented in Table 4.

#### 5.2.1. Signal joint diagnosis

The applications combining text and signal modalities are similar to Text + Vision (Section 5.1.1), primarily focused on disease diagnosis and detection. Typically, these methods first process each modality separately, then combine their information using fusion strategies. For instance, Hernandez et al. [133] developed a predictive model for hemodynamic instability in post-cardiac surgery patients. The model extracts and refines physiological signal features (such as ECG, arterial blood pressure, and pulse oximetry), integrates them with clinical records and laboratory data, and then applies traditional machine learning algorithms. Instead of adopting a strict early-fusion scheme, it concatenates multimodal features at an intermediate stage before modeling. To mitigate the high dimensionality of physiological signals, the features are reformatted to align with EHR text, thereby reducing potential information loss. ECG-LM [137] is a multimodal LLM that enhances cardiovascular disease detection by combining ECG signals with textual features. Due to limited paired datasets, the authors synthesized a large corpus of ECG-text pairs from clinical guidelines for model pretraining.

Another approach, C-MELT [139], employs multimodal self-supervised learning. It pretrains a contrastive-enhanced autoencoder architecture on ECG-text data, significantly improving cardiac arrhythmia classification and patient identification. Lalam et al. [134] proposed an ECG representation learning method based on multimodal contrastive learning, integrating ECG signals with structured and textual EHR data. This approach effectively addresses challenges arising from limited single-modality data. Contrastive learning-based CLIP fusion methods align multimodal data into a unified representation space, thereby enhancing representation quality and discriminative performance in downstream tasks.

#### 5.2.2. Search system

Search systems that integrate signals with medical reports significantly improve retrieval efficiency by automatically identifying similar records. Qiu et al. [142] introduced an ECG-text retrieval framework that combines multimodal information via a cross-attention mechanism, employing machine learning and image-text alignment techniques to accurately retrieve relevant ECG signals and expert reports, thereby supporting clinical interpretation and diagnosis. Interestingly, their findings reveal that smaller models (ViT-base) can outperform larger ones (ViT-large) on limited medical datasets, highlighting the importance of aligning model scale with data availability.

The fusion of text and physiological signals closely resembles strategies in text-image fusion, with Intermediate Fusion being the most widely adopted. Some approaches [142] even convert time-series signals into image-like representations, making it possible to apply advanced computer vision models for richer spatial feature extraction. This strategy strengthens the representational power of signals and enhances the discriminative capability of multimodal frameworks, ultimately improving overall retrieval performance. Image-based transformations exploit mature vision models to capture stronger spatial features, whereas direct signal-text fusion preserves temporal dynamics that may be lost in image conversion. Consequently, the choice of method should be guided by whether spatial abstraction or temporal fidelity is of greater importance for diagnostic accuracy.

### 5.3. Other multimodal tasks

#### 5.3.1. Text + Audio

Text data, such as EHR or clinical narratives, often contain detailed patient information, while audio signals capture respiratory or vocal patterns relevant to diagnosis. Zhang et al. [143] developed a multimodal language model that integrates audio and EHR text for comprehensive respiratory health prediction by employing cross-modal attention mechanisms to enable effective audio-text fusion, and demonstrated through experiments on five real-world datasets that RespLLM outperforms state-of-the-art baselines by an average of 4.6 % on trained tasks, 7.9 % on unseen datasets, while also supporting zero-shot prediction on novel tasks.

#### 5.3.2. Visual + Audio

Visual data such as facial expressions captured during patient assessments, combined with voice recordings, can reflect emotional and psychological states. Audio and visual modalities are commonly combined in emotion recognition systems to improve accuracy. Speech signals provide emotional cues through tone and speech rate, while visual signals convey emotional expressions. Hossain et al. [144] proposed an audio-visual emotion recognition system using separate CNN models for speech (2D CNN) and visual data (3D CNN), where the features are fused





**表 4** 总结 不同 方法 在 文本 + 生理信号 模态 医学 任务 中的表现（早期 融合： 数据级， 中间 融合： 特征级， 晚期 融合： 决策级）。 *表示 非公开 数据集。

| 任务 Type | M方法 | 融合 阶段 | Model Archi架构 | 融合 算法 | 目标 get Tasks | D数据集 | 性能 表现 |
|---|---|---|---|---|---|---|---|
| 联合 诊断 | Hernandez 等人 [133] | Data-level (Early) | 简单 融合 | – | 不良 事件 预测 | 密歇根 医 学* | AUC: 87.0 |
| | sEHR-BERT [134] | 特征级（中间） 自监督 学习 | | CLIP | 心血管 疾病 Classification | PhysioNet2020 [135] | AUC: 91.5 |
| | | | | | 心血管 疾病 Classification | Chapman [136] | AUC: 99.0 |
| | ECG-LM [137] | F特征级 （中间 层） | 预训练ained Architectures | BioMedGPT-LM | Cardiovascular Diseases Classification | PTB-XL [138] | ACC: 69.3, F1: 0.647 |
| | C-MELT [139] | F特征级 （中间 层） | Atte基于注意力 融合 | 交叉注意力 n | 心脏 心律失常 a Classification | MIMIC-IV-ECG [140], PhysioNet 2021 [141] | ACC: 85.7 |
| Search System | Qiu 等人 。 [142] | F特征级 （Interm 介） | Atte基于注意力的 融合 | 交叉注意力 n | 跨模态 检索 | PTBXL [138] | RSUM: 133.5 |

其将视觉特征集成到预训练语言模型中。这种集成增强了模态之间的语义对齐，并且与之前的方法相比，更加强调时间答案定位，要求精确识别答案在视频中的位置。

总体而言，中间融合已成为医学视觉-语言任务的主流策略。其中，基于注意力的多模态交互被最广泛采用。其主要优势在于促进深度语义对齐和动态特征选择，从而提升模型捕捉医学图像与文本之间复杂关系的能力。

### 5.2. 文本 + 信号

本节按照上一节的结构，回顾了关于医学文本与生理信号融合的先前研究。这些研究及其结果的总结见表4。

#### 5.2.1. 信号 联合 诊断

结合文本与信号模态的应用与文本+ 视觉（第5.1.1节）类似，主要聚焦于疾病诊断与检测。通常，这些方法首先分别处理每个模态，然后通过融合策略将其信息组合起来。例如，Hernandez等人[133] 开发了一种用于心脏手术后患者血流动力学不稳定的预测模型。该模型提取并精炼生理信号特征（如ECG、动脉血压和脉搏血氧饱和度），将其与临床记录和实验室数据整合，然后应用传统机器学习算法。该模型并非采用严格的早期融合方案，而是在建模前于中间阶段拼接多模态特征。为缓解生理信号的高维度问题，特征被重新格式化以与EHR文本对齐，从而减少潜在的信息损失。ECG-LM[137] 是一种多模态大语言模型，通过结合ECG信号与文本特征来增强心血管疾病检测。由于配对数据集有限，作者从临床指南中合成了大量ECG-文本对语料库用于模型预训练。

另一种方法C-MELT [139]，采用多模态自监督学习。它在ECG-文本数据上预训练一种对比增强的自编码器架构，显著提升了心律失常分类和患者识别性能。Lalam等人[134] 提出了一种基于多模态对比学习的心电图表征学习方法，将ECG信号与结构化和文本电子健康记录数据相结合。该方法有效应对了单模态数据有限所带来的挑战。基于对比学习的CLIP融合方法将多模态数据对齐到统一表征空间，从而提升了下游任务中的表示质量和判别性能。

#### 5.2.2. 检索 系统

将信号与医学报告相集成的搜索系统通过自动识别相似记录，显著提高了检索效率。Qiu等人[142] 提出了一种ECG-文本检索框架，通过交叉注意力机制结合多模态信息，采用机器学习和图像-文本对齐技术准确检索相关ECG信号和专家报告，从而支持临床解读和诊断。有趣的是，他们的研究结果表明，在有限的医学数据集上，较小的模型（ViT-base）可以优于较大的模型（ViT-large），凸显了模型规模与数据可用性相匹配的重要性。

文本与生理信号的融合与文本-图像融合策略高度相似，其中中间融合是最广泛采用的方式。一些方法[142] 甚至将时间序列信号转换为类图像表示，使得能够应用先进的计算机视觉模型进行更丰富的空间特征提取。该策略增强了信号的表示能力，并提升了多模态框架的判别能力，最终改善了整体检索性能。基于图像的变换利用成熟的视觉模型来捕获更强的空间特征，而直接的信号-文本融合则保留了在图像转换中可能丢失的时间动态。因此，方法的选择应取决于空间抽象还是时间保真度对诊断准确性更为重要。

### 5.3. 其他 多模态 任务

#### 5.3.1. 文本 + 音频

文本数据（如电子健康记录或临床叙述）通常包含详细的患者信息，而音频信号则捕获与诊断相关的呼吸或声音模式。Zhang等人[143] 开发了一种多模态语言模型，通过采用跨模态注意力机制实现有效的音频-文本融合，将音频与电子健康记录文本相结合以进行全面呼吸健康预测，并通过在五个真实世界数据集上的实验证明，RespLLM在训练任务上平均优于最先进基线4.6%，在未见数据集上优于7.9%，同时还支持对新任务的零样本预测。

#### 5.3.2. 视觉 + 音频

在患者评估过程中采集的面部表情等视觉数据，结合语音记录，可以反映情绪和心理状态。音频和视觉模态通常在情感识别系统中被结合使用以提高准确率。语音信号通过语调和语速提供情绪线索，而视觉信号则传达情绪表达。Hossain等[144] 提出了一种视听情感识别系统，分别使用CNN模型处理语音（二维CNN）和视觉数据（三维CNN），其中特征通过

through extreme learning machines in a late fusion strategy, enhancing the accuracy and efficiency of emotion recognition.

#### 5.3.3. Signal + Audio

Signal data such as EGG or ECG recordings, when linked with audio inputs like patient speech, are useful in detecting speech pathologies. In the EHR context, these signals help track vocal function and its relation to underlying conditions. Hossain et al. [145] proposed an intelligent healthcare monitoring framework that diagnoses speech disorders by analyzing audio and EGG signals from patients, achieving a voice pathology classification accuracy of 93 %. An early fusion strategy is used to integrate multiple data sets from both modalities into a unified input. This strategy is structurally simple, easy to implement, and enables effective multimodal fusion at the data level, making it well-suited for low-resource, small-data.

#### 5.3.4. Multimodal integration

EHRs contain diverse multimodal data with high variability, dimensionality, and irregularity, which makes integration particularly challenging. Beyond simple pairwise modality combinations, EHRs encompass richer configurations involving multiple modalities simultaneously, offering opportunities for more comprehensive patient modeling.

The HAIM framework [146] integrates image, text, time-series, and tabular data into a unified AI model, using consistent preprocessing, modality contribution quantification to mitigate heterogeneity, and feature learning to address high dimensionality, thereby improving predictive performance across diverse healthcare tasks. The multimodal fusion subtyping framework [147] adopts a hybrid fusion strategy. It first integrates radiological, pathological, genomic, transcriptomic, and proteomic features through intermediate fusion. After that, it performs late fusion of 11 clustering outputs using consensus clustering. This two-stage design helps overcome modality heterogeneity and high dimensionality, ultimately enabling the identification of glioma subtypes with distinct prognostic and therapeutic potential. Compared with early and intermediate fusion, the late fusion strategy avoids feature incompatibility and enhances robustness under missing modality conditions by fusing independent models at the decision-level without the need for unified modeling at the feature or score level. Boehm et al. [148] applied a late fusion strategy to integrate clinicogenomic, radiologic, and histopathologic machine-learning models, leveraging complementary imaging features such as nuclear size and omental texture to improve risk stratification of high-grade serous ovarian cancer and achieve significantly better prediction of overall and progression-free survival in real-world cohorts. Das et al. [149] proposed a posterior-probability-based decision-level (late) fusion strategy for multimodal eye recognition in the visible spectrum, integrating iris, sclera, and periocular features. When the iris or sclera is occluded, or when the eyes are closed, the system can still rely on periocular features for recognition. Compared with single-modal approaches, the proposed method improved recognition accuracy by 7.6-17.6 % and verification accuracy by 1.0-7.7 %.

### 5.4. Comparison of methods in discriminative applications

In this section, our analysis focuses on the strengths and limitations of different fusion strategies in medical discriminative tasks, aiming to inform practical applications. Among these, the intermediate fusion strategy demonstrates the best overall performance and is suitable for most medical scenarios. It aligns multimodal information at the feature level, often using shared attention or gating mechanisms. This approach enables efficient integration of complementary information with moderate parameter complexity. It has achieved strong results across multiple clinical benchmarks and shows a lower risk of overfitting. In contrast, early fusion is more appropriate for tasks with limited data, constrained computational resources, or real-time requirements. It performs well when modalities are well-aligned and noise levels are low, but is more sensitive to noise and modality mismatch. Late fusion, on the other hand, is more suitable for scenarios with high modality heterogeneity, difficulty in unified feature representation, or missing modalities. For example, as shown in [149], it has been effectively applied in sclera recognition, where certain modalities may be partially occluded. By allowing each modality to be processed independently before decision-level integration, it enhances robustness in the presence of complex or incomplete data.

## 6. Generative applications

This section discusses the application of multimodal learning in generative tasks. We first introduce the main generative networks, and then classify these applications according to the specific combinations of data modalities. The generative directions between different modalities are shown in Table 5.

### 6.1. Generative networks

Before exploring specific application tasks, we introduce three widely adopted backbone models for general generative tasks, extensively used in multimodal generative research.

#### 6.1.1. VAE-based models

Variational Auto-Encoders (VAEs) [188] build on auto-encoders [189], adding probabilistic encoding. The encoder outputs a Gaussian distribution, from which latent codes are sampled. VAEs optimize the evidence lower bound (ELBO), balancing reconstruction and regularization:

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - D_{\text{KL}}[q(z|x)\|p(z)] \tag{3}$$

Here, $p(z)$ is typically $\mathcal{N}(0,1)$.

VAE-based models are well suited for medical tasks requiring controllability and interpretability, such as structured image synthesis, lesion generation, and cross-modal alignment [190,191]. Optimizing the ELBO allows VAEs to balance reconstruction and regularization, enabling generalization in low-resource or explainability-focused scenarios. However, they often produce blurrier outputs than GANs and are limited by Gaussian assumptions, making them less ideal for high-fidelity synthesis but effective for structured, interpretable generation.

#### 6.1.2. GAN-based models

Generative Adversarial Networks (GANs) [192] train a generator $G$ against a discriminator $D$, where $G$ tries to fool $D$ by producing realistic samples. The standard objective is:

$$\min_G \max_D \; V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p_Z}[\log(1 - D(G(z)))] \tag{4}$$

GAN-based models are well suited for multimodal medical generation tasks that demand high visual fidelity [193–195]. Through adversarial training, GANs effectively model complex data distributions without relying on explicit likelihoods, making them powerful for generating realistic outputs from unlabeled or weakly labeled data. GANs face persistent challenges, including training instability, mode collapse, and limited latent space interpretability, all of which reduce controllability. Consequently, these models are better suited for applications that prioritize output realism over structural accuracy or semantic consistency.

#### 6.1.3. DPM-based models

Diffusion Probabilistic Models (DPMs) [196] gradually add and then remove noise through forward and reverse Markov processes. Training minimizes the variational bound:

$$\mathcal{L}_{\text{vb}} = \mathbb{E}_q\Big[D_{\text{KL}}(q(x_T|x_0)\|p(x_T))$$





通过 极限学习机 的 晚期 融合策略， 提升 情绪 识别的准确率 与 效率。

#### 5.3.3. 信号 + 音频

信号数据（如EGG或ECG记录）与音频输入（如患者语音）相关联时，有助于检测语音病理。在电子健康记录的背景下，这些信号有助于追踪发声功能及其与潜在病症的关系。Hossain等[145] 提出了一种智能医疗健康监测框架，通过分析患者的音频与EGG信号来诊断言语障碍，实现了93%的嗓音病理分类准确率。该框架采用早期融合策略，将来自两种模态的多个数据集整合为统一输入。该策略结构简单、易于实现，能够在数据层面实现有效的多模态融合，非常适合低资源、小数据场景。

#### 5.3.4. 多模态 集成

电子健康记录 包含 多样的 多模态 数据， 具有 高 变异性、 高维度和 不规则性， 这 使得 集成 尤为 具有挑战性。 除了 简单的 成对 模态 组合之外， 电子健康记录 还包含 更丰富的 配置， 涉及 多种 模态 同时存在， 为 更全面的 患者 建模提供了机会。

HAIM框架[146] 将图像、文本、时间序列和表格数据整合为一个统一AI模型，采用一致的数据预处理、模态贡献量化来缓解异质性，并通过特征学习来解决高维度问题，从而提升跨多种医疗任务的预测性能。多模态融合亚型分型框架[147] 采用混合融合策略。它首先通过中间融合整合放射学、病理学、基因组学、转录组学和蛋白质组学特征。之后，利用共识聚类对11个聚类输出进行晚期融合。这种两阶段设计有助于克服模态异质性和高维度问题，最终能够识别具有不同预后和治疗潜力的胶质瘤亚型。与早期融合和中间融合相比，晚期融合策略通过在决策级融合独立模型，无需在特征或分数层面进行统一建模，从而避免了特征不兼容性，并增强了模态缺失条件下的鲁棒性。Boehm等[148] 应用晚期融合策略整合临床基因组学、影像学和组织病理学机器学习模型，利用细胞核大小和大网膜纹理等互补影像特征，改进高级别浆液性卵巢癌的风险分层，并在真实世界队列中实现对总生存期和无进展生存期的显著更优预测。Das等[149] 提出了一种基于后验概率的决策级（晚期）融合策略，用于可见光谱下的多模态眼部识别，整合虹膜、巩膜和眼周特征。当虹膜或巩膜被遮挡，或眼睛闭合时，系统仍可依赖眼周特征进行识别。与单模态方法相比，所提方法将识别准确率提高了7.6-17.6%，验证准确率提高了1.0-7.7%。

### 5.4. 判别式应用中 方法 的 比较

在本节中，我们的分析聚焦于医学判别任务中不同融合策略的优势与局限性，旨在为实际应用提供指导。其中，中间融合策略展现出最佳的整体性能，适用于大多数医学场景。该策略在特征级对多模态信息进行对齐，通常采用共享注意力或门控机制。这种方法能够以中等的参数复杂度高效整合互补信息。它已在多个临床基准上取得优异结果，并表现出较低的过拟合风险。相比之下，早期融合更适合数据有限、计算资源受限或具有实时性要求的任务。当模态对齐良好且噪声水平较低时，它表现良好，但更容易受到噪声和模态不匹配的影响。另一方面，晚期融合更适合模态异质性高、难以实现统一特征表示或存在模态缺失的场景。例如，如[149]，所示，它已被有效应用于巩膜识别中，其中某些模态可能被部分遮挡。通过允许每个模态在决策级整合之前独立处理，它增强了在复杂或不完整数据情况下的鲁棒性。

## 6. 生成式 应用

本节讨论多模态学习在生成任务中的应用。我们首先介绍主要的生成网络，然后根据数据模态的具体组合对这些应用进行分类。不同模态之间的生成方向如表5所示。

### 6.1. 生成网络

在探索具体应用任务之前,我们介绍三种广泛采用的骨干模型,用于通用生成任务,这些模型在多模态生成研究中被广泛使用。

#### 6.1.1. 基于VAE的 模型

变分 自编码器 (VAE) [188] 基于 自编码器 构建 [189]，，加入了 概率化 编码。 该 编码器 输出 一个 高斯分布， 并从中 采样 得到 潜在编码。 VAE 优化 证据下界 (ELBO)， 在重建 与正则化之间取得平衡:

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - D_{\text{KL}}[q(z|x)\|p(z)] \tag{3}$$

其中，$p(z)$ 通常为$\mathcal{N}(0,1)$。

基于VAE的模型非常适合需要可控性和可解释性的医学任务，例如结构化图像合成、病灶生成和跨模态对齐[190,191]。优化ELBO使VAE能够平衡重建与正则化，从而在低资源或侧重可解释性的场景中实现泛化。然而，它们生成的输出通常比GAN更模糊，并受限于高斯假设，因此不太适合高保真度合成，但在结构化、可解释的生成方面效果显著。

#### 6.1.2. 基于GAN的模型

生成对抗网络（GAN）[192] 训练一个生成器$G$对抗一个判别器$D$，其中$G$ 试图通过生成逼真的样本来欺骗$D$ 。标准目标是:

$$\min_G \max_D \; V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p_Z}[\log(1 - D(G(z)))] \tag{4}$$

基于GAN的模型非常适合需要高视觉保真度的多模态医学生成任务[193–195]。通过对抗训练，GAN能够在不依赖显式似然的情况下有效建模复杂的数据分布，使其在从无标注或弱标注数据生成逼真输出方面具有强大能力。GAN面临持续存在的挑战，包括训练不稳定、模式崩溃和潜在空间可解释性有限，这些都会降低可控性。因此，这些模型更适合优先考虑输出逼真度而非结构准确性或语义一致性的应用。

#### 6.1.3. 基于DPM的 模型

扩散 概率 模型 (DPM) [196] 通过 正向和反向 马尔可夫 过程逐步添加 并 随后 去除 噪声 。 训练 最小化 变分下界 ：

$$\mathcal{L}_{\text{vb}} = \mathbb{E}_q\Big[D_{\text{KL}}(q(x_T|x_0)\|p(x_T))$$

**Table 5**
Summary of generation directions from the modality perspective: T (Text), V (Vision), S (Physiological signal), A (Audio).

| Modality Mapping | Task | Example Methods |
|---|---|---|
| T → V | Medical image generation | Chen et al. [150], Xu et al. [151], Akrout et al. [152], Lee et al. [153], Huang et al. [154], Chambon et al. [155] Bluethgen et al. [156], Kim et al. [157] |
| T → S | Physiological signal simulation | Chung et al. [158], Delaney et al. [159], Chen et al. [160] |
| V → T | Medical image report generation | Zeng et al. [161], Cao et al. [162], Tanwani et al. [163], Hou et al. [164], Zhou et al. [165], Liu et al. [166] Jin et al. [167], Wang et al. [168], Tanida et al. [169], Bu et al. [170], Zhang et al. [171], You et al. [172] Gu et al. [173] |
| V + T → T | Medical visual question answering (MVQA) | Moor et al. [174], Yuan et al. [175], Chen et al. [176], Chen et al. [177] |
| V + A + T → T | Audiovisual question answering (AVQA) | Li et al. [178], Neto et al. [179] |
| S → T | Physiological signal report generation | Biswal et al. [180], Tung et al. [181], Tian et al. [182] |
| S → T | Physiological signal caption generation | Bartels et al. [183], Tran et al. [184] |
| S + T → T | Physiological signal QA | Yang et al. [185], Tang et al. [186] |
| S → V | Signal-driven video generation | Liu et al. [187] |

$$+\sum_{t>1} D_{\mathrm{KL}}(q(x_{t-1}|x_t,x_0)\|p_\theta(x_{t-1}|x_t)) - \log p_\theta(x_0|x_1)\Big] \quad (5)$$

DPM-based models generate high-fidelity outputs by gradually denoising data through forward and reverse diffusion processes, optimizing a likelihood-based variational bound. They offer stable training and precise control, making them suitable for complex multimodal medical tasks requiring fine-grained cross-modal alignment [197]. However, their iterative sampling is computationally intensive, limiting efficiency in real-time scenarios. DPMs are advantageous in settings where generation quality and modality consistency are prioritized over sampling speed.

### 6.2. Text + Vision

#### 6.2.1. Text report generation from medical images

*Medical image report generation:* Medical image report generation is a typical cross-modal task that aims to transform visual data into structured or free-text diagnostic reports. Its goal is to support doctors in clinical decision-making, improve diagnostic efficiency, and minimize human oversight. High-quality report generation relies on three key techniques: contextual understanding, fine-grained feature extraction, and integration of medical knowledge. Contextual information enables accurate localization of abnormalities and establishes their relationship with surrounding anatomical structures.

Zeng et al. [161] developed AERMNet, an attention-enhanced memory network that improves report accuracy by capturing word relationships. Cao et al. [162] proposed Kdtnet, which integrates image features with expert knowledge to enhance contextual reasoning. Tanwani et al. [163] applied semantic attention to highlight informative image regions and gather contextual details. Hou et al. [164] designed a visual recalibration module to emphasize lesion features and a context-aware module to fuse them into reports. Zhou et al. [165] introduced a model that aligns text with images by combining visual features, clinical data, and semantic embeddings.

Another research direction emphasizes fine-grained alignment between image details and medical terminology. These methods involve either single-grain alignment, which links specific image regions to textual descriptions, or multi-grain alignment, which considers the global image. Liu et al. [166] proposed a framework that analyzes individual report sentences to match their themes and content with image features. You et al. [172] developed AlignTransformer, which predicts disease labels and refines visual representations to improve reporting. Jin et al. [167] introduced a mechanism for detecting both coarse and subtle anomalies prior to report generation. Wang et al. [168] designed AGFNet, adapting global and local features according to the presence of abnormalities.

A third line of work focuses on anatomy-based approaches that explicitly model anatomical structures to improve localization and interpretability. Tanida et al. [169] proposed RGRG, which employs object detection to identify anatomical regions in X-ray images and generates reports by analyzing each region individually. Bu et al. [170] combined expert knowledge with attention mechanisms to better capture organ- and lesion-specific information. Gu et al. [173] used image segmentation to extract organ-level features and associate them with disease-related keywords. Zhang et al. [171] introduced a weakly supervised approach that highlights key regions through attention mechanisms, thereby improving reporting quality.

These three categories of methods differ in focus and applicability. Context-aware models emphasize global semantics and are well suited for complex cases with multiple lesions. Fine-grained alignment approaches improve precision by linking subtle image cues to text, making them effective for detecting nuanced abnormalities. Anatomy-based methods incorporate structural priors, enabling more accurate localization and enhancing interpretability, which is particularly valuable for region-specific analysis.

*Medical visual question answering (MVQA):* . The Visual Question Answering (VQA) task began with Antol et al. in 2015 [198]. It focuses on answering natural language questions using visual content. Medical Visual Question Answering (MVQA) adapts this idea to clinical contexts, where models respond to questions about diagnosis or procedures based on medical images. Such tasks usually involve large-scale question-answer pairs. This requires models to integrate both visual and textual information with high precision. Most MVQA systems follow an encoder-decoder framework. In this design, encoders process images and text independently, while the decoder generates natural language answers, which facilitates effective cross-modal interaction.

For instance, Med-Flamingo [174] employs a gated cross-attention mechanism to connect visual and textual features. It is further enhanced by few-shot prompting, which enables the model to provide clinically grounded explanations. In contrast, RAMM [175] adopts a retrieval-enhanced fine-tuning strategy. By drawing on medical literature and image-text pairs, it strengthens the input representations and improves downstream performance. From a methodological perspective, prompt-driven approaches highlight flexibility, retrieval-based pipelines emphasize contextual enrichment, and fine-tuning strategies enhance domain adaptation. However, these paradigms also reveal distinct weaknesses: prompt methods risk superficial reasoning, retrieval pipelines depend heavily on external sources, and fine-tuning often entails significant computational cost. To further adapt models to medical data, cross-domain fine-tuning has become common. Surgical VQA illustrates this difficulty clearly. Surgical tools and environments vary widely, which makes generalization across datasets challenging. Chen et al. [176] proposed a framework that combines multi-teacher distillation with LLMs. In this framework, the language model functions as a teacher to miti-





**表 5** 从模态视角看生成方向 的 总结： T （文本）、 V （视觉）、 S （生理 信号）、 A （音频）。

| M模态 M映射 | Task | 示例 方法 |
|---|---|---|
| T → V | Medical image generation | Chen et al. [150], Xu et al. [151], Akrout et al. [152], Lee et al. [153], Huang et al. [154], Chambon et al. [155] Bluethgen et al. [156], Kim et al. [157] |
| T → 生理信号 | Physiological signal simulation | Chung et al. [158], Delaney et al. [159], Chen et al. [160] |
| V → 文本 | 医学 图像 影像 报告 生成 | Zeng 等人[161], Cao 等人[162], Tanwani 等人[163], Hou 等人[164], Zhou 等人[165], Liu et al. [166] 金 等人[167], Wang 等 人[168], Tanida 等 人, [169], Bu et al. [170], Zhang et al. [171], 你 等 人[172] Gu est al. [173] |
| V + 文本 → 文本 | Medical visual 问答 （MVQA） | Moor 等 人[174], Yuan 等人[175], Chen 等 人[176], Chen 等 人[177] |
| V + A + 文本 → 文本 | 视听 问答 任务 （AVQA） | Li 等人[178], Neto 等 人[179] |
| S → T | Physiological signal report 生成 | Biswal et al. [180], Tung et al. [181], Tian 等人[182] |
| S → 文本 | Physiological signal caption关于 生成ion | Bartels et al. [183], Tran et al. [184] |
| S + T → T | Physiological signal QA | Yang 等人[185], Tang 等 人。[186] |
| S → V | 信号驱动的 vi 报告 生成 | Liu 等 人[187] |

$$+\sum_{t>1} D_{\mathrm{KL}}(q(x_{t-1}|x_t,x_0)\|p_\theta(x_{t-1}|x_t)) - \log p_\theta(x_0|x_1)\Big] \quad (5)$$

基于DPM的模型通过正向和反向扩散过程逐步去噪数据，优化基于似然的变分界，从而生成高保真度输出。它们提供稳定的训练和精确的控制，使其适用于需要细粒度跨模态对齐的复杂多模态医学任务[197]。然而，其迭代采样计算量大，限制了实时场景中的效率。在优先考虑生成质量和模态一致性而非采样速度的场景中，DPM具有优势。

### 6.2. 文本 + 视觉

#### 6.2.1. 基于 医学 图像 的 文本 报告生成

医学图像报告生成：医学图像报告生成是一项典型的跨模态任务，旨在将视觉数据转化为结构化或自由文本的诊断报告。其目标是支持医生进行临床决策，提高诊断效率，并减少人工监督。高质量的报告生成依赖于三项关键技术：上下文理解、细粒度特征提取以及医学知识的整合。上下文信息能够准确定位异常并建立其与周围解剖结构的关系。

Zeng等人[161] 开发了AERMNet，这是一种注意力增强记忆网络，通过捕获词语关系来提高报告准确率。Cao等人[162] 提出了Kdtnet，它将图像特征与专家知识相结合以增强上下文推理。Tanwani等人[163] 应用语义注意力来突出信息丰富的图像区域并收集上下文细节。Hou等人[164] 设计了一个视觉重校准模块以强调病变特征，以及一个上下文感知模块将其融合到报告中。Zhou等人[165] 引入了一个模型，通过结合视觉特征、临床数据和语义嵌入来对齐文本与图像。

另一个研究方向强调图像细节与医学术语之间的细粒度对齐。这些方法涉及单粒度对齐，即将特定图像区域与文本描述相关联，或多粒度对齐，即考虑全局图像。Liu等人[166] 提出了一个框架，通过分析单个报告句子来将其主题和内容与图像特征进行匹配。You等人[172] 开发了AlignTransformer，该模型预测疾病标签并优化视觉表示以改进报告生成。Jin等人[167] 引入了一种在报告生成之前检测粗粒度和细微异常的机制。Wang等人[168] 设计了AGFNet，根据异常的存在自适应调整全局与局部特征。

第三类工作聚焦于基于解剖结构的方法，这些方法显式地建模解剖结构以改进定位和可解释性。Tanida等人[169] 提出了RGRG，该方法采用目标检测来识别X射线图像中的解剖区域，并通过分析每个区域单独生成报告。Bu等人[170] 将专家知识与注意力机制相结合，以更好地捕获器官和病灶特异性信息。Gu等人[173] 使用图像分割提取器官级特征，并将其与疾病相关关键词相关联。Zhang等人[171] 引入了一种弱监督方法，通过注意力机制突出关键区域，从而改进报告质量。

这三类方法在侧重点和适用性上有所不同。上下文感知模型强调全局语义，非常适合处理涉及多个病灶的复杂病例。细粒度对齐方法通过将细微的图像线索与文本相关联来提高精确率，使其能够有效检测细微的异常。基于解剖学的方法引入了结构先验，能够实现更精确的定位并增强可解释性，这对于区域特定分析尤为有价值。

医学视觉问答（MVQA）：。视觉问答（VQA）任务始于Antol等人在2015 [198]中的工作。它侧重于利用视觉内容回答自然语言问题。医学视觉问答（MVQA）将这一思想应用于临床情境，模型根据医学图像回答有关诊断或操作的问题。此类任务通常涉及大规模的问答对。这要求模型以高精确率整合视觉和文本信息。大多数MVQA系统遵循编码器-解码器框架。在该设计中，编码器分别处理图像和文本，而解码器生成自然语言答案，从而促进有效的跨模态交互。

例如，Med-Flamingo[174] 采用门控交叉注意力机制来连接视觉与文本特征。它进一步通过少样本提示得到增强，使模型能够提供临床依据解释。相比之下，RAMM[175] 采用检索增强的微调策略。通过利用医学文献和图像-文本对，它增强了输入表示并提升了下游性能。从方法论角度看，提示驱动方法强调灵活性，基于检索的流程强调上下文丰富，而微调策略增强领域适应。然而，这些范式也暴露出各自的弱点：提示方法存在肤浅推理的风险，检索流程严重依赖外部来源，微调往往需要大量计算成本。为了进一步使模型适应医疗数据，跨域微调已变得普遍。外科视觉问答清晰地说明了这一困难。手术工具和环境差异很大，这使得跨数据集的泛化具有挑战性。Chen等人[176] 提出了一个将多教师蒸馏与大语言模型结合的框架。在该框架中，语言模型充当教师，以减轻

gate catastrophic forgetting during adaptation. It achieved an average accuracy of approximately 0.52 on cross-dataset tasks, demonstrating its ability to generalize across multiple surgical types.

Meanwhile, MedBLIP [177] introduces a query network into a 2D image encoder. This addition transforms 3D image features into a representation compatible with language models. By enabling question answering on 3D medical data, MedBLIP broadens the scope of MVQA. Nevertheless, it also underscores the persistent tension between richer multimodal integration and the associated computational demands.

*Audio-visual question answering (AVQA):* . Most AVQA methods build upon the traditional VQA framework by incorporating audio information to handle questions involving visual objects, sounds, and their relationships in videos. For example, Li et al. [178] proposed a three-branch self-supervised encoder that independently processes video, audio, and text data, fuses them via an attention mechanism, and leverages masked language modeling, image-text matching, and contrastive losses during pretraining. This approach achieved state-of-the-art performance on three medical VQA datasets: VQA-RAD with an accuracy improvement of 2.2%, PathVQA with 14.7%, and Slake with 1.7%. While fusion-oriented methods improve accuracy by capturing fine-grained cross-modal relations, interpretability-focused designs offer clearer outputs. Studies have shown that audiovisual reports are easier to comprehend than text-only ones [179], enabling faster analysis of imaging results by orthopedic surgeons. Real-world trials further indicate that most attending physicians are willing to adopt this technology in similar future scenarios.

#### 6.2.2. Medical vision generation from text

*Medical image generation:* Medical image generation focuses on creating synthetic medical images from textual descriptions. In contrast to medical image report generation, it emphasizes transforming language into realistic medical visual data. In anatomy-guided generation methods, Chen et al. [150] proposed an anatomical pathology prompt module that utilizes GPT-4 to generate detailed descriptions of anatomical structures and pathological conditions. Visual prompts (key patches) are then extracted through fine-grained image-text alignment to guide the generation of medical images. Xu et al. [151] employed a pretrained medical BERT model to encode radiology reports (Medsyn). The generation process initiates with low-resolution images and progressively produces high-resolution CT images using a cross-attention mechanism combined with anatomical structure segmentation maps. Akrout et al. [152] introduced a stable diffusion model that enhances fine-grained control over the generated image based on input text prompts. ViewXGen [153] is a typical multi-view generation model that uses a GAN model as image view tokens. An improved Transformer is employed to enhance computational and memory efficiency while incorporating multi-view information. ViewXGen is capable of generating images that closely resemble real-world images and can also create views that do not exist in reality. Anatomy-guided approaches are often preferred for lesion localization and surgical planning because they integrate structural priors. By contrast, multi-view models excel in visual realism and diversity, yet may fall short in scenarios where strict pathological consistency is essential.

With the rise of diffusion models in recent years, their basic principle involves gradually adding noise to data, disrupting it into a random distribution, and then progressively denoising this data in a reverse process to reconstruct the original input. Diffusion models offer superior controllability and generation quality, making them particularly suitable for synthesizing complex medical images with high fidelity. In chest X-ray generation, Chester diffusion [154] employs a transformer within a diffusion model to achieve a lightweight architecture and incorporates BiomedCLIP for contrastive learning between generated images and the corresponding medical reports. Chambon et al. [155] explored a stable diffusion framework consisting of a VAE, U-Net, and a text encoder, fine-tuning the U-Net with medical-specific text prompts to generate chest X-ray images with pathological features. Bluethgen et al. [156] introduced a VLM with conditional mechanisms into a diffusion model, enabling the generation of high-quality X-ray images based on free-form medical text prompts (e.g., lesion size and location). Kim et al. [157] developed a structure-guided diffusion model, where the structure-guided network copies parameters from the encoder and central block of the denoising network, updating them with structural mask conditions. This approach incorporates both global and local structural information to facilitate multimodal MRI synthesis. Diffusion models thus prove valuable for producing high-fidelity pathological images, particularly in training and simulation contexts, whereas structure-guided variants are more advantageous for cross-modal synthesis

### 6.3. Text + Signal

#### 6.3.1. Text report generation from physiological signals

Text report generation from physiological data seeks to transform raw recordings into clinically meaningful narratives, providing a bridge between complex signal patterns and interpretable diagnostic information. The process is shown in Fig. 4(B). EEGtoText [180] extracts temporal patterns from EEG data by stacking CNNs and RNNs, subsequently classifying key phenotypes based on these extracted patterns. A hierarchical LSTM network combined with an attention mechanism is then employed to generate detailed explanatory EEG reports. Experimental results show that this method performs well in phenotype classification and report generation, and the effectiveness of the generated reports is verified by clinical review.

Compared with task-specific architectures, LLM-based approaches leverage large-scale pretraining to produce fluent and contextually appropriate clinical language. This enables robust generalization across related tasks with only limited domain-specific supervision. Tung et al. [181] proposed an LLM-based approach for the automatic interpretation of EEG background activity and corresponding report generation, exploiting the language understanding capabilities of LLMs to enhance diagnostic reporting. In parallel, the knowledge-enhanced automated ECG diagnosis system (KED) [182] exemplifies a foundation model for ECG analysis. It is capable of detecting diverse disease conditions as well as morphological and rhythm abnormalities, while simultaneously generating explanatory diagnostic reports. Foundation models therefore offer advantages in domain alignment, multimodal integration, and controllable outputs, but these benefits come at the cost of requiring larger and higher-quality training data. From the perspective of task support, LLM-based methods are particularly effective in generating coherent textual reports, while deep learning-based multimodal fusion methods prioritize modeling cross-signal dependencies, which are indispensable in clinically complex scenarios.

#### 6.3.2. Caption generation from physiological signals

Caption generation aims to automatically translate raw signals into human-readable textual descriptions, thereby bridging low-level signal patterns with high-level clinical semantics. Fig. 4(D) illustrates the process. Different from standard diagnostic reports, Bartels et al. [183] proposed an approach based on transformer, where the generation process is autoregressive, to automatically produce precise ECG subtitles and improve ECG interpretability, such as "Sinus rhythm 52.0/min, left axis deviation, left bundle branch block, QRS 146.0 ms". Tran et al. [184] introduced a diffusion probabilistic model DPM and employed a co-attention Siamese network (COSNet) architecture on top of it to capture the deep correlations between ECG features and semantic data, thereby generating accurate medical captions for ECG signals. Experiments on multiple datasets validated the effectiveness of the proposed method and further highlighted the superior generation quality of the DPM-based approach.





门控 灾难性 遗忘 在 自适应期间。 它 在 跨数据集 任务上 达到了 约 0.52的 平均 准确率， 展示了 其 在 多种 手术类型间 泛化的 能力。

与此同时，MedBLIP[177] 将查询网络引入二维图像编码器。这一添加将三维图像特征转换为与语言模型兼容的表示。通过支持对三维医学数据的问答，MedBLIP拓宽了MVQA的范围。然而，它也凸显了更丰富的多模态集成与相关计算需求之间持续存在的矛盾。

音视频问答（AVQA）：。大多数AVQA方法建立在传统VQA框架之上，通过纳入音频信息来处理涉及视频中视觉对象、声音及其关系的问题。例如，Li等[178] 提出了一种三分支自监督编码器，独立处理视频、音频和文本数据，通过注意力机制将它们融合，并在预训练期间利用掩码语言建模、图文匹配和对比损失。该方法在三个医学VQA数据集上取得了最先进的性能：VQA-RAD准确率提升2.2%，PathVQA提升14.7%，Slake提升1.7%。虽然面向融合的方法通过捕获细粒度跨模态关系来提高准确率，但面向可解释性的设计提供了更清晰的输出。研究表明，音视频报告比纯文本报告更容易理解，[179]，使骨科医生能够更快地分析影像结果。真实世界试验进一步表明，大多数主治医生愿意在类似的未来场景中采用这项技术。

#### 6.2.2. 基于 文本的 医学 视觉 生成

医学图像生成：医学图像生成侧重于根据文本描述创建合成医学图像。与医学图像报告生成不同，它强调将语言转化为真实感医学视觉数据。在解剖引导的生成方法中，Chen等人[150] 提出了一种解剖病理提示模块，利用GPT-4生成解剖结构和病理状况的详细描述。随后通过细粒度图像-文本对齐提取视觉提示（关键图像块），以引导医学图像的生成。Xu等人[151] 采用预训练的医学BERT模型对放射学报告进行编码（Medsyn）。生成过程从低分辨率图像开始，结合交叉注意力机制与解剖结构分割图，逐步生成高分辨率CT图像。Akrout等人[152] 引入了一种稳定扩散模型，可根据输入文本提示增强对生成图像的细粒度控制。ViewXGen[153] 是一种典型的多视角生成模型，使用GAN模型作为图像视角标记。该模型采用改进的Transformer来提升计算与内存效率，同时融入多视角信息。ViewXGen能够生成与真实世界图像高度相似的图像，还可以创建现实中不存在的视角。解剖引导方法通常更适用于病灶定位和手术规划，因为它们整合了结构先验。相比之下，多视角模型在视觉真实性和多样性方面表现优异，但在需要严格病理一致性的场景中可能有所不足。

近年来随着扩散模型的兴起，其基本原理是逐步向数据中添加噪声，将其扰乱为随机分布，然后在逆过程中逐步去噪以重建原始输入。扩散模型具有优越的可控性和生成质量，使其特别适合合成高保真度的复杂医学图像。在胸部X光生成中，Chester扩散[154] 在扩散模型内采用Transformer以实现轻量级架构，并引入BiomedCLIP对生成图像与相应医学报告进行对比学习。Chambon等[155] 探索了一种稳定的扩散框架，由VAE、U-Net和文本编码器组成，通过医学专用文本提示微调U-Net以生成具有病理特征的胸部X射线图像。Bluethgen等[156] 将具有条件机制的VLM引入扩散模型，使得能够基于自由形式的医学文本提示（例如病灶大小和位置）生成高质量的X光图像。Kim等[157] 开发了一种结构引导扩散模型，其中结构引导网络从去噪网络的编码器和中心模块复制参数，并通过结构掩码条件进行更新。该方法结合全局和局部结构信息以促进多模态MRI合成。因此，扩散模型在生成高保真病理图像方面具有重要价值，特别是在训练和模拟场景中，而结构引导变体则更有利于跨模态合成

### 6.3. 文本 + 信号

#### 6.3.1. 基于 生理信号的 文本报告 生成

基于生理数据的文本报告生成旨在将原始记录转化为具有临床意义的叙述，为复杂的信号模式与可解释的诊断信息之间搭建桥梁。该流程如图4(B)所示。EEGtoText通过堆叠CNN和RNN从EEG数据中提取时间模式，随后基于这些提取的模式对关键表型进行分类。接着，采用结合注意力机制的层次化LSTM网络生成详细的解释性EEG报告。实验结果表明，该方法在表型分类和报告生成方面表现良好，且所生成报告的有效性经临床评审验证。

与任务特定架构相比，基于LLM的方法利用大规模预训练来生成流畅且符合语境的临床语言。这使其仅需有限的领域特定监督即可在相关任务间实现稳健泛化。Tung等[181] 提出了一种基于LLM的方法，用于自动解读EEG背景活动并生成相应报告，利用LLM的语言理解能力增强诊断报告。与此同时，知识增强的自动心电图诊断系统（KED）[182] 展示了一种用于ECG分析的基础模型。它能够检测多种疾病状况以及形态学和节律异常，同时生成解释性诊断报告。因此，基础模型在领域对齐、多模态集成和可控输出方面具有优势，但这些优势的代价是需要更大规模、更高质量的训练数据。从任务支持的角度来看，基于LLM的方法在生成连贯文本报告方面尤为有效，而基于深度学习的多模态融合方法则优先建模跨信号依赖关系，这在临床复杂场景中不可或缺。

#### 6.3.2. 基于 生理信号 的 描述生成

描述生成旨在自动将原始信号转换为人类可读的文本描述，从而将低级信号模式与高级临床语义相连接。图4(D)展示了该过程。与标准诊断报告不同，Bartels等人[183] 提出了一种基于Transformer的方法，其生成过程为自回归方式，可自动生成精确的ECG字幕并提高ECG可解释性，例如“窦性心律52.0/min，电轴左偏，左束支传导阻滞，QRS 146.0 ms”。Tran等人[184] 引入了扩散概率模型DPM，并在其之上采用了共同注意力孪生网络（COSNet）架构，以捕获ECG特征与语义数据之间的深层相关性，从而为ECG信号生成准确的医学描述。在多个数据集上的实验验证了所提方法的有效性，并进一步凸显了基于DPM的方法在生成质量上的优越性。

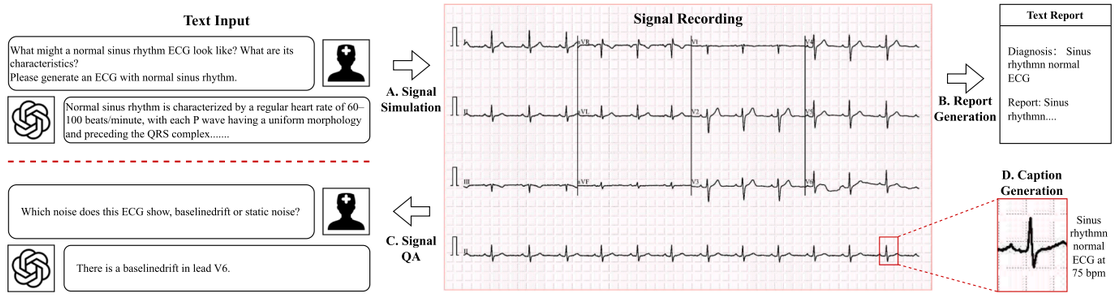



**Fig. 4.** Diagram of applications for physiological signal (ECG) generation tasks, illustrating the following functions: A. Signal Simulation: Used to synthesize electrocardiogram signals; B. Report Generation: Automatically generates diagnostic reports based on input signals; C. Signal QA: Conducts automated question-answering based on signal content; D. Caption Generation: Extracts key information from signals to generate concise captions.

#### 6.3.3. *Physiological signals simulated from text*

signal simulation is a task that generates synthetic physiological signals from textual clinical information, connecting descriptive medical records with quantitative physiological data, and facilitating applications in data augmentation, clinical simulation, and privacy-preserving analysis. As shown in Fig. 4(A), the process is outlined. Chung et al. [158] proposed Auto-TTE, a novel text-to-ECG generation method that produces 12-lead signals from clinical reports through an autoregressive model. Their results demonstrated superior performance over representative baselines in both quantitative metrics and qualitative assessments, with validation from multiple cardiologists confirming the fidelity and semantic consistency of the outputs. Delaney et al. [159] introduced a GAN-based strategy to generate synthetic ECG data, primarily addressing the challenge of safeguarding patient privacy. To mitigate typical GAN limitations such as training instability and the absence of robust evaluation criteria, the authors applied maximum mean discrepancy and dynamic time warping as quality assessment measures. Chen et al. [160] explored diffusion models for text-to-ECG generation, incorporating clinical reports together with patient-specific multimodal information. This design yielded signals with strong semantic alignment and high fidelity, ensuring accurate reflection of the conditions described in the source text.

#### 6.3.4. *Physiological signals QA*

Interpreting physiological signal data and patient information requires a lot of medical expertise and resources. The model can automatically extract key physiological features, realize intelligent question-answering and auxiliary diagnosis of physiological signals, and greatly improve medical efficiency and accessibility. The process is depicted in Fig. 4(C). ECG-LM [185] is a multimodal LLM capable of processing both natural language and electrocardiogram (ECG) signals. It effectively captures the complex physiological patterns within ECG data and demonstrates strong potential for diverse medical applications such as disease prediction and advanced question answering. Tang et al. [186] proposed ECG-ReGen, a retrieval-based method for ECG-to-text report generation and question answering. By leveraging self-supervised learning in the ECG encoder, the method enables efficient similarity search and report retrieval, allowing for accurate analysis of ECG data and automated responses to related queries. In comparison, ECG-LM is more suitable for tasks emphasizing prediction and complex reasoning, whereas ECG-ReGen offers advantages in transparency and traceability.

### 6.4. *Other multimodal collaborative generation*

Beyond conventional text-vision and text-signal generation paradigms, recent research has begun to explore other forms of multimodal collaborative generation. Liu et al.[187] explored the decoding of dynamic visual perception from EEG signals, a neuroimaging technique that records brain activity with high temporal resolution (1000 Hz). This indicates that integrating neurophysiological signals with other modalities has the potential to drive multimodal generation.

### 6.5. *Comparison of methods in generative applications*

This section analyzes the suitability of different generation methods across various tasks to highlight their performance differences and application advantages. For text-to-vision generation tasks, prompt-based methods or few-shot prompting are preferred when the focus is on rapid adaptation and cross-domain generalization. When the task requires structured integration of domain knowledge or external evidence, retrieval-augmented fine-tuning is more appropriate. If the goal is to maximize performance within a specific domain, direct fine-tuning combined with cross-domain strategies and knowledge distillation is recommended [199]. When large amounts of unlabeled data are available, self-supervised pretraining can serve as a robust foundation to enhance downstream stability and transferability. In vision-to-text generation, diffusion models are more suitable when high fidelity and controllability are required. When the task emphasizes anatomical consistency or precise localization, incorporating structural priors can improve performance. If the goal is to produce diverse, multi-perspective outputs, multi-view generation paradigms are a better fit.

For signal-to-text applications, when data is limited and there is a strong emphasis on clinical language quality and task generalization, base models or large language models offer better cost-effectiveness. When modeling fine-grained physiological phenomena with a need for interpretability, traditional temporal modeling architectures remain a more reliable choice. In text-to-signal generation, the most common approaches are GANs and diffusion models. GANs are advantageous in generation speed and output diversity, while diffusion models provide better stability and detail fidelity.

## 7. Existing challenges

This section describes the typical challenges and limitations of applying Multimodal Learning in the EHR field, focusing on three aspects: data (Data Challenges, Label Scarcity), processing techniques (Multimodal Fusion and Alignment, Reliability of Models), and applications (Privacy and Ethics, Real-time and Computational Cost).

### 7.1. *Data challenges*

Data challenges in EHRs mainly stem from unavoidable issues in the data collection process. These challenges can be grouped into three categories: systematic missingness, heterogeneity, and inconsistency.





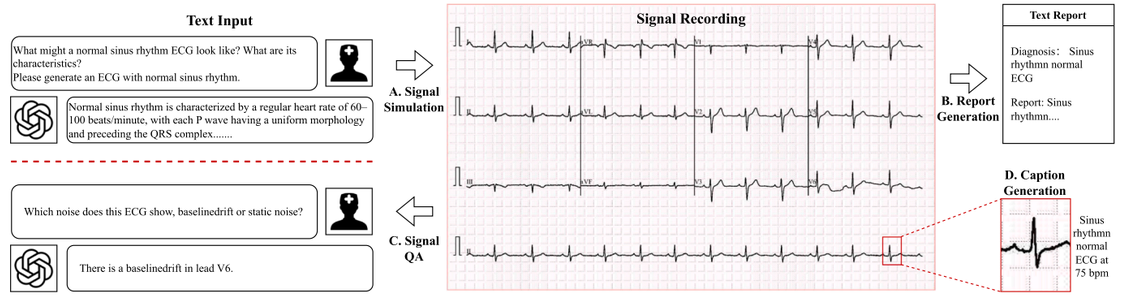



图4. 生理信号（心电图）生成任务的应用示意图，展示以下功能：A. 信号模拟：用于合成心电图信号；B. 报告生成：根据输入信号自动生成诊断报告；C. 信号问答：基于信号内容进行自动问答；D. 描述生成：从信号中提取关键信息以生成简洁描述。

#### 6.3.3. 从文本模拟生理信号

信号模拟是一项从文本临床信息生成合成生理信号的任务，将描述性医疗记录与定量生理数据相连接，并促进其在数据增强、临床模拟和隐私保护分析中的应用。如图4(A)所示，该过程已被概述。Chung等人[158] 提出了Auto-TTE，一种新颖的文本到心电图生成方法，通过自回归模型从临床报告生成12导联信号。他们的结果表明，在定量指标和定性评估方面均优于代表性基线，且多位心脏病专家的验证确认了输出的保真度和语义一致性。Delaney等人[159] 引入了一种基于GAN的策略来生成合成心电图数据，主要解决保护患者隐私的挑战。为缓解GAN的典型局限性，如训练不稳定和缺乏稳健评估标准，作者采用最大均值差异和动态时间规整作为质量评估指标。Chen等人[160] 探索了扩散模型用于文本到心电图生成，将临床报告与患者特定的多模态信息相结合。该设计产生了具有强语义对齐和高保真度的信号，确保准确反映源文本中所描述的状况。

#### 6.3.4. 生理信号问答

解读生理信号数据和患者信息需要大量医学专业知识和资源。该模型能够自动提取关键生理特征，实现生理信号的智能问答和辅助诊断，大幅提升医疗效率和可及性。该过程如图4(C)所示。ECG-LM [185] 是一种多模态大语言模型，能够同时处理自然语言和心电图（ECG）信号。它有效捕捉心电图数据中复杂的生理模式，并在疾病预测和高级问答等多种医疗应用中展现出强大潜力。Tang等人[186] 提出了ECG-ReGen，一种基于检索的方法，用于心电图到文本报告生成和问答。通过在心电图编码器中利用自监督学习，该方法实现了高效的相似性搜索和报告检索，从而能够准确分析心电图数据并自动响应相关查询。相比之下，ECG-LM更适合强调预测和复杂推理的任务，而ECG-ReGen在透明度和可追溯性方面具有优势。

### 6.4. 其他 多模态 协同 生成

超越 传统的 文本-视觉 和 文本-信号 生成 范式， 近期 研究 已 开始 探索 其他 形式的 多模态 协同 生成。 Liu 等 人 [187] 探索了 从 解码 脑电图 信号中进行 动态 视觉 感知 ， 这是一种 以 高 时间 分辨率（1000 Hz） 记录 大脑 活动 的 神经影像 技术 。 这 表明 将 神经 生理 信号 与 其他 模态 相融合 具有 推动 多模态 生成的 潜力 。

### 6.5. 生成 应用中 各 方法 的 比较

本节分析不同生成方法在各种任务中的适用性，以突出其性能差异与应用优势。对于文本到视觉生成任务，当关注点在于快速自适应和跨域泛化时，基于提示的方法或少样本提示更受青睐。当任务需要对领域知识或外部证据进行结构化整合时，检索增强微调更为合适。如果目标是在特定领域内最大化性能，建议采用直接微调结合跨域策略和知识蒸馏[199]。当有大量无标注数据可用时，自监督预训练可以作为增强下游稳定性和可迁移性的坚实基础。在视觉到文本生成中，当需要高保真度和可控性时，扩散模型更为合适。当任务强调解剖一致性或精确定位时，引入结构先验可以提升性能。如果目标是产生多样化的多视角输出，多视角生成范式更为适合。

对于信号到文本应用，当数据有限且高度重视临床语言质量和任务泛化时，基础模型或大语言模型具有更好的成本效益。当需要对细粒度生理现象进行建模且要求可解释性时，传统时序建模架构仍然是更可靠的选择。在文本到信号生成中，最常见的方法是GAN和扩散模型。GAN在生成速度和输出多样性方面具有优势，而扩散模型则提供更好的稳定性和细节保真度。

## 7. 现有 挑战

本节描述了在EHR领域应用多模态学习的典型挑战和局限性，重点关注三个方面：数据（数据挑战、标签稀缺）、处理技术（多模态融合与对齐、模型可靠性）以及应用（隐私与伦理、实时性与计算成本）。

### 7.1. 数据 挑战

数据 挑战 在 电子健康记录 主要 源于 不可避免的 问题 在 数据 收集 过程。 这些 挑战 可以 被 归为 三 类： 系统性 缺失、 异质性，以及 不一致性。

Systematic missingness is prevalent in longitudinal patient records, often resulting from incomplete diagnostic tests or missed follow-up appointments. This type of missing data not only impairs the accurate modeling of individual modalities by introducing gaps in temporal sequences, but also complicates the integration across multiple data types. Incomplete information may propagate biases, reduce predictive accuracy, and create instability during model training, manifesting as overfitting to available data or failure to capture underlying clinical patterns.

Heterogeneity emerges from the inherently diverse nature of EHR data. Structured components include standardized diagnostic codes, laboratory values, and vital signs, while unstructured sources cover free-text clinical notes, radiological images, and genomic profiles. The fusion of these modalities requires sophisticated alignment techniques capable of harmonizing diverse representations. Structured numerical features may be embedded into vector spaces to interact with textual or visual embeddings, while preserving modality-specific properties such as the temporal dynamics in physiological signals or the semantic richness of clinical narratives.

Inconsistency arises across healthcare providers and institutions. Variations in clinical standards, divergent coding systems (e.g., ICD-9 vs. ICD-10), and differences in documentation practices frequently produce fragmented or incompatible datasets. Such inconsistencies hinder the creation of unified large-scale datasets that are indispensable for robust model training. They also undermine generalizability, as multimodal models often exhibit performance degradation when applied to data from institutions with unfamiliar protocols.

In addition, the reliability of research in this field depends heavily on accurate and complete written summaries for model training and validation. Yet, clinical narratives are frequently compromised by ambiguities, inconsistencies, or even factual errors, stemming from clinician fatigue, variable documentation practices, or transcription mistakes. These deficiencies exacerbate challenges in ensuring data quality and undermine model reliability. Addressing these multifaceted issues is therefore critical for advancing multimodal learning in clinical applications.

### 7.2. Label scarcity

The scarcity of high-quality labeled data represents a fundamental challenge for the practical deployment of artificial intelligence in the medical domain. Annotating medical data requires substantial domain expertise and typically relies on well-trained professionals, such as clinicians, radiologists, or pathologists. This dependence on specialized human resources leads to high costs and low efficiency, particularly in the case of rare diseases or complex clinical scenarios, where cross-validation by multiple experts is often necessary to ensure labeling accuracy. Furthermore, discrepancies among annotators in interpreting clinical content, identifying pathological features, or analyzing medical images introduce subjectivity into the annotation process, which compromises the consistency and reproducibility of labeled datasets. The heterogeneity of EHRs further exacerbates this challenge, as EHRs commonly include structured entries, unstructured clinical narratives, and cross-modal medical images. The lack of standardized annotation protocols across these modalities makes it difficult to generate coherent labels of sufficient quality. Ambiguity in clinical terminology, the demand for fine-grained lesion identification in imaging, and semantic misalignment across modalities further increase the complexity and labor intensity of annotation.

From a modeling perspective, insufficient labeled data can significantly hinder the development of robust and generalizable multimodal learning systems. Deep learning models, in particular, require large volumes of accurately annotated data to effectively learn complex patterns across modalities. When labeled data are scarce, models are prone to overfitting, show degraded performance on unseen data, and risk producing biased predictions that fail to generalize across patient populations or clinical settings. Limited annotation also constrains the integration of multimodal information, as noisy or incomplete supervision disrupts the alignment and fusion of heterogeneous data. This bottleneck directly undermines scalability, reliability, and ultimately the clinical applicability of AI systems trained on EHRs.

### 7.3. Multimodal fusion and alignment

From a data processing perspective, the core challenge lies in the fusion and alignment of multi-source, heterogeneous EHR data. Different modalities create semantic and scale gaps. Structured data are clean and regular but have limited expressive power. Unstructured text and medical images carry rich information, yet contain redundancy, implicit meaning, and inconsistent formats. Differences in device models, report templates, and departmental practice further amplify these gaps. Effective fusion requires a shared representation that is comparable and composable, while preserving modality-specific signals. Examples include spatial lesion morphology in images, causal cues and negations in text, and subtle inflection points in structured trends. Excessive compression causes modality collapse, whereas loose alignment impedes cross-modal inference.

The temporal dimension adds further difficulty. Data arrive at irregular intervals: imaging is sporadic, labs are periodic, and medications and clinical events are stochastic. Missing entries and delayed charting are common and can misalign observations for the same clinical state over time. Ignoring timestamps, inter-event intervals, and missingness mechanisms can induce spurious disease trajectories, especially in chronic follow-up and in settings with alternating acute and chronic states.

Additionally, scale and computation impose practical constraints. Integrating high-frequency time series, lengthy clinical notes, and high-resolution images expands the feature space dramatically. This expansion not only raises the risk of the curse of dimensionality but also increases sample complexity, memory consumption, and computational demand. For instance, an EEG sampled at 500 Hz with 128 channels generates 64,000 data points per second. When combined with high bit-depth storage, time-frequency transformations, and multi-scale modeling, the computational and I/O burden grows rapidly, complicating real-world deployment. While downsampling and feature extraction are commonly used to mitigate this issue, they often entail the loss of clinically relevant information.

Finally, noise and inconsistency pervade data collected across modalities and institutions. Typical problems include transcription and typing errors, imaging artifacts and variable quality, heterogeneous coding schemes and vocabularies, as well as demographic and device-related shifts. These inconsistencies substantially complicate the task of achieving reliable data alignment.

### 7.4. Reliability of models

Ensuring the reliability and interpretability of multimodal models is crucial yet challenging, primarily due to the complexity and opacity of deep learning architectures. A core challenge to their reliability stems from their inherent "black box" nature.

In stark contrast to traditional machine learning models like decision trees or logistic regression, where decision paths are transparent and based on traceable feature weights, deep learning models operate differently. They are composed of millions or even billions of interconnected parameters that work together through complex, non-linear computations. The final output is an emergent result of these massive parameter interactions, making it nearly impossible to precisely trace and explain why a model arrived at a specific conclusion. For clinicians, this lack of interpretability poses a fundamental barrier to building trust, as it contradicts the evidence-based standards of medicine. Medical practice requires every diagnostic or treatment decision to be founded on verifi-





系统性缺失在纵向患者记录中普遍存在，通常由不完整的诊断测试或错过的随访预约引起。这类缺失数据不仅通过在时间序列中引入缺口而损害单个模态的准确建模，还使多种数据类型之间的整合变得复杂。不完整的信息可能传播偏见、降低预测准确率，并在模型训练期间造成不稳定性，表现为对可用数据的过拟合或无法捕捉潜在的临床模式。

异质性源于电子健康记录数据本身固有的多样性。结构化组成部分包括标准化的诊断代码、实验室检验值和生命体征，而非结构化来源则涵盖自由文本临床笔记、放射影像和基因组谱。这些模态的融合需要能够协调多样表示的复杂对齐技术。结构化数值特征可以被嵌入向量空间，以与文本或视觉嵌入进行交互，同时保留模态特定属性，例如生理信号中的时间动态或临床叙述的语义丰富性。

不一致性在医疗服务提供者和机构之间普遍存在。临床标准的差异、编码系统的分歧（例如，ICD-9与ICD-10），以及文档实践的不同，经常产生碎片化或不兼容的数据集。此类不一致性阻碍了统一大规模数据集的创建，而这些数据集对于稳健的模型训练不可或缺。它们还削弱了泛化能力，因为多模态模型在应用于来自陌生协议机构的数据时，往往表现出性能退化。

此外，该领域研究的可靠性在很大程度上取决于用于模型训练和验证的准确且完整的书面摘要。然而，临床叙述经常因歧义、不一致甚至事实错误而受到损害，这些问题的根源在于临床医生的疲劳、多变的文档实践或转录错误。这些缺陷加剧了确保数据质量的挑战，并削弱了模型的可靠性。因此，解决这些多方面的问题对于推进临床应用中的多模态学习至关重要。

### 7.2. 标签 稀缺

高质量标注数据的稀缺是人工智能在医学领域实际部署中面临的一项根本性挑战。医学数据标注需要大量的领域专业知识，且通常依赖于训练有素的专业人员，如临床医生、放射科医生或病理学家。这种对专业人力资源的依赖导致成本高昂且效率低下，尤其是在罕见病或复杂临床场景中，通常需要多位专家进行交叉验证以确保标注准确率。此外，标注者之间在解读临床内容、识别病理特征或分析医学图像方面存在差异，给标注过程引入了主观性，从而影响了标注数据集的一致性和可重复性。电子健康记录的异质性进一步加剧了这一挑战，因为电子健康记录通常包含结构化条目、非结构化临床叙述以及跨模态医学图像。这些模态之间缺乏标准化的标注方案，使得生成足够质量的连贯标签变得困难。临床术语的歧义性、影像中细粒度病变识别的需求，以及跨模态间的语义错位，进一步增加了标注的复杂性和劳动强度。

从建模的角度来看，标注数据不足会严重阻碍稳健且具有泛化能力的多模态学习系统的开发。尤其是深度学习模型，需要大量准确标注的数据才能有效学习跨模态的复杂模式。当标注数据稀缺时，模型容易过拟合，在未见数据上性能下降，并可能产生有偏置的预测，无法泛化到不同的患者群体或临床环境。有限的标注还制约了

### 7.3. 多模态 融合 与 对齐

从数据处理的角度来看，核心挑战在于多源、异构电子健康记录数据的融合与对齐。不同模态之间会产生语义和尺度差距。结构化数据干净且规整，但表达能力有限。非结构化文本和医学图像携带丰富信息，却包含冗余、隐含意义以及格式不一致的问题。设备型号、报告模板和科室实践方面的差异进一步放大了这些差距。有效的融合需要一种可比较且可组合的共享表示，同时保留模态特定信号。示例包括图像中的空间病变形态、文本中的因果线索和否定，以及结构化趋势中的细微拐点。过度压缩会导致模态坍塌，而松散的对齐则会阻碍跨模态推理。

时间维度进一步增加了难度。数据以不规则的时间间隔到达：影像是零星的，实验室检查是周期性的，而药物和临床事件则是随机发生的。缺失条目和记录延迟很常见，并且可能使同一临床状态在不同时间点的观测发生错位。忽略时间戳、事件间间隔和缺失机制，可能会诱导出虚假疾病轨迹，尤其是在慢性随访以及急性和慢性状态交替出现的情形中。

此外，规模与计算带来了实际约束。整合高频时间序列、冗长的临床笔记和高分辨率图像会急剧扩展特征空间。这种扩展不仅增加了维度灾难的风险，还提高了样本复杂度、内存消耗和计算需求。例如，以500 Hz采样的128通道脑电图每秒生成64,000个数据点。当与高位深存储、时频变换和多尺度建模相结合时，计算和I/O负担迅速增长，使实际部署变得复杂。虽然下采样和特征提取常被用来缓解这一问题，但它们往往会导致临床相关信息的丢失。

最后，噪声和不一致性普遍存在于跨模态和跨机构收集的数据中。典型问题包括转录和输入错误、成像伪影和质量变异、异构编码方案和词汇表，以及人口统计和设备相关变化。这些不一致性极大地增加了实现可靠数据对齐的难度。

### 7.4. 模型的 可靠性

确保 多模态 模型 的 可靠性 和 可解释性 至关重要 但 颇具挑战，这主要 是 由于 深度学习 架构的 复杂性 和 不透明性 。 其 可靠性 面临的一个 核心 挑战 源于 其 固有的 “黑 箱” Nature 。

与决策树或逻辑回归等传统机器学习模型形成鲜明对比的是，后者的决策路径是透明的，并基于可追溯的特征权重，而深度学习模型的运作方式则截然不同。它们由数百万甚至数十亿个相互连接的参数组成，这些参数通过复杂的非线性计算协同工作。最终输出是这些海量参数交互的涌现结果，因此几乎不可能精确追溯和解释模型为何得出某个特定结论。对于临床医生而言，这种可解释性的缺失构成了建立信任的根本障碍，因为它与医学的循证标准相矛盾。医疗实践要求每一项诊断或治疗决策都必须建立在可验证的

able evidence, clear pathophysiological logic, and peer-reviewed clinical guidelines.

Furthermore, this challenge is exacerbated by the inherent complexity and noise of EHR data. Real-world clinical data is often messy, filled with inconsistencies, missing values, and individual variations, unlike the standardized datasets used for model training. When this "dirty" data is fed into an already opaque model, the model may struggle to handle unexpected variations or learn spurious correlations from the noise. This can lead to serious errors when dealing with rare or atypical cases.

### 7.5. Privacy and ethics

The handling of sensitive medical data in multimodal learning raises critical ethical and privacy concerns. Training effective models requires large, diverse datasets from multiple institutions, but these datasets contain highly sensitive patient information, subject to strict legal regulations like HIPAA and GDPR. Balancing the need for data sharing with privacy protection is a complex challenge. While techniques such as federated learning, differential privacy, and secure multi-party computation offer potential solutions, their practical implementation in healthcare remains limited and requires further development and validation. Addressing these privacy and ethical issues is crucial for the responsible deployment of multimodal models in clinical practice.

### 7.6. Real-time and computational cost

In clinical applications, the real-time performance and computational cost of models are critically important, especially in scenarios involving emergency medical decisions, the need for immediate results, or limited computational resources. However, multimodal medical data are often high-dimensional, making their analysis computationally intensive. As a result, both training and inference can be costly, which poses significant challenges for their effective implementation in real-world clinical practice. Furthermore, the adoption of large-scale models such as LLMs and LVMs imposes even higher demands on computational resources and system configurations.

## 8. Research directions

In this section, we explore several promising research directions driven by the challenges outlined in Section 7, with the aim of analyzing how these directions may help address the current challenges. The six existing challenges and twelve future research directions of multimodal learning driven by EHR, along with their interrelationships, as well as the corresponding potential research questions and possible avenues for improvement, are shown in Fig. 5.

### 8.1. Data augmentation

To address incomplete or limited data, various augmentation methods can be used. For single-modality data, techniques like probabilistic or model-based imputation can estimate missing values, improving training outcomes. In multimodal settings, complementary modalities can fill gaps when one is missing, enhancing data completeness and prediction accuracy. Additionally, generating synthetic data using models like GANs or diffusion models can increase dataset variety, especially for rare diseases or underrepresented groups. This direction can alleviate the systematic missing and heterogeneous in **Data Challenges.**

### 8.2. Cross-modal learning and shared representations

To handle diverse EHR data, cross-modal learning and shared representations can align structured and unstructured data. Methods like joint embeddings, co-attention, or multimodal transformers bridge differences between modalities, improving model stability. Future work could focus on adaptable representation spaces that adjust modality weights based on clinical needs. By harmonizing diverse data types, this approach tackles heterogeneity in **Data Challenges** and enhances modality integration in **Multimodal Fusion and Alignment**.

### 8.3. Domain transfer and adaptation

EHR data collected from different institutions often vary in structure, distribution, and semantics. These differences, caused by inconsistencies in clinical practice, documentation, and data collection, make modality alignment and fusion more difficult. To address these challenges, transfer learning and domain adaptation are widely used to improve model robustness and generalizability across diverse clinical settings.

Transfer learning focuses on reusing knowledge learned from a source domain to support tasks in a target domain with limited labeled data. In EHR applications, models are typically pre-trained on large, multi-institution datasets that include various modalities such as structured data and clinical text. The learned parameters are then fine-tuned on target domain data using techniques like selective layer tuning or cross-modal guidance. This process helps preserve general semantic patterns while adapting to new environments. It also offers stable initialization for downstream tasks, which is particularly helpful in low-resource or institution-specific contexts. Domain adaptation, on the other hand, deals directly with differences between source and target data distributions. This is important in multimodal EHR settings where modality-specific variations can disrupt joint learning. Adaptation methods aim to reduce such distribution gaps by learning domain-invariant representations. Adversarial training introduces a domain classifier to encourage shared feature extraction, while statistical matching techniques help align feature distributions. These methods are often integrated with modality alignment objectives to support consistent fusion across domains.

Together, transfer learning and domain adaptation offer complementary approaches. Transfer learning improves efficiency and reuse of learned knowledge, while domain adaptation ensures alignment across varying data sources. Such strategies resolve inconsistency in **Data Challenges** and support effective data alignment in **Multimodal Fusion and Alignment**.

### 8.4. Multi-level feature alignment strategies

Multi-level feature alignment captures both broad context and fine details, boosting performance in precise tasks like detecting small lesions. Current rigid alignment methods may limit flexibility. Future research should explore adaptable, learnable alignment structures that automatically optimize detail levels, improving accuracy in complex imaging tasks. This direction addresses the heterogeneity in data challenges by aligning different data features and improves feature alignment in **Multimodal Fusion and Alignment** to achieve precise task execution.

### 8.5. Fusion of pre-training and knowledge graphs

Medical knowledge graphs, which map clinical entities and relationships, can enhance multimodal learning. Integrating these graphs into VLM pre-training uses medical knowledge to align modalities. This approach helps models interpret complex data by linking images and text to medical frameworks, tackling data diversity and inconsistency. By leveraging structured medical knowledge, this method alleviates heterogeneity and inconsistency in **Data Challenges** while strengthening modality alignment in **Multimodal Fusion and Alignment**.

### 8.6. Interpretability through attention mechanisms

Attention mechanisms improve model interpretability by assigning different levels of importance to input features during the prediction





证据 、 清晰的 病理生理 逻辑 以及 经过同行评审的 临床指南之上。

此外，电子健康记录数据固有的复杂性和噪声加剧了这一挑战。现实世界的临床数据往往杂乱无章，充满了不一致、缺失值和个体差异，与用于模型训练的标准化数据集截然不同。当这些“脏”数据被输入到一个本就不透明的模型中时，模型可能难以处理意外的变化，或从噪声中学习到虚假相关。这可能在处理罕见或非典型病例时导致严重错误。

### 7.5. 隐私 与 伦理

在多模态学习中处理敏感医疗数据引发了关键的伦理与隐私问题。训练有效的模型需要来自多机构的大型多样化数据集，但这些数据集包含高度敏感的患者信息，受到HIPAA和GDPR等严格法律法规的约束。在数据共享需求与隐私保护之间取得平衡是一项复杂的挑战。虽然联邦学习、差分隐私和安全多方计算等技术提供了潜在的解决方案，但它们在医疗健康领域的实际部署仍然有限，需要进一步的开发和验证。解决这些隐私和伦理问题对于在临床实践中负责任地部署多模态模型至关重要。

### 7.6. 实时性 与 计算 成本

在临床应用中，模型的实时性能和计算成本至关重要，尤其是在涉及急诊医疗决策、需要即时结果或计算资源有限的情况下。然而，多模态医疗数据往往是高维的，这使得其分析在计算上非常密集。因此，训练和推理都可能代价高昂，这为其在真实世界临床实践中的有效实施带来了重大挑战。此外，采用大语言模型和大型视觉模型等大规模模型对计算资源和系统配置提出了更高的要求。

## 8. 研究 方向

在本节中，我们探讨由第7节中所概述的挑战驱动的若干有前景的研究方向，旨在分析这些方向如何可能帮助解决当前的挑战。电子健康记录驱动的多模态学习的六项现有挑战和十二个未来研究方向，及其相互关系，以及相应的潜在研究问题和可能的改进途径，如图5所示。

### 8.1. 数据 增强

**为了解决不完整或有限数据的问题，可以采用多种增强方法。对于单模态数据，概率插补或基于模型的插补等技术可以估计缺失值，从而改善训练效果。在多模态情境中，当某一模态缺失时，互补模态可以填补空白，从而提高数据完整性和预测准确率。此外，使用GAN或扩散模型等模型生成合成数据可以增加数据集的多样性，尤其针对罕见病或代表性不足群体。这一方向可以缓解数据挑战中系统性的缺失和异质性问题。**

### 8.2. 跨模态 学习 与 共享 表示

为了 处理 多样化的 EHR 数据， 跨模态 学习 与 共享 表示 能够 对齐 结构化 与 非结构化 数据。 联合 嵌入、 共注意力 或 多模态 Transformer 等方法 弥合了 模态间的 差 异， 提升了 模型 稳定性。 未来 工作 可以 聚焦于 可适配的 表示 空间 ， 根据 临床 需求 调整 模态 权重。 通过 协调 多样化的 数据 类型， 该方法 解决了 数据 挑战 中的 异质性， 并增强了 模态 集成 在 多模态 融合 与 对齐中的效果。

### 8.3. 领域 迁移 与 自适应

从不同机构收集的电子健康记录数据在结构、分布和语义上往往存在差异。这些差异由临床实践、文档记录和数据收集中的不一致性引起，使得模态对齐和融合更加困难。为应对这些挑战，迁移学习和领域适应被广泛用于提高模型在不同临床环境中的鲁棒性和泛化能力。

迁移学习侧重于复用从源领域学到的知识，以支持标注数据有限的目标领域中的任务。在电子健康记录应用中，模型通常在包含结构化数据和临床文本等多种模态的大型多机构数据集上进行预训练。随后，利用选择性层调整或跨模态引导等技术，将学到的参数在目标领域数据上进行微调。这一过程有助于在适应新环境的同时保留通用语义模式。它还为下游任务提供了稳定的初始化，这在低资源或机构特定情境中尤为有益。另一方面，领域适应直接处理源数据和目标数据分布之间的差异。这在多模态电子健康记录环境中非常重要，因为模态特定变异可能会干扰联合学习。自适应方法旨在通过学习领域不变表示来缩小此类分布差距。对抗训练引入领域分类器以促进共享特征提取，而统计匹配技术则有助于对齐特征分布。这些方法通常与模态对齐目标相结合，以支持跨领域的一致融合。

迁移学习和领域适应共同提供了互补的方法。迁移学习提高了效率和已学知识的复用，而领域适应确保了不同数据源之间的对齐。这些策略解决了数据挑战中的不一致性，并支持多模态融合与对齐中有效的数据对齐。

### 8.4. 多层级 特征 对齐 策略

多层级特征对齐能够同时捕捉广泛的上下文信息和精细细节，从而提升检测微小病变等精确任务的性能。当前的刚性对齐方法可能限制灵活性。未来研究应探索可适应、可学习的对齐结构，自动优化细节层级，提高复杂影像任务的准确率。这一方向通过对齐不同的数据特征来解决数据挑战中的异质性，并改善多模态融合与对齐中的特征对齐，以实现精确的任务执行。

### 8.5. 预训练 与 知识图谱 的 融合

医学知识图谱通过映射临床实体及其关系，可以增强多模态学习。将这些图谱集成到视觉语言模型预训练中，利用医学知识来对齐模态。这种方法通过将图像和文本链接到医学框架，帮助模型理解复杂数据，应对数据多样性和不一致性。通过利用结构化医学知识，该方法缓解了数据挑战中的异质性和不一致性，同时增强了多模态融合与对齐中的模态对齐。

### 8.6. 通过 注意力 机制实现的可解释性

注意力 机制 通过 在 预测 过程 中 为 输入 特征 分配 不同 的 重要 程度 来 提升模型的

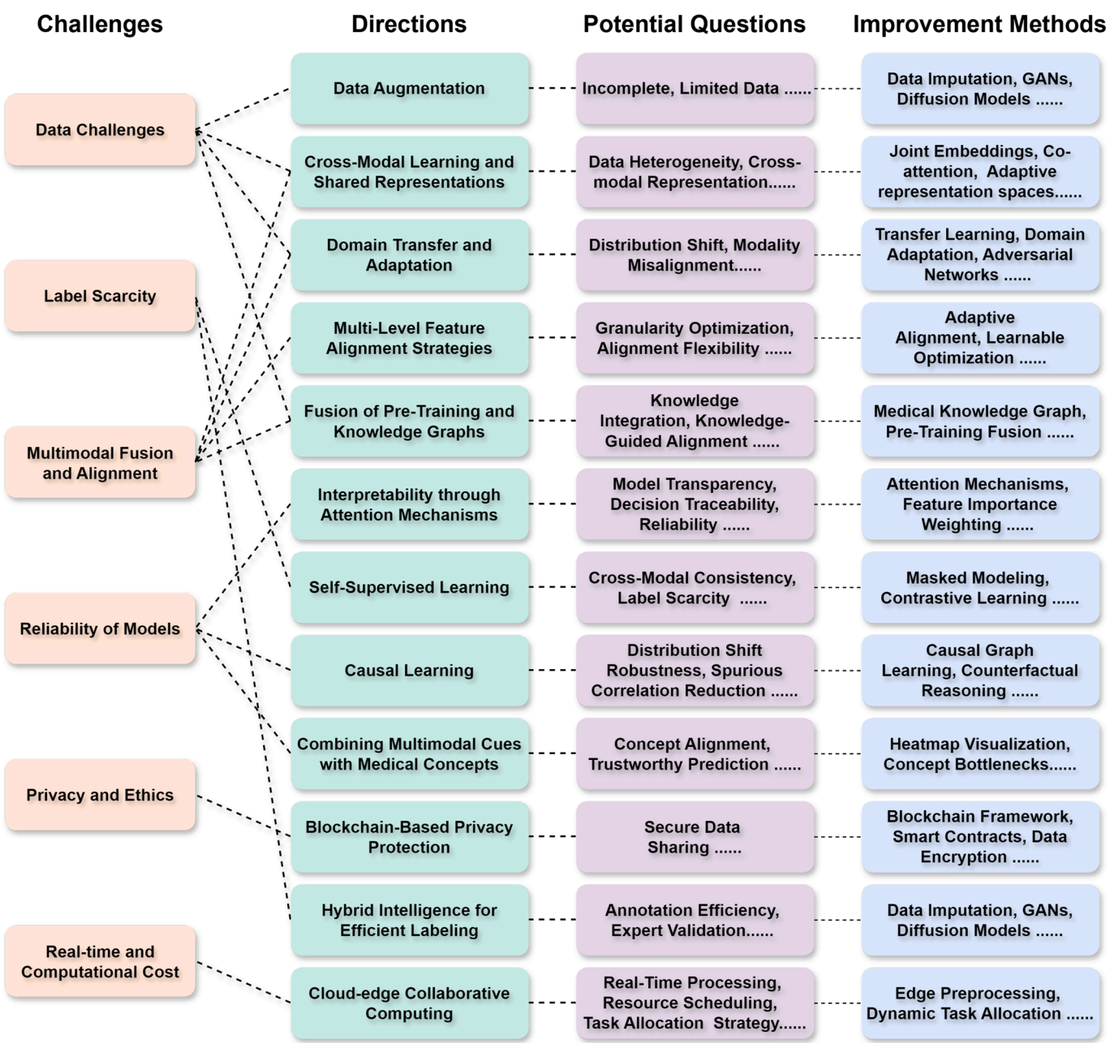



**Fig. 5.** The relationship between the summarized existing challenges and future research directions.

process. This allows the model to highlight which parts of the input contribute most to its decisions. In medical tasks, where reliability and transparency are essential, this property helps uncover how the model processes complex inputs such as images or patient records. From an architectural perspective, attention introduces a structured way of weighting features, enabling the model to focus selectively on relevant information while suppressing less informative signals.

By learning these importance weights during training, the model creates a mapping between input features and prediction outcomes that can be examined and understood. This supports traceability in the decision-making process, as attention scores reflect the internal reasoning path of the model. In high-stakes clinical applications such as diagnosis or risk assessment, such transparency makes it easier to align model behavior with expert knowledge and clinical guidelines. As a result, attention mechanisms not only enhance the performance of predictive models but also make their outputs more understandable and trustworthy to human users. By making model decisions more transparent, this approach enhances transparency in **Reliability of Models**.

### *8.7. Self-supervised learning*

Self-supervised learning reduces reliance on manual annotation and enhances model generalization by constructing pretext tasks that do not require labeled data. From an algorithmic standpoint, models learn latent structural information from large-scale unlabeled multimodal EHR data by solving carefully designed surrogate objectives. Representative approaches include masked modeling and contrastive learning, both of which exploit intra- and inter-modal redundancies to encourage the learning of semantically meaningful representations. Masked modeling randomly masks portions of the input and requires the model to reconstruct them based on contextual information, enabling the capture of modality-specific distributional structures without external supervision.





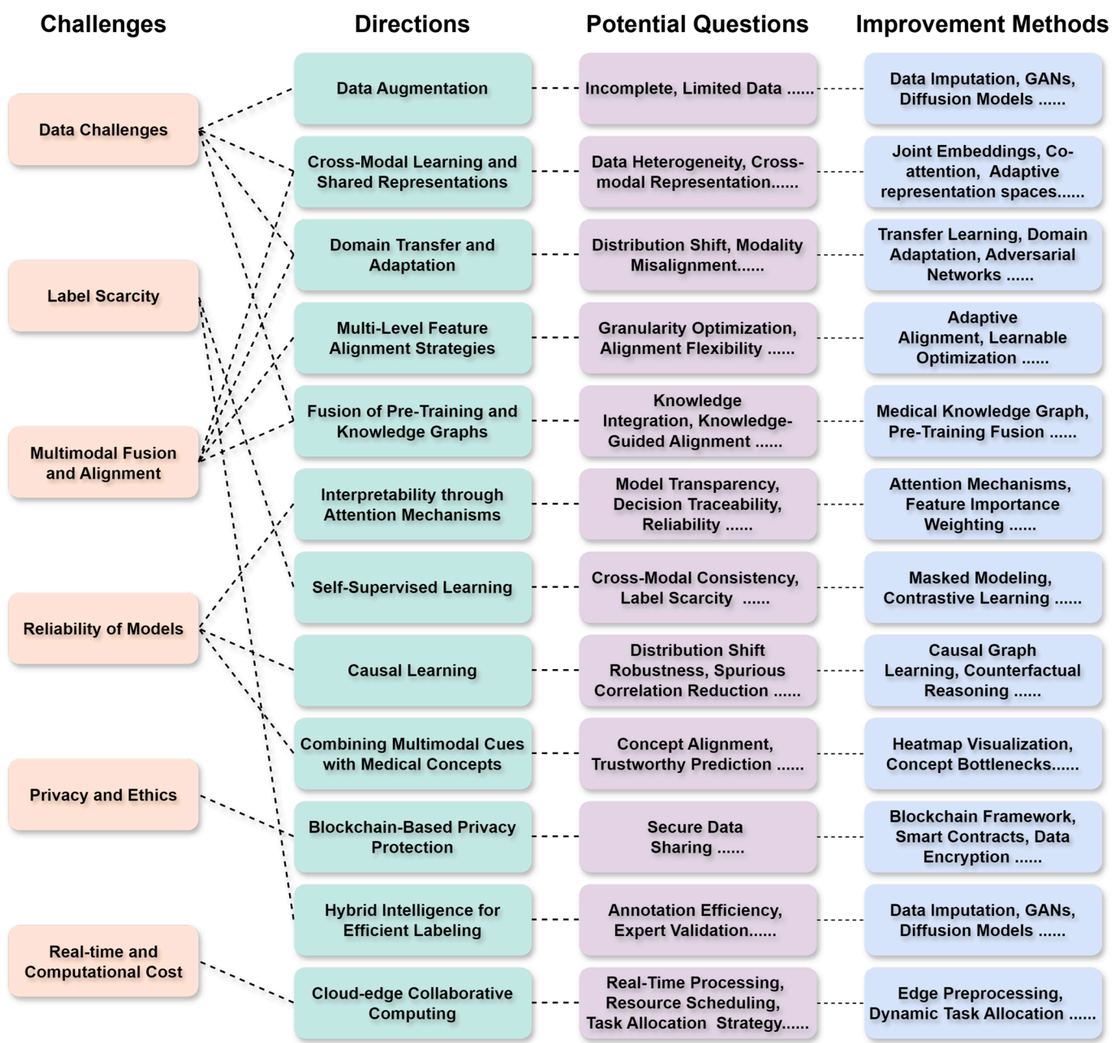



**图 5.** 已总结的现有挑战与未来研究方向之间的关系。

可解释性。这使得模型能够突出显示输入的哪些部分对其决策贡献最大。在可靠性和透明度至关重要的医学任务中，这一特性有助于揭示模型如何处理图像或患者记录等复杂输入。从架构的角度来看，注意力引入了一种结构化的特征加权方式，使模型能够选择性地关注相关信息，同时抑制信息量较少的信号。

通过在训练过程中学习这些重要性权重，模型在输入特征与预测结果之间建立起一种可被审视和理解的映射关系。这支持了决策过程中的可追溯性，因为注意力分数反映了模型的内部推理路径。在诊断或风险评估等高风险临床应用中，这种透明度使得模型行为更容易与专家知识和临床指南保持一致。因此，注意力机制不仅提升了预测模型的性能，还使其输出对人类用户而言更加可理解和可信。

用户。通过使模型决策更加透明，这一方法增强了**模型可靠性的透明度**。

### *8.7.* 自监督学习

自监督学习通过构建不需要标注数据的前置任务，减少了对人工标注的依赖并提升了模型泛化能力。从算法角度来看，模型通过求解精心设计的代理目标，从大规模无标注多模态EHR数据中学习潜在的结构信息。代表性的方法包括掩码建模和对比学习，二者均利用模态内和模态间冗余来促进语义有意义的表征的学习。掩码建模随机掩盖输入的部分内容，并要求模型基于上下文信息对其进行重构，从而能够在无外部监督的情况下捕获模态特定的分布结构。

Contrastive learning, in contrast, optimizes the representation space by bringing semantically similar samples closer while pushing dissimilar ones apart, thereby promoting consistency across modalities. These techniques allow the model to acquire transferable representations during pretraining, which can be effectively adapted to downstream predictive tasks with only limited labeled data. Compared to traditional supervised paradigms, self-supervised learning better addresses the challenges of data heterogeneity and annotation scarcity common in healthcare, ultimately improving the robustness and applicability of multimodal systems in real-world clinical settings.This direction can overcome the problem of limited dataset availability in the **Label Scarcity** by effectively utilizing unlabeled data.

### *8.8. Causal learning*

In multimodal learning based on EHRs, causal learning improves model interpretability and generalization by explicitly modeling the underlying causal mechanisms. The core principle is to move beyond simple correlations and identify variables that have a direct causal impact on clinical outcomes. Structural causal models are used to represent these relationships across modalities. The underlying method involves optimizing a continuous objective function under acyclicity constraints, enabling efficient learning of causal dependencies from high-dimensional inputs. The learned causal graph then serves as prior knowledge to guide representation learning: during feature encoding and fusion, the model gives priority to variables that lie on the causal paths to the target outcome, thereby reducing the influence of spurious or confounding signals.

In tasks such as diagnosis or risk scoring, causal contrastive learning aligns sample representations based on shared causal factors rather than superficial similarity, using a contrastive loss that reflects causal relevance. Counterfactual reasoning further strengthens the framework by generating hypothetical scenarios (e.g., simulating a treatment or a change in risk factor) to assess the model's response to interventions. These principles enhance the model's robustness under distribution shifts and ensure that predictions are grounded in clinically meaningful causal relationships, leading to more reliable and interpretable decision support. This research direction is capable of addressing **Reliability of Models**.

### *8.9. Combining multimodal cues with medical concepts*

Building clinically interpretable healthcare systems requires integrating multimodal information with medical concepts to support transparent and clinically meaningful decisions. By combining data from images, text, speech, and physiological signals, and mapping them to structured medical knowledge, models can better capture key features and align with clinical reasoning.

Multimodal interpretability methods visualize or model intermediate representations to reveal what the model attends to. For example, heatmaps can highlight lesions in images, while text features aligned with image data enhance semantic clarity. Concept bottlenecks map features to interpretable medical concepts such as shape, boundary, and color, enabling more transparent predictions. Combining these two approaches forms a unified framework where modality-specific features are aligned and linked to clinical concepts, enhancing interpretability and trust in real-world applications. This research avenue can effectively respond to **Reliability of Models**

### *8.10. Blockchain-based privacy protection*

Blockchain offers a secure, decentralized way to share medical data, addressing privacy concerns. With smart contracts, encryption, and immutable records, it ensures data security and regulatory compliance. In multimodal learning, blockchain supports secure, cross-institution data sharing, enabling effective model training while protecting patient privacy. This secure framework safeguards privacy in **Privacy and Ethics** and mitigates inconsistency in **Data Challenges** by enabling standardized data sharing.

### *8.11. Hybrid intelligence for efficient labeling*

Hybrid intelligence combines AI and human expertise to tackle labeled data shortages. AI processes and annotates large datasets, flagging key findings, while experts review and refine outputs. This approach speeds up annotation, lowers costs, and improves accuracy, blending computational power with clinical precision. This collaborative approach streamlines annotation in **Label Scarcity** and boosts transparency in Reliability of Models by ensuring high-quality, expert-validated outputs.

### *8.12. Cloud-edge collaborative computing*

Cloud-edge collaborative computing is a computational paradigm in which tasks are jointly distributed between cloud servers and edge devices. This approach enables a rational allocation of workloads: complex and resource-intensive tasks are processed in the cloud, while tasks that require low latency or involve smaller data volumes are handled locally at the edge. For example, in image processing applications, raw image data can be initially preprocessed on the edge device. Only the extracted key features or important information are then transmitted to the cloud for further analysis. This strategy effectively reduces the demand for data transmission bandwidth and lowers computational costs, while significantly improving system response time.

The system dynamically determines where to execute a given task based on factors such as task urgency, data sensitivity and type, and computational complexity. In doing so, it achieves efficient resource scheduling and timely responses [200]. This mechanism is particularly effective in addressing the issues identified in **Real-time and Computational Cost**.

## 9. Conclusion

Multimodal learning based on EHRs has advanced significantly in recent years within the medical field. Future research can focus on two main directions tied to data modalities. First, researchers aim to build a unified general model that captures all relevant data types, such as text, images, and physiological signals, to support various medical applications. Second, there is a growing need to improve performance on specific healthcare tasks as these applications become more detailed and complex.

This survey explores multimodal learning from the perspective of EHR. It starts by examining the features of common EHR data types, including medical texts, visual data, and physiological signals. Then, the review also covers multimodal representation learning, groups existing studies by their learning approaches, and discusses trends in algorithm development. We also discuss key methods for combining and integrating these different data types. Next, it highlights specific healthcare applications, including discriminative tasks like prediction and generative tasks like data generation. Many applications pair text with other data types, and these are further sorted by the combinations used. For discriminative tasks, the review assesses current work, explores data features, and explains how methods connect to modality properties. For generative tasks, it introduces popular generative models before detailing specific examples. Finally, the review identifies challenges in multimodal learning and suggests paths for future research.

## CRediT authorship contribution statement

**Yutao Dou:** Writing – review & editing, Writing – original draft, Conceptualization; **Yaoyu Liu:** Writing – review & editing; **Haitao Zou:**





相比之下，对比学习通过将语义相似样本拉近、将不相似样本推远来优化表示空间，从而促进跨模态的一致性。这些技术使模型能够在预训练期间获得可迁移表示，并且仅需有限的标注数据即可有效适应下游预测任务。与传统的监督范式相比，自监督学习更好地应对了医疗领域中常见的数据异质性和标注稀缺挑战，最终提升了多模态系统在真实临床环境中的鲁棒性和适用性。这一方向可以通过有效利用无标注数据来克服标签稀缺中数据集可用性有限的问题。

### *8.8.* 因果 学习

在基于电子健康记录的多模态学习中，因果学习通过显式建模潜在的因果机制来提高模型可解释性和泛化能力。其核心原则是超越简单的相关性，识别对临床结局具有直接因果影响的变量。结构因果模型被用于表示跨模态的这些关系。其底层方法涉及在无环约束下优化连续目标函数，从而能够从高维输入中高效学习因果依赖。学习得到的因果图随后作为先验知识指导表示学习：在特征编码和融合过程中，模型优先考虑位于通向目标结局的因果路径上的变量，从而减少虚假或混杂信号的影响。

在诊断或风险评分等任务中，因果对比学习基于共享因果因子而非表面相似性来对齐样本表示，使用反映因果相关性的对比损失。反事实推理通过生成假设场景（例如模拟治疗或风险因素变化）来评估模型对干预的响应，从而进一步增强该框架。这些原则增强了模型在分布偏移下的鲁棒性，并确保预测建立在具有临床意义的因果关系之上，从而实现更可靠、更可解释的决策支持。这一研究方向能够解决模型的可靠性问题。

### *8.9.* 将 多模态 线索 与 医学 概念相结合

构建临床可解释的医疗系统，需要将多模态信息与医学概念相结合，以支持透明且具有临床意义的决策。通过整合来自图像、文本、语音和生理信号的数据，并将其映射到结构化的医学知识，模型能够更好地捕捉关键特征并与临床推理保持一致。

**多模态可解释性方法通过可视化或建模中间表示来揭示模型关注的内容。例如，热力图可以突出显示图像中的病灶，而与图像数据对齐的文本特征则能增强语义清晰度。概念瓶颈将特征映射到可解释的医学概念（如形状、边界和颜色），从而实现更透明的预测。将这两种方法相结合，形成一个统一框架，使模态特定特征得以对齐并链接到临床概念，从而增强实际应用中的可解释性和信任。这一研究方向能够有效回应模型可靠性**

### *8.10.* 基于 区块链 的隐私保护

区块链 提供 一 种安全、 去中心化 的 方式 来 共享 医疗数据， 解决 隐私 问题。 通过 智能 合约、 加密 和 不可篡改 记录， 它 确保 数据 安全 和 法规 合规。 在 多模态 学习中， 区块链 支持 安全的、 跨机构 数据 共享， 从而 实现 有效的 模型 训练 同时 保护 患者隐私。 该 安全 框架 在 隐私 方面 提供 保障 **并 伦理考量**， 并 通过 实现 标准化 数据 共享 来 缓解 数据 挑战中 的 不一致性。

### *8.11.* 用于 高效 标注的混合智能

混合智能将人工智能与人类专业知识相结合，以应对标注数据短缺的问题。人工智能处理和标注大型数据集，标记关键发现，而专家则审查并优化输出。这种方法加快了标注速度，降低了成本，并提高了准确率，将计算能力与临床精度融为一体。这种协作方法简化了标签稀缺场景下的标注工作，并通过确保高质量的、经专家验证的输出，提升了模型可靠性中的透明度。

### *8.12.* 云边协同计算

云边协同计算是一种计算范式，其中任务在云服务器和边缘设备之间联合分配。这种方法能够实现工作负载的合理分配：复杂且资源密集型的任务在云端处理，而需要低延迟或涉及较小数据量的任务则在边缘本地处理。例如，在图像处理应用中，原始图像数据可以首先在边缘设备上进行预处理。随后，只有提取的关键特征或重要信息才会被传输到云端进行进一步分析。这一策略有效降低了对数据传输带宽的需求，并降低了计算成本，同时显著改善了系统响应时间。

该系统根据任务紧急性、数据敏感性和类型以及计算复杂度等因素，动态决定在何处执行给定任务。通过这种方式，它实现了高效的资源调度和及时的响应[200]。这一机制在解决实时性和计算成本中识别出的问题方面尤为有效。

## 9. 结论

近年来，基于电子健康记录的多模态学习在医学领域取得了显著进展。未来研究可以聚焦于与数据模态相关的两个主要方向。第一，研究人员旨在构建一个统一通用模型，能够捕获所有相关数据类型，如文本、图像和生理信号，以支持各种医疗应用。第二，随着这些应用变得更加细致和复杂，改进特定医疗任务性能的需求日益增长。

本综述从电子健康记录的视角探讨多模态学习。首先考察常见EHR数据类型的特征，包括医学文本、视觉数据和生理信号。然后，综述还涵盖多模态表示学习，按学习方法对现有研究进行分组，并讨论算法开发的趋势。我们还讨论了组合和整合这些不同数据类型的关键方法。接下来，重点介绍了具体的医疗健康应用，包括预测等判别任务和数据生成等生成任务。许多应用将文本与其他数据类型配对，并按所使用的组合进一步分类。对于判别任务，综述评估当前工作，探索数据特征，并解释方法如何与模态属性相关联。对于生成任务，在详述具体示例之前介绍了流行的生成模型。最后，综述指出了多模态学习中的挑战，并提出了未来研究的路径。

## CRediT 作者贡献 声明

**Yutao Dou:** 写作 – 审阅 & 编辑， 写作 – 原始 草稿， 概念化； **Yaoyu Liu:** 写作 – 审阅 & 编辑； **Haitao Zou:**

Data curation; **Wenwu Zeng:** Methodology; **Kele Xu:** Writing – review & editing, Validation; **Shaoliang Peng:** Funding acquisition.

## Data availability

No data was used for the research described in the article.

## Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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数据 管理； **Wenwu Zeng:** 方法学； **Kele Xu:** 写作 – 审阅 & 编辑，验证； **Shaoliang Peng:** 资金 获取。

## 数据 可用性

未 数据 被 用于 本文 所述 研究 中 的 分析 。

## 利益冲突声明 的 声明

作者 声明 他们 没有 任何 已知的 可能影响 本论文所报告工作的 竞争性经济利益或 个人 关系 。

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## 参考文献

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